

Western MA Per Pupil Expenditure Report

With selected districts for comparison

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Executive Summary

"What gets measured gets managed — even when it's pointless to measure and manage it, and even if it harms the purpose of the organisation to do so"

— V. F. Ridgway, 'Dysfunctional Consequences of Performance Measurements', 1956

"By the time it can be captured in numbers, it's too late."

— Peter Drucker, 'The Effective Executive', 1967

How to use this report:

The report is organized from broad regional patterns down to district-level details. As a descriptive report, it makes no forecasts and offers no recommendations. Since most of this report is descriptive statistical reference material, one way to use it would be as a companion to discussions that depend on meaningful cost comparisons. You could read straight through to build understanding from high-level trends to selected district details. Or you could use this **quick-start guide** to answer two specific questions:

- How do my school district finances compare to Western MA baselines?
- How does the growth rate of my district finances compare?

Quick-start guide:

1. Review the **Cohort comparison: Western MA enrollment cohorts and districts of interest** on the next page.
2. Refer to **Section 1: Western MA traditional district trends—Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2024)** to see how all districts are grouped into cohorts using 2024 enrollment numbers.
3. Locate your district in **Section 3: Specific districts compared to cohorts** to see details of district expense categories, enrollment, Chapter 70 aid, and required net school spending (NSS) and actual NSS compared to cohort benchmarks.

Draft Status: As of **October 26, 2025**, this report is under review and has not been presented to any committee. Questions and comments should be directed to the author, Tim Shores (Leverett and Amherst-Pelham Regional School Committee member) at shores@arps.org.

Understanding Per-Pupil Expenditure in Context

Per-pupil expenditure (PPE) provides an imperfect lens for understanding school budgets. While this single metric obscures the complexity of school systems — shaped by student needs, facility requirements, governance structures, and historical commitments — it remains an accessible standard for state reporting and public discussion. We might as well try to understand it in full. This report examines PPE patterns across Western Massachusetts districts, organizing them into enrollment-based cohorts to identify trends and provide context for budget deliberations.

The Challenge of Interpretation

In February 2025, an Amherst Town Councilor asked why regional school spending was "increasing faster than the town's revenue, especially given the significant long-term reduction in the number of students we teach."¹ This question reflects a common assumption or hope that school budgets should track with municipal revenue and scale proportionally with enrollment. However, evidence of school budget trends suggests different dynamics at work.

As a social service, education is fundamentally labor-intensive. Roughly four-fifths of costs are tied to personnel — salaries, health insurance, and retirement contributions.² National data shows school districts dedicate nearly 90% of instructional budgets to staff and benefits, more than double the 35-40% typical in other organizations.³ These costs respond to regional and sectoral labor markets rather than local tax revenues, with the cost of education service historically rising faster than general inflation.⁴ Eligible students have a legal right to special education services regardless of overall enrollment changes. Fixed and semi-fixed costs — debt service, insurance, transportation contracts, utilities, minimum staffing requirements, and facility operations — cannot be reduced proportionally with enrollment. Charter school transfers result in a net loss of state aid to tuition payments and often concentrate higher-need student populations in the traditional schools.

These factors are essential for understanding school budgets, but they can distract us from two key questions: What are our actual growth rates over time, and what was our starting point?

Key Finding: Consistent Growth from Elevated Baselines

Analysis of spending patterns in Amherst-Pelham, Amherst, Leverett, Pelham, and Shutesbury reveals an important pattern: per-pupil expenditure in these districts has grown at rates comparable to — and often lower than — their Western Massachusetts peers over the 2009-2024 period. Their current high PPE reflects not exceptional recent growth, but the persistence of historically elevated spending levels. Given typical budget growth, a significant driver of today's high PPE is yesterday's high PPE.

In 2009, the selected districts already maintained PPE significantly above regional averages. With typical growth rates applied to above-average baselines, the spending gap has remained relatively constant. This finding should shift our focus from addressing perceived runaway costs to understanding how communities can respond to spending levels established by educational commitments made by previous generations living in different economic and policy contexts.

Implications

This report breaks down PPE and compares expenses by category, state aid, and local contributions across enrollment-based district cohorts. These groupings allow comparison of districts facing similar scale challenges. The goal is to support informed deliberation about education funding within the broader context of regional economic conditions and policy decisions.

Understanding whether education costs have grown predictably or exceptionally is a question we can answer by looking at data collected by the Commonwealth. If we learn that school budgets tend to grow at 4%, then we should plan for 4% school budget growth and recognize that less growth means accepting risk to education adequacy and student outcomes. Schools are not like businesses, factories, nonprofit organizations or hospitals. Schools are like schools, and we expect to fund them like schools. If local revenue capacity cannot support predictable school budget growth, communities face difficult choices. School committees can identify educational needs and purposeful spending within available resources, but their principal concern is to act in the best interests of the district and the students. Leadership on deciding what resources are available falls to appropriating bodies and, ultimately, to voters.

In the end, we are likely to find that our structural funding challenges will persist until the Commonwealth finds the courage to deliver policy changes that no region, town or school district can achieve alone.

For a good start on how to advocate for meaningful change, see the [July 2022 report A Sustainable Future for Rural Schools](#), the steadfast work of the champions at [Rural Schools MA](#), and the September 2025 draft report by the Amherst-Pelham Regional School Committee's Fiscal Sustainability Subcommittee.⁵

References

1. [Amherst Indy, "Letter: Correcting Councilor Ryan on the Regional School Budget"](#)
<https://www.amherstindy.org/2025/02/07/letter-correcting-councilor-ryan-on-the-regional-school-budget/>
2. [4 Towns Meeting & Special Meeting of the Regional School Committee, "School Budgeting Basics and FAQ", November 2021](#)
<https://go.boarddocs.com/ma/arps/Board.nsf/goto?open&id=C8SLXU58B56B>
3. [American Association of School Administrators, "School Budgets 101"](#)
<https://www.aasa.org/docs/default-source/resources/reports/school-budgets-101.pdf>
4. [Where's the Money Gone? Changes in the Level and Composition of Education Spending \(1995\)](#) and [Inflation and the Measurement of School Spending \(1996\)](#)
https://www.epi.org/publication/books_where_money_gone/
<https://nces.ed.gov/pubs97/97535/97535jx1.asp>
5. [Rep. Blais - Rural Schools](#), [Rural Schools Advocacy in Massachusetts](#), and [RSC FSS Draft Report](#).
<https://www.repblais.org/ruralschools>
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<https://go.boarddocs.com/ma/arps/Board.nsf/goto?open&id;=DLM3SC083619>

All data in this report comes from the Massachusetts Department of Elementary and Secondary Education (DESE). See Appendix A for detailed source information.

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Executive Summary (continued)

Total PPE comparison: Western MA enrollment cohorts and selected districts

Shading vs baseline: $|\Delta\$/\text{pupil}| \geq 5.0\%$, $|\Delta\text{CAGR}| \geq 0.5\text{pp}$

Above baseline

Baseline

$\text{CAGR} = (\text{End}/\text{Start})^{(1/\text{years})} - 1$

Below baseline

Table 1

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA (all, excl. Springfield)	\$12,234	+4.0%	+4.6%	+6.0%	\$22,065
Western MA Tiny (0-200 FTE)	\$14,738	+4.2%	+4.4%	+6.0%	\$27,504
Western MA Small (201-800 FTE)	\$12,523	+4.6%	+5.1%	+6.7%	\$24,622
Western MA Medium (801-1600 FTE)	\$13,032	+4.1%	+4.7%	+6.1%	\$23,785
Western MA Large (1601-10K FTE)	\$11,775	+3.9%	+4.5%	+5.9%	\$20,841
Outliers (Springfield)	\$14,608	+4.1%	+5.7%	+9.4%	\$26,615

Table 2

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA Medium (801-1600 FTE)	\$13,032	+4.1%	+4.7%	+6.1%	\$23,785
Amherst-Pelham Regional	\$16,211	+3.8%	+3.8%	+4.8%	\$28,233
Amherst	\$16,029	+4.6%	+4.7%	+5.7%	\$31,267

Table 3

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA Tiny (0-200 FTE)	\$14,738	+4.2%	+4.4%	+6.0%	\$27,504
Leverett	\$15,156	+2.5%	+0.5%	+2.0%	\$21,910
Pelham	\$14,733	+3.0%	+3.3%	+5.9%	\$23,040
Shutesbury	\$14,011	+4.4%	+5.3%	+5.0%	\$26,908

Table 1 compares PPE and cost growth of Western MA school district cohorts with the benchmark of all Western MA districts. **Tables 2 and 3** benchmark districts against their cohort averages.

Cohorts are organized by enrollment size. Grouping districts by enrollment size enables more meaningful cost comparisons—districts of similar size face similar challenges with administration, staffing, facilities, and program requirements. Detailed information about cohorts follow in **sections 1 and 2**. Detailed comparisons of individual districts to cohorts follow in **sections 2 and 3**.

Enrollment size is just one factor. Districts in the same cohort can differ significantly in wealth, demographics, geography, facility age, and governance — differences that PPE figures can obscure. This report examines revenue and expenditure subcategories over time rather than relying on top-line comparisons, but these breakdowns alone cannot determine which districts are 'efficient' or 'wasteful.' Such judgments require local knowledge of community priorities, student needs, and historical context. What this analysis can do is reveal patterns and anomalies that invite well-grounded investigation of education costs across Western Massachusetts.

Note: The following districts are omitted from this analysis: Chesterfield-Goshen (missing expenditure data), Hampshire (missing expenditure data), Southampton (missing expenditure data), Warwick (no enrollment data), Westhampton (missing expenditure data), Williamsburg (missing expenditure data).

Executive Summary (continued)

Chapter 70 Aid (per foundation pupil): Western MA enrollment cohorts and selected districts

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta CAGR| \geq 0.5pp$

Above baseline

Baseline

$CAGR = (End/Start)^{(1/years)} - 1$

Below baseline

Table 4

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA (all, excl. Springfield)	\$5,444	+3.6%	+4.4%	+6.6%	\$9,313
Western MA Tiny (0-200 FTE)	\$3,297	+2.2%	+2.8%	+5.0%	\$4,547
Western MA Small (201-800 FTE)	\$3,672	+2.7%	+3.1%	+3.8%	\$5,465
Western MA Medium (801-1600 FTE)	\$5,698	+3.2%	+4.0%	+5.8%	\$9,202
Western MA Large (1601-10K FTE)	\$5,742	+3.8%	+4.6%	+7.0%	\$10,065
Outliers (Springfield)	\$10,452	+4.5%	+5.6%	+8.1%	\$20,138

Table 5

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA Medium (801-1600 FTE)	\$5,698	+3.2%	+4.0%	+5.8%	\$9,202
Amherst-Pelham Regional	\$5,773	+2.3%	+2.5%	+2.5%	\$8,068
Amherst	\$4,518	+2.2%	+2.4%	+3.1%	\$6,271

Table 6

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA Tiny (0-200 FTE)	\$3,297	+2.2%	+2.8%	+5.0%	\$4,547
Leverett	\$1,782	+3.1%	+2.3%	+5.5%	\$2,798
Pelham	\$1,983	-0.1%	+1.4%	+2.5%	\$1,963
Shutesbury	\$3,808	+2.9%	+4.6%	+3.5%	\$5,862

Tables 4-6 compare Chapter 70 Aid (per foundation pupil) across Western MA school districts organized by enrollment size.

Executive Summary (continued)

Actual NSS above Required NSS (per foundation pupil): Western MA enrollment cohorts and selected districts

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta CAGR| \geq 0.5pp$

Above baseline

Baseline

$CAGR = (End/Start)^{(1/years)} - 1$

Below baseline

Table 7

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA (all, excl. Springfield)	\$1,394	+5.9%	+8.0%	+2.6%	\$3,301
Western MA Tiny (0-200 FTE)	\$4,523	+5.6%	+6.3%	+4.1%	\$10,285
Western MA Small (201-800 FTE)	\$2,054	+7.0%	+9.1%	+5.9%	\$5,656
Western MA Medium (801-1600 FTE)	\$2,318	+4.2%	+8.8%	+3.3%	\$4,315
Western MA Large (1601-10K FTE)	\$934	+6.7%	+7.7%	+1.4%	\$2,455
Outliers (Springfield)	-\$169	-59.5%	-52.4%	+24.0%	-\$0

Table 8

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA Medium (801-1600 FTE)	\$2,318	+4.2%	+8.8%	+3.3%	\$4,315
Amherst-Pelham Regional	\$3,584	+3.9%	+4.5%	+2.1%	\$6,383
Amherst	\$6,217	+4.3%	+5.8%	+3.9%	\$11,670

Table 9

Cohort/District	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Western MA Tiny (0-200 FTE)	\$4,523	+5.6%	+6.3%	+4.1%	\$10,285
Leverett	\$5,429	+2.4%	+0.6%	-1.9%	\$7,799
Pelham	\$5,561	+4.0%	+6.6%	+7.9%	\$10,086
Shutesbury	\$4,173	+6.4%	+5.7%	+2.5%	\$10,526

Tables 7-9 compare Actual NSS above Required NSS (per foundation pupil) - showing local funding effort beyond state requirements.

Section 1 — Western MA traditional public school district trends

Section 1 examines per-pupil expenditure (PPE) trends across all Western Massachusetts traditional school districts, organized by enrollment-based cohorts.

This section provides a regional overview before diving into cohort-specific details (Section 2) and individual district comparisons (Section 3).

Contents

• **Per-pupil expenditure and growth 2009-2024** compares 2009 to 2024 PPE for Western MA districts. • **Year-over-Year (YoY) growth rates** shows the rate of PPE change from 2009-2024 for Amherst-Pelham, Amherst, Leverett, Pelham, Shutesbury, and their cohorts. • **5-year and 15-year CAGR by district and cohort** shows how compound annual growth rate (CAGR) smooths out the volatility of YoY rates of change, making it easier to understand and compare change in PPE over time. • **Distribution of 2024 enrollment and proposed cohort grouping** shows a histogram of Western MA districts by enrollment with cohorts determined by statistics visualized by the histogram. • **Scatterplot analysis** shows the relationship between district enrollment and PPE for 2024, with a table of cohort members. • **Maps** of district locations and 2024 enrollment cohorts, 2024 PPE vs enrollment cohort baseline, and PPE growth (2009-2024) vs enrollment cohort baseline. • **Statistical associations** between enrollment and PPE.

All districts are grouped into enrollment cohorts based on 2024 in-district FTE using IQR (Interquartile Range) analysis. This grouping enables meaningful peer comparisons while acknowledging that enrollment size is just one factor influencing costs. Districts in the same cohort can differ significantly in wealth, demographics, geography, facility age, and governance—all of which shape costs and outcomes.

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Section 1 — Western MA traditional public school district trends

Per-pupil expenditure and growth: 2009 PPE (lighter) to 2024 PPE (darker segment)

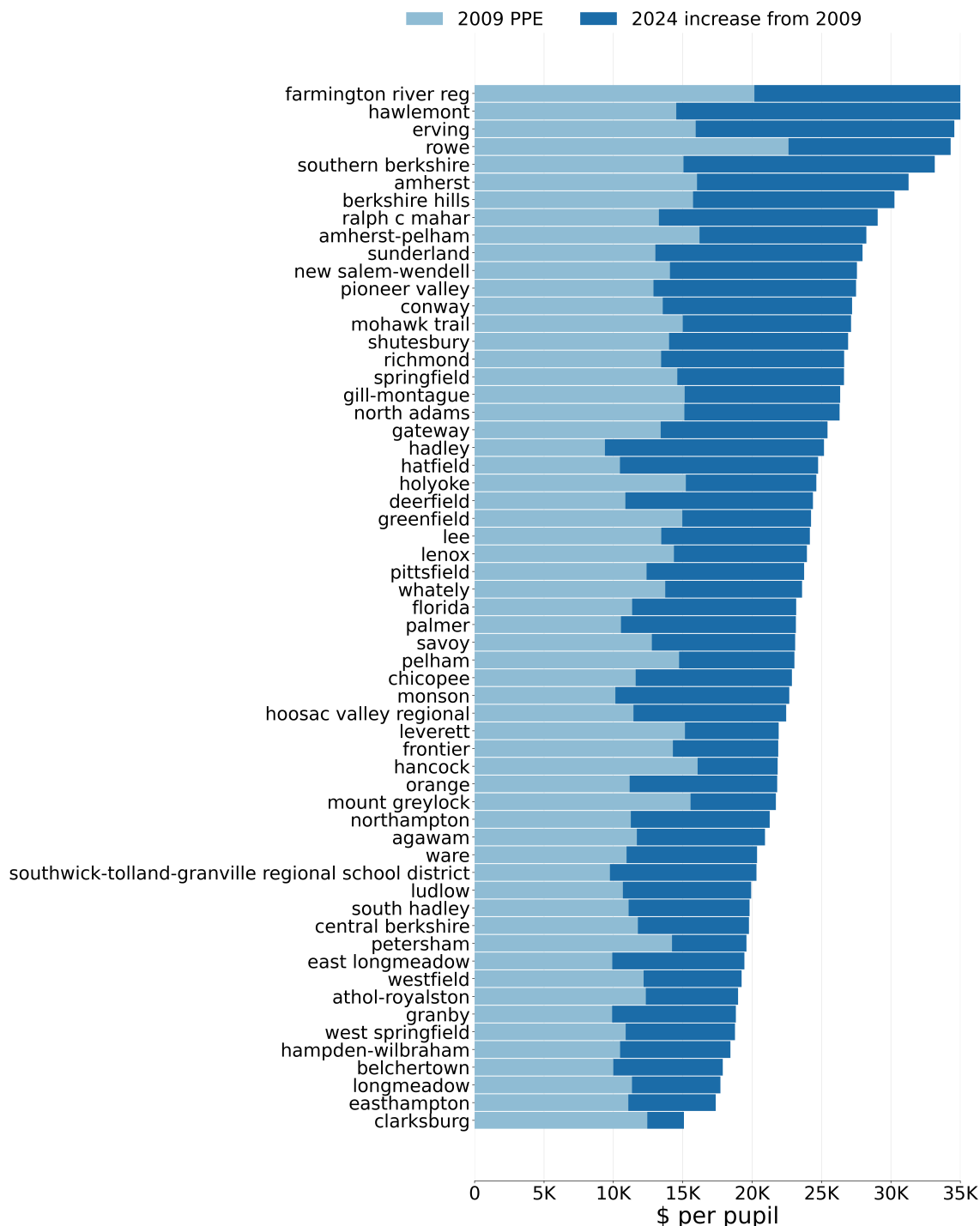


Figure 1

Each bar represents a district's per-pupil expenditures (PPE) in 2024, with growth from 2009 shown by a darker color. Districts are sorted by 2024 spending. While the spread is visible, this alone is nearly meaningless: we don't know to what degree the difference between \$15K and \$30K per student reflects demographic change, education quality, cost-of-living adjustments, special education demand, economies of scale, administrative decisions, ballot box decisions, or legacy cost structures. The next page zeroes in on rates of change in selected districts.

Note: The following districts are omitted from this analysis: Chesterfield-Goshen (missing expenditure data), Hampshire (missing expenditure data), Southampton (missing expenditure data), Warwick (no enrollment data), Westhampton (missing expenditure data), Williamsburg (missing expenditure data).

Section 1 — Western MA traditional public school district trends

Year-over-Year (YoY) growth rates by district and cohort

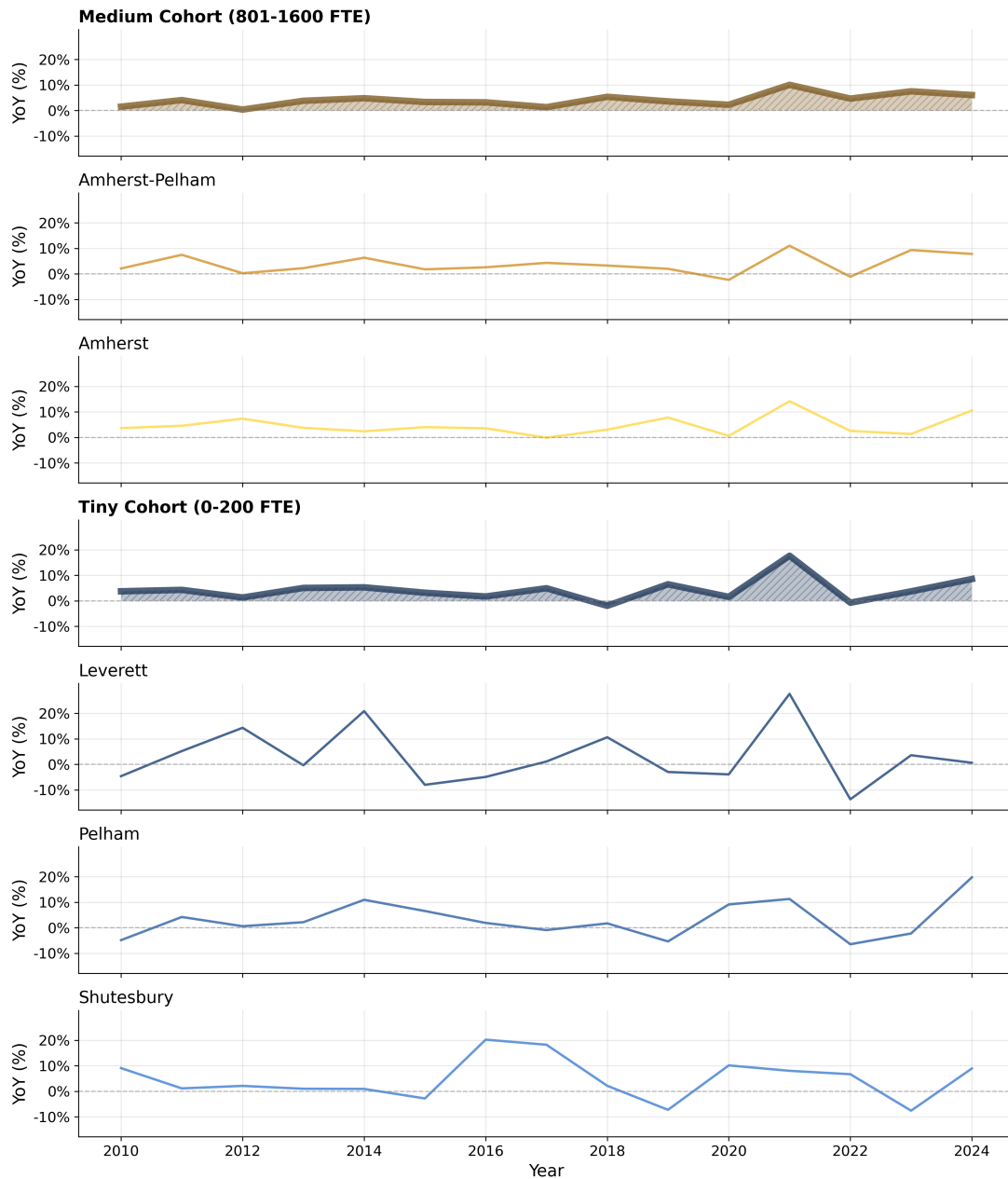


Figure 2

These plots show year-over-year (YoY) growth rates in per-pupil expenditure (PPE) for districts of interest and their enrollment cohorts. Thicker lines represent cohort aggregates, thinner lines show individual districts. Cohort determination is detailed in Appendix A. While patterns emerge, no clear, actionable signal appears across all districts and cohorts over the 2009-2024 period. On the next page, we'll see how a compound growth calculation makes it easier to see similarities and differences between districts.

Section 1 — Western MA traditional public school district trends

5-year and 15-year CAGR by district and cohort

Compound annual growth rates (CAGR) smooth out YoY volatility to reveal long-term trends. Below, districts are color-coded by enrollment cohort (blues for Tiny, golds for Medium), with white diagonal lines marking cohort baselines. Comparing three 5-year periods (2009-2014, 2014-2019, 2019-2024) shows how growth rates have shifted over time. The second chart displays overall 15-year CAGR from 2009 to 2024.

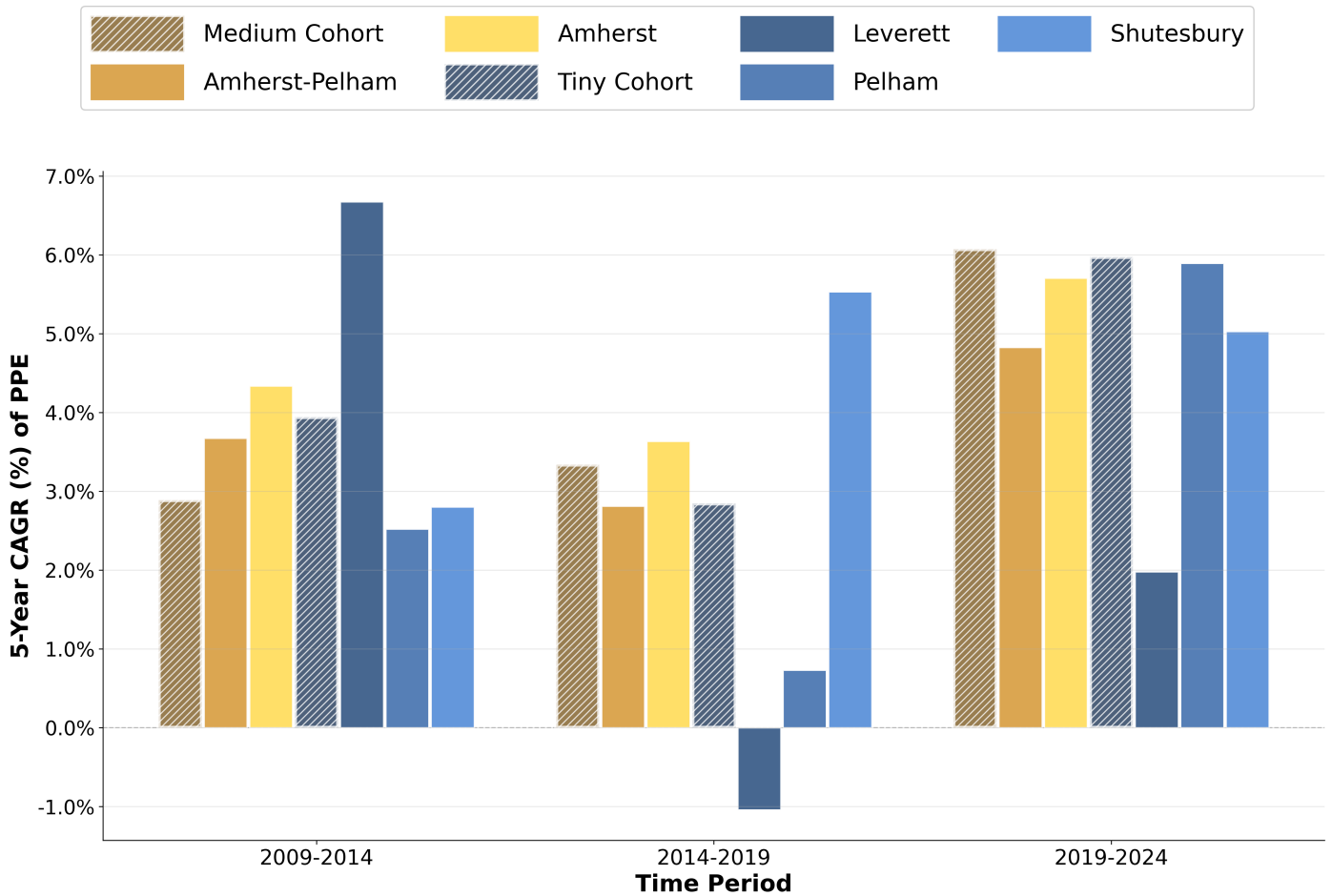


Figure 3

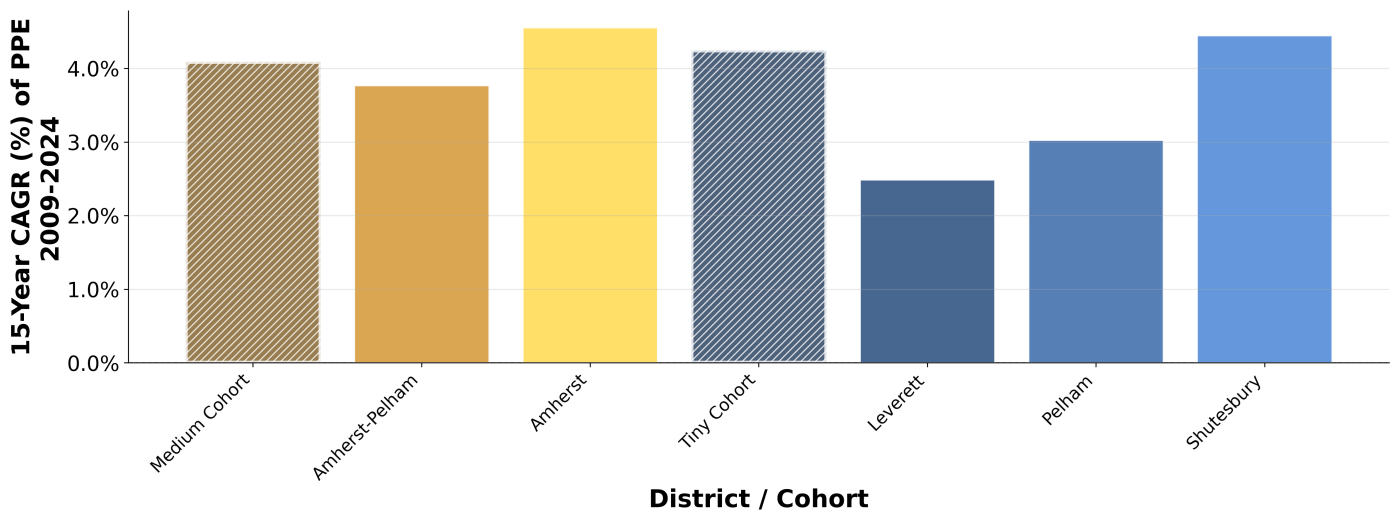


Figure 4

Section 1 — Western MA traditional public school district trends

Distribution of 2024 enrollment and proposed cohort grouping

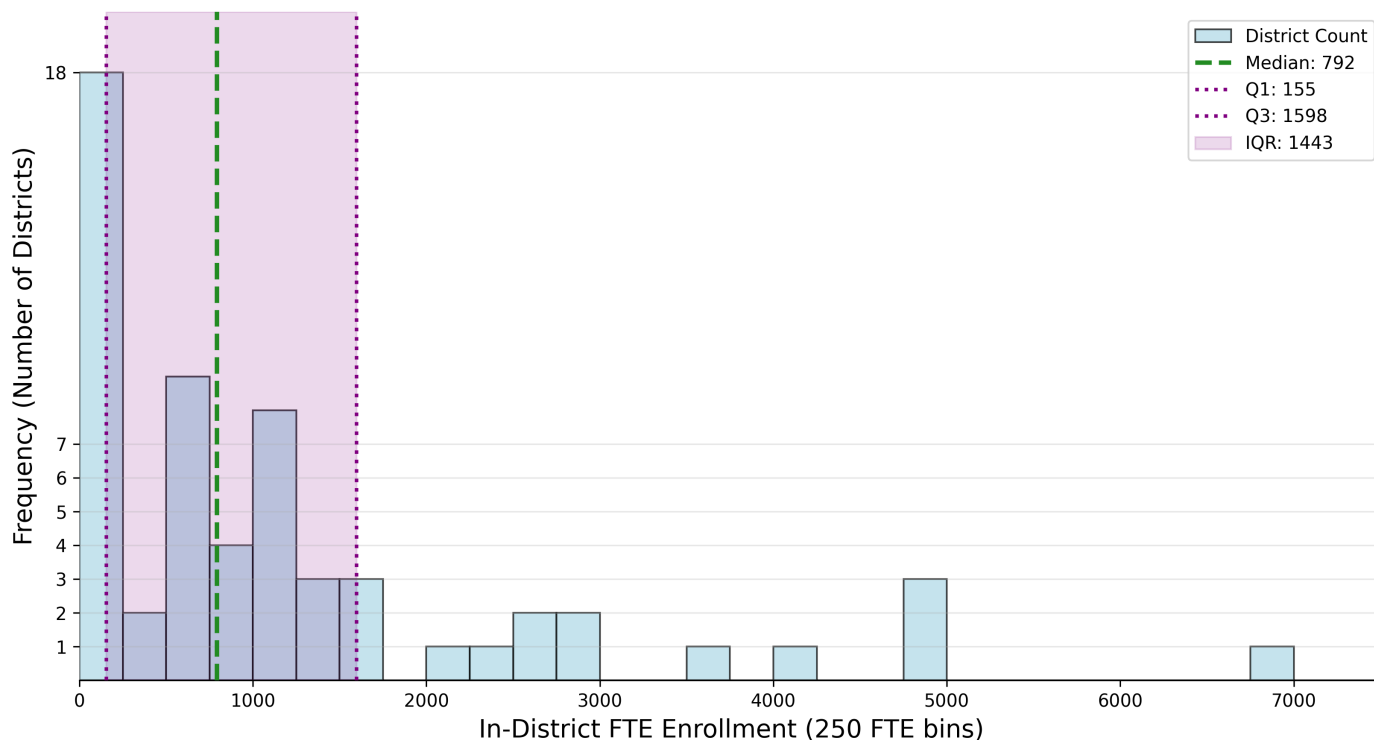


Figure 5

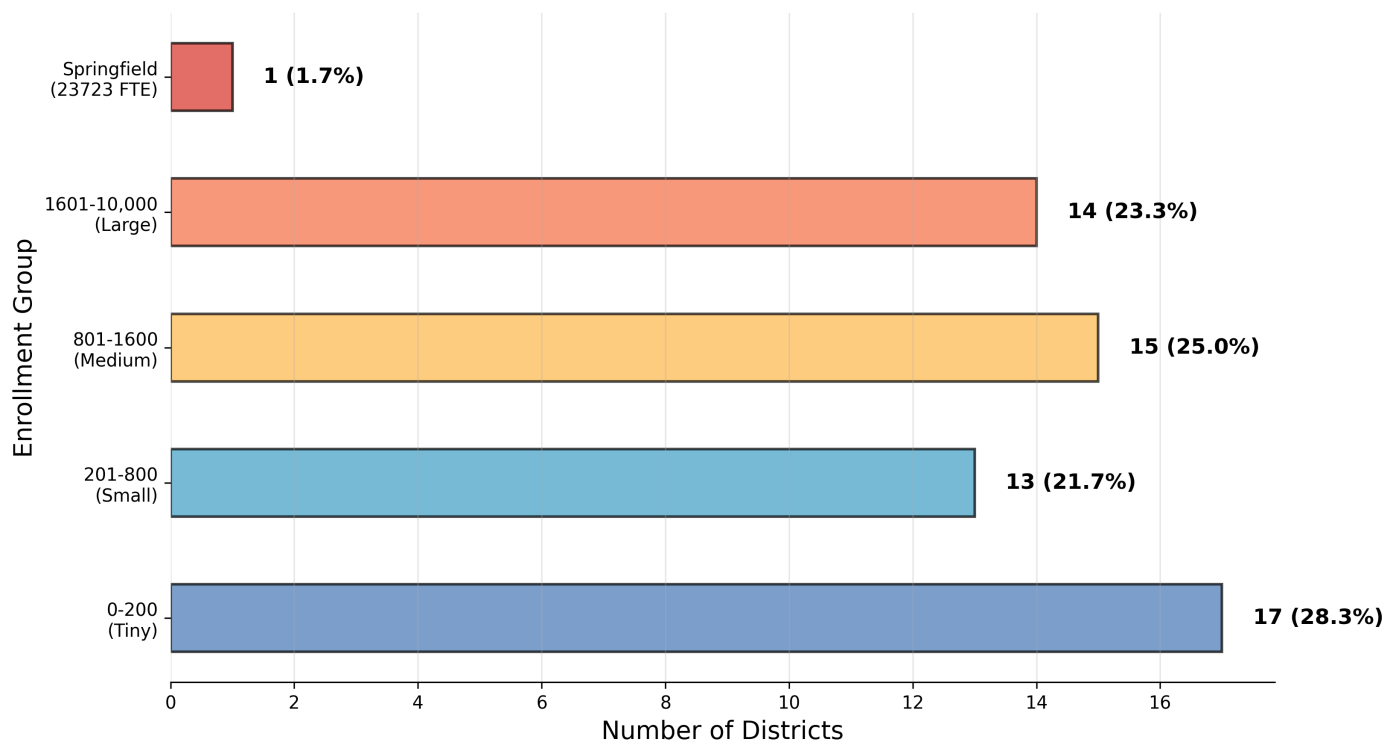


Figure 6

District enrollment cohorts need clear boundaries. To avoid arbitrary cutoffs, this approach uses the natural spread of enrollments. The histogram (top) shows a right-skewed distribution of 2024 enrollments with a long tail of larger districts. Based on interquartile range analysis, districts are grouped into four cohorts (bottom): Tiny (0-200 FTE), Small (201-800 FTE), Medium (801-1600 FTE), and Large (1601-10,000 FTE). Springfield (>10,000 FTE) is excluded as a statistical outlier. These cohorts enable comparison by scale, not to imply spending norms.

The next page presents each district's enrollment and PPE in both scatterplot and reference table format.

Section 1 — Western MA traditional public school district trends

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2024)

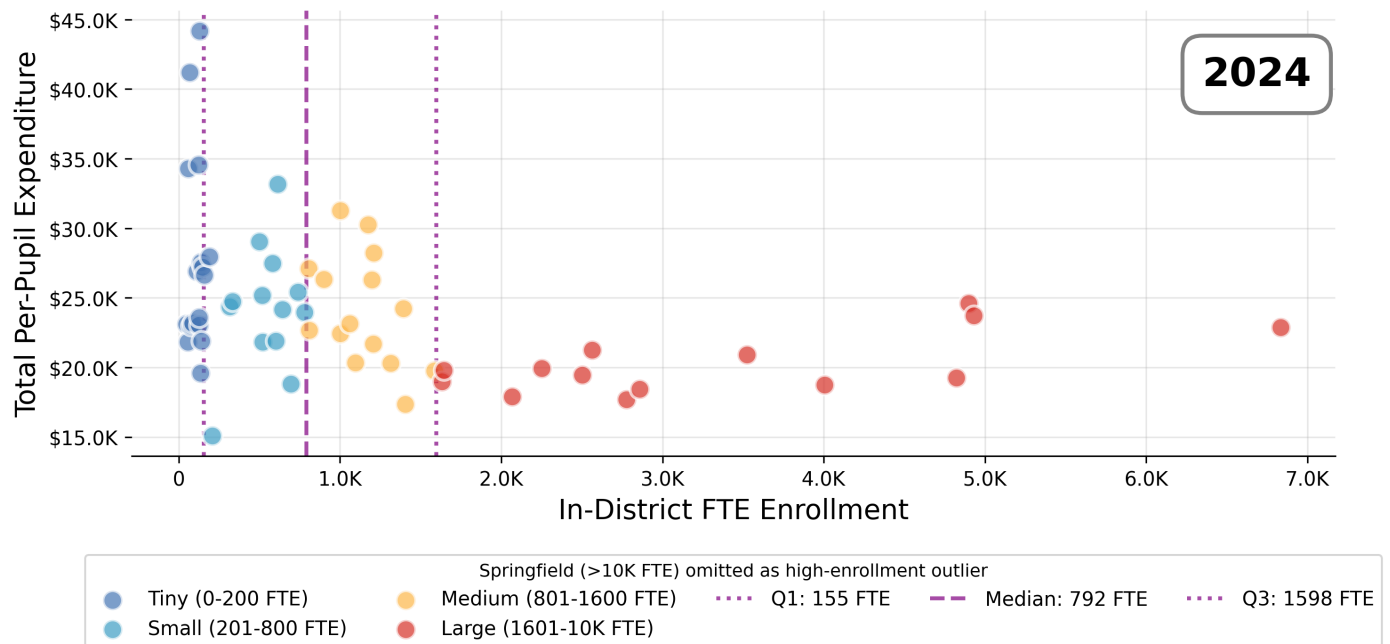


Figure 7

Table 10

District	2024 FTE	2024 PPE ▼	District	2024 FTE	2024 PPE ▼
● Hawlemont	108	\$41,189	● Amherst-Pelham	1,356	\$28,233
● Rowe	86	\$34,297	● Mohawk Trail	951	\$27,124
● New Salem-Wendell	148	\$27,553	● Gill-Montague	1,127	\$26,345
● Conway	155	\$27,200	● North Adams	1,447	\$26,300
● Shutesbury	129	\$26,908	● Gateway	847	\$25,430
● Whately	136	\$23,593	● Lenox	814	\$23,944
● Florida	94	\$23,175	● Palmer	1,234	\$23,154
● Savoy	62	\$23,099	● Monson	910	\$22,675
● Pelham	136	\$23,040	● Hoosac Valley Regional	1,240	\$22,450
● Worthington	140	\$22,985	● Mount Greylock	1,269	\$21,705
● Leverett	145	\$21,910	● Ware	1,310	\$20,348
● Hancock	106	\$21,836	● Southwick-Tolland-Granville Regional School District	1,432	\$20,311
● Petersham	141	\$19,591	● Holyoke	6,165	\$24,619
● Farmington River Reg	254	\$44,178	● Greenfield	1,869	\$24,237
● Erving	208	\$34,566	● Pittsfield	5,806	\$23,739
● Southern Berkshire	758	\$33,162	● Chicopee	7,530	\$22,866
● Ralph C Mahar	622	\$29,040	● Northampton	2,846	\$21,267
● Sunderland	209	\$27,953	● Agawam	3,691	\$20,920
● Pioneer Valley	664	\$27,490	● Ludlow	2,377	\$19,928
● Richmond	210	\$26,631	● South Hadley	1,950	\$19,810
● Hadley	624	\$25,173	● Central Berkshire	1,787	\$19,773
● Hatfield	412	\$24,752	● East Longmeadow	2,600	\$19,444
● Deerfield	355	\$24,382	● Westfield	5,162	\$19,236
● Lee	746	\$24,155	● Athol-Royalston	1,887	\$18,981
● Frontier	656	\$21,881	● West Springfield	4,245	\$18,760
● Orange	593	\$21,809	● Hampden-Wilbraham	2,950	\$18,436
● Granby	781	\$18,824	● Belchertown	2,265	\$17,887
● Clarksburg	262	\$15,078	● Longmeadow	2,826	\$17,710
● Amherst	1,135	\$31,267	● Easthampton	1,712	\$17,371
● Berkshire Hills	1,291	\$30,257	● Springfield	29,468	\$26,615

Section 1 — Western MA traditional public school district trends

Geographic map showing district locations and enrollment cohorts (2024)

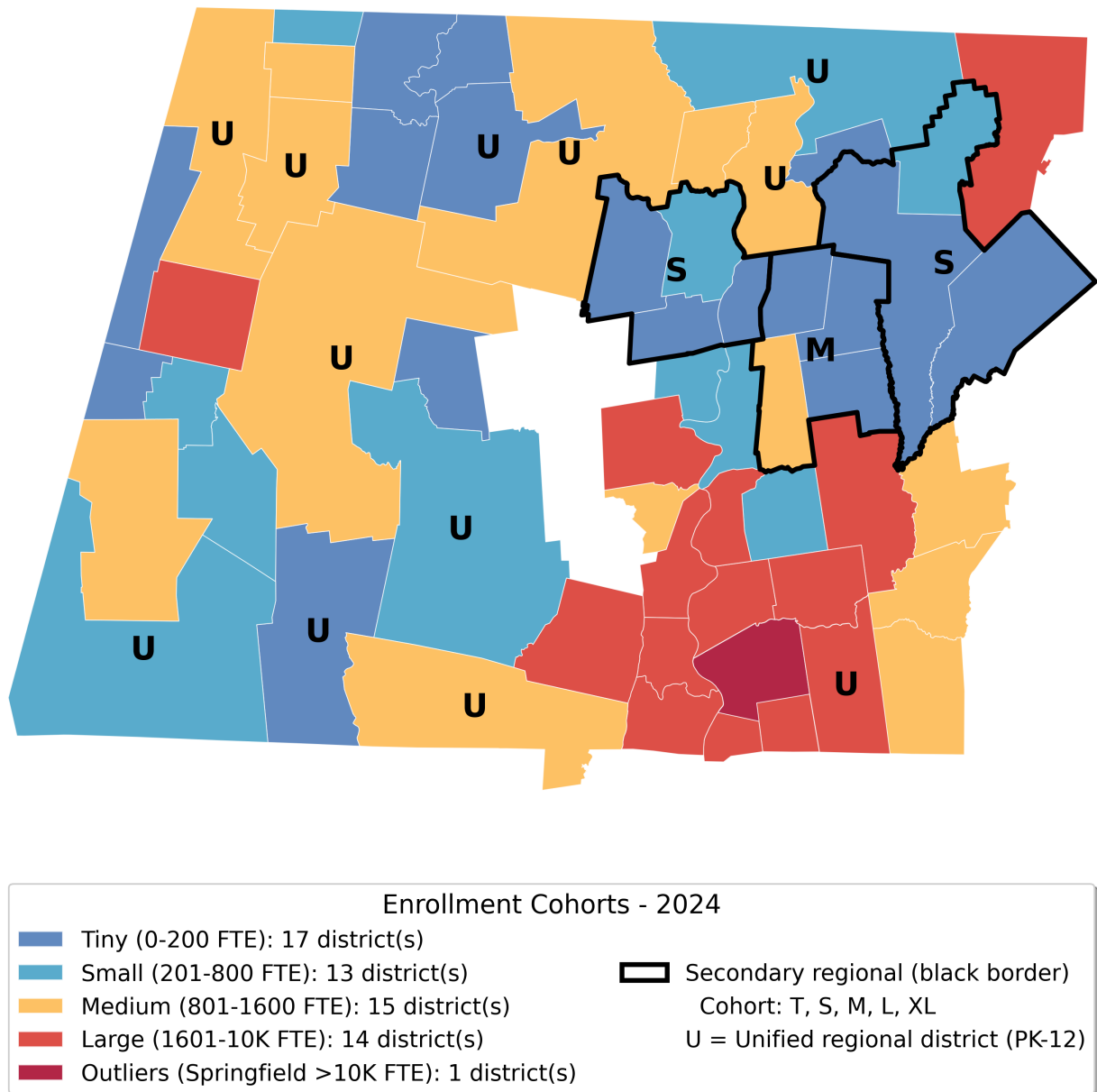


Figure 8

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Section 1 — Western MA traditional public school district trends

Geographic map showing 2024 PPE vs enrollment cohort baseline

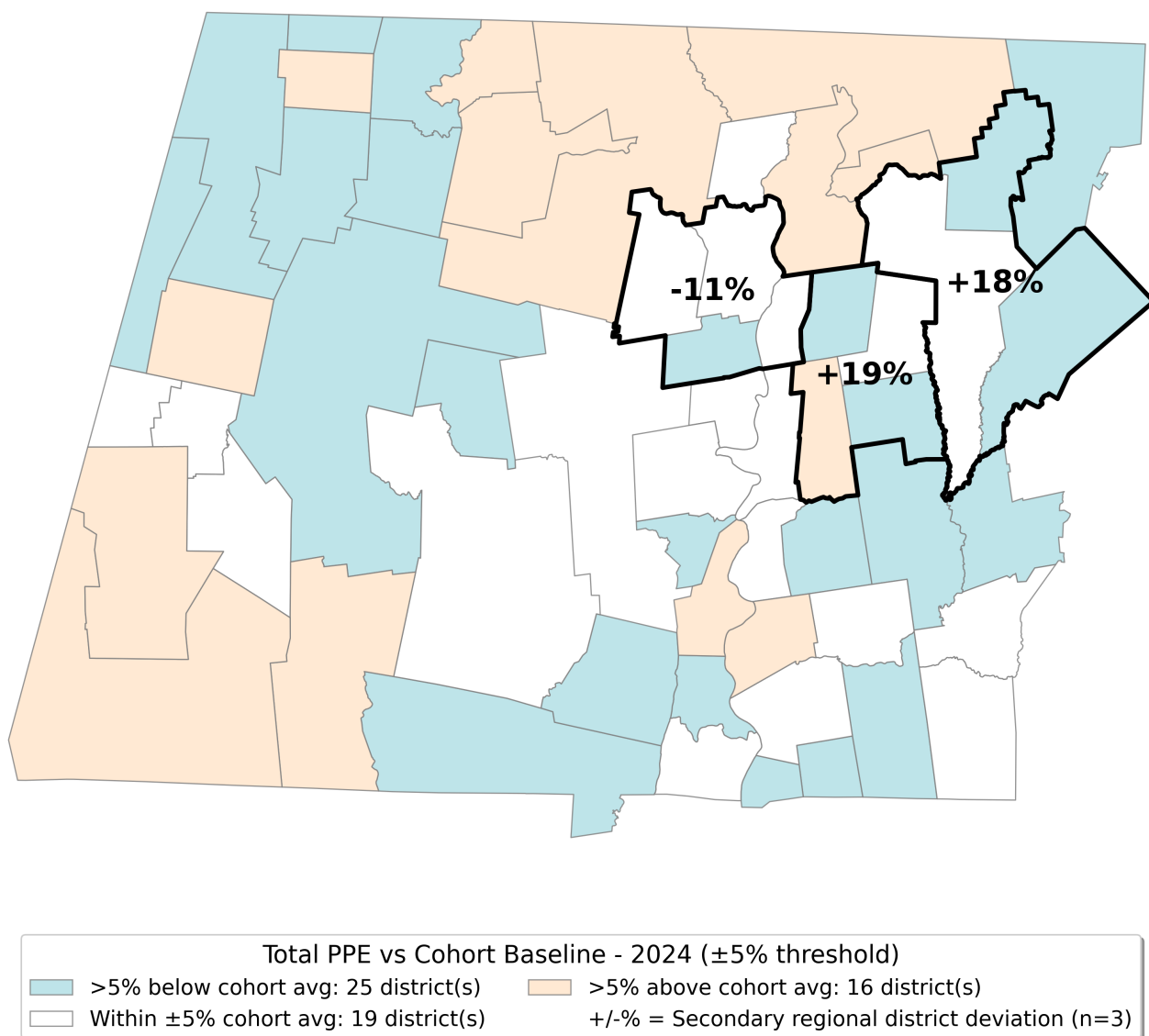


Figure 9

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort).

Section 1 — Western MA traditional public school district trends

Geographic map showing 15-year PPE growth (2009-2024) vs enrollment cohort baseline

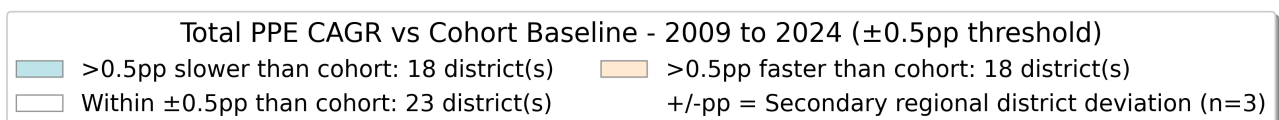
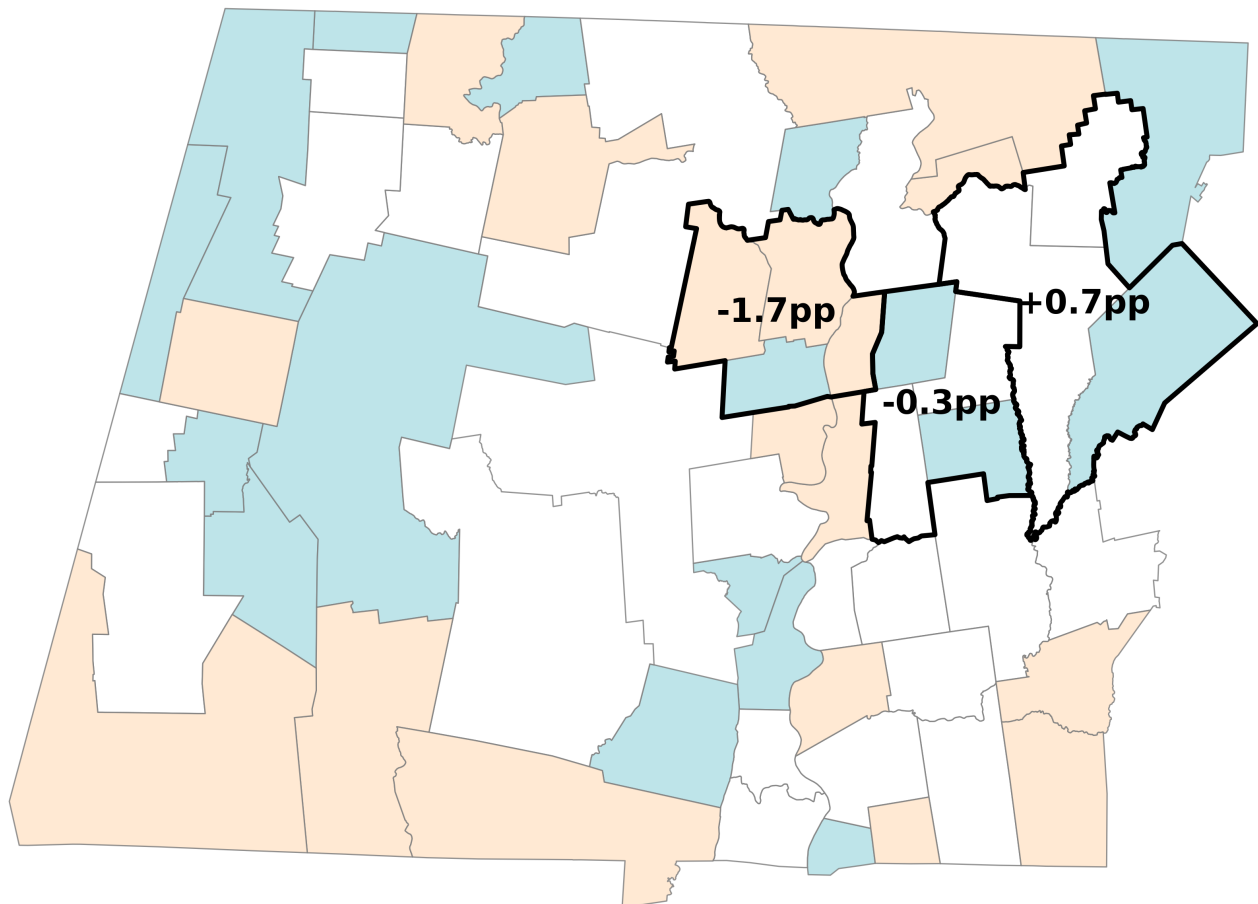


Figure 10

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate.

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Per-Pupil Expenditures

Key Finding

District size and enrollment trends explain very little about spending patterns. Analysis of Western Massachusetts traditional districts (2009-2024) reveals that enrollment cohort (district size) explains only 22% of variation in current spending levels and has no association with spending growth rates. Enrollment growth rates show even weaker associations. This is evidence that knowing a district's size or enrollment trajectory tells us little about its per-pupil expenditure.

Distribution of PPE by Enrollment Cohort

The five-number summaries and box plots below show two critical dimensions: where districts are now (2024 PPE) and how they got there (2009-2024 PPE growth rates).

2024 PPE by Cohort

Box plots show the spread of current spending within each size cohort. Wide boxes indicate greater spending variation; narrow boxes suggest similar spending patterns. The median line shows typical spending for each cohort.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	17	\$19,591	\$23,040	\$26,631	\$27,953	\$44,178	
Small (201-800 FTE)	13	\$15,078	\$21,881	\$24,382	\$25,430	\$33,162	
Medium (801-1600 FTE)	15	\$17,371	\$21,026	\$23,154	\$26,734	\$31,267	
Large (1601-10K FTE)	14	\$17,710	\$18,815	\$19,627	\$21,180	\$24,619	
Outliers (Springfield >10K FTE)	1	\$26,615	\$26,615	\$26,615	\$26,615	\$26,615	

2009-2024 PPE Growth Rates by Cohort

These box plots reveal whether districts of similar size grew at similar rates (narrow boxes) or diverged over time (wide boxes). Comparing medians shows which cohorts experienced faster or slower growth.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	16	2.1%	3.0%	4.5%	5.0%	7.2%	
Small (201-800 FTE)	13	1.3%	4.0%	4.6%	5.4%	6.8%	
Medium (801-1600 FTE)	15	2.2%	3.6%	4.0%	4.6%	5.5%	
Large (1601-10K FTE)	14	2.9%	3.4%	3.9%	4.3%	4.6%	
Outliers (Springfield >10K FTE)	1	4.1%	4.1%	4.1%	4.1%	4.1%	

Pattern Recognition: A cohort with high 2024 PPE but typical growth rates suggests historically elevated spending maintained over time — exactly what we see in our selected districts.

Section 1 — Western MA traditional public school district trends (continued)

Statistical Test Results for Per-Pupil Expenditures

Statistical Test Results for Per-Pupil Expenditures

What Associates with Current Spending (2024 PPE)?

District size (cohort) — WEAK ASSOCIATION

- ANOVA: $F(3,36) = 3.85$, $p = 0.014$, $\eta^2 = 0.216$
- Enrollment cohort explains ~22% of PPE variation
- Effect size: small to medium (substantial variation within cohorts)

Enrollment growth rate — NO ASSOCIATION

- Linear regression: $r = 0.226$, $R^2 = 0.051$, $p = 0.160$
- Explains only 5% of PPE variation
- Not statistically significant at $\alpha = 0.05$

Combined model — MODEST PREDICTIVE POWER

- ANCOVA: $R^2 = 0.232$, Adjusted $R^2 = 0.155$, $F(4,35) = 3.02$, $p = 0.023$
- Size + growth together explain 23% of variation
- Enrollment growth adds minimal predictive value once cohort is accounted for

What Associates with Spending Growth (2009-2024 CAGR)?

District size (cohort) — NO ASSOCIATION

- ANOVA: $F(3,36) = 0.61$, $p = 0.611$, $\eta^2 = 0.036$
- Effect size: negligible
- All cohorts show similar growth rate distributions

Enrollment growth rate — NO ASSOCIATION

- Linear regression: $r = -0.175$, $R^2 = 0.031$, $p = 0.283$
- Weak negative correlation (not significant)
- Declining enrollment doesn't predict higher PPE growth

Combined model — NO PREDICTIVE POWER

- ANCOVA: $R^2 = 0.069$, Adjusted $R^2 = -0.027$, $F(4,35) = 0.74$, $p = 0.569$
- Negative adjusted R^2 indicates model performs worse than sample mean
- Enrollment characteristics don't predict spending trajectories

Interpretation

These analyses examine statistical associations, not causal relationships. The weak associations demonstrate that:

- Districts of similar size have widely varying spending levels
- Enrollment change doesn't systematically drive PPE change
- Other factors — wealth, demographics, facility age, governance, historical commitments — matter more than enrollment characteristics

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Chapter 70 Aid (per foundation pupil)

Key Finding

District size and enrollment cohort show associations with Chapter 70 Aid patterns. This section examines the distribution of Chapter 70 state aid (per foundation pupil) across Western Massachusetts traditional districts organized by enrollment size.

Distribution of Chapter 70 Aid by Enrollment Cohort

The five-number summaries below show two dimensions: current aid levels (2024 Ch70 Aid) and how they grew over time (2009-2024 growth rates).

2024 Chapter 70 Aid per Foundation Pupil by Cohort

Box plots show the spread of state aid within each size cohort. The median line shows typical aid levels for each cohort. Note: Foundation enrollment includes weighted adjustments for low-income, ELL, and special education students.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	\$1,963	\$3,056	\$4,531	\$6,273	\$11,113	
Small (201-800 FTE)	10	\$1,727	\$2,863	\$4,262	\$7,733	\$13,208	
Medium (801-1600 FTE)	8	\$6,173	\$7,619	\$9,099	\$11,187	\$12,345	
Large (1601-10K FTE)	11	\$2,557	\$5,371	\$7,103	\$12,669	\$19,792	
Outliers (Springfield >10K FTE)	1	\$20,138	\$20,138	\$20,138	\$20,138	\$20,138	

2009-2024 Chapter 70 Aid Growth Rates by Cohort

These box plots reveal whether districts of similar size received similar aid growth (narrow boxes) or diverged over time (wide boxes). Comparing medians shows which cohorts experienced faster or slower aid increases.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	\$-1	\$1	\$2	\$4	\$13	
Small (201-800 FTE)	10	\$0	\$1	\$3	\$3	\$6	
Medium (801-1600 FTE)	8	\$2	\$2	\$3	\$4	\$4	
Large (1601-10K FTE)	11	\$1	\$2	\$3	\$4	\$5	
Outliers (Springfield >10K FTE)	1	\$4	\$4	\$4	\$4	\$4	

Section 1 — Western MA traditional public school district trends (continued)

Statistical Test Results for Chapter 70 Aid (per foundation pupil)

Statistical Test Results for Chapter 70 Aid

What Associates with Current Chapter 70 Aid (2024 Ch70 per foundation pupil)?

District size (cohort) — MODERATE ASSOCIATION

- ANOVA: $F(3,36) = 6.12$, $p = 0.002$, $\eta^2 = 0.338$
- Enrollment cohort explains ~34% of Ch70 Aid variation
- Effect size: medium (smaller districts receive higher per-pupil aid due to foundation formula minimum)

Enrollment growth rate — MINIMAL ASSOCIATION

- Linear regression: $r = -0.142$, $R^2 = 0.020$, $p = 0.389$
- Explains only 2% of Ch70 Aid variation
- Not statistically significant at $\alpha = 0.05$

Combined model — MODEST PREDICTIVE POWER

- ANCOVA: $R^2 = 0.351$, Adjusted $R^2 = 0.277$, $F(4,35) = 5.46$, $p = 0.002$
- Size + growth together explain 35% of variation
- Enrollment growth adds minimal predictive value once cohort is accounted for

What Associates with Chapter 70 Aid Growth (2009-2024 CAGR)?

District size (cohort) — WEAK ASSOCIATION

- ANOVA: $F(3,36) = 1.28$, $p = 0.297$, $\eta^2 = 0.096$
- Effect size: small
- All cohorts show similar aid growth rate distributions

Enrollment growth rate — NO ASSOCIATION

- Linear regression: $r = -0.084$, $R^2 = 0.007$, $p = 0.616$
- Weak negative correlation (not significant)
- Declining enrollment doesn't predict higher or lower Ch70 growth

Combined model — NO PREDICTIVE POWER

- ANCOVA: $R^2 = 0.103$, Adjusted $R^2 = 0.000$, $F(4,35) = 1.13$, $p = 0.358$
- Near-zero adjusted R^2 indicates model performs no better than sample mean
- Enrollment characteristics don't predict aid growth trajectories

Interpretation

These analyses examine statistical associations, not causal relationships. The findings demonstrate that:

- Chapter 70 Aid levels associate with district size due to foundation budget formula minimums
- Aid growth rates show no association with enrollment characteristics
- State funding policy changes and formula adjustments affect all districts similarly

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Actual NSS (per foundation pupil)

Key Finding

District size and enrollment cohort show associations with local funding effort patterns. This section examines Actual Net School Spending above Required NSS (per foundation pupil) - a measure of spending beyond state-mandated minimums.

Distribution of Actual NSS above Required NSS by Enrollment Cohort

The five-number summaries below show: current local funding effort (2024 values) and how this effort changed over time (2009-2024 growth rates).

2024 Actual NSS above Required NSS per Foundation Pupil by Cohort

Box plots show the spread of spending above requirements within each size cohort. Positive values indicate spending above the state-required minimum; negative values indicate spending below required levels. The median line shows typical local effort for each cohort.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	\$1,085	\$7,274	\$10,086	\$12,440	\$23,329	
Small (201-800 FTE)	10	\$165	\$1,980	\$6,233	\$8,230	\$11,848	
Medium (801-1600 FTE)	8	\$629	\$1,475	\$3,563	\$6,467	\$11,670	
Large (1601-10K FTE)	11	\$-526	\$1,198	\$2,875	\$4,983	\$5,171	
Outliers (Springfield >10K FTE)	1	\$-0	\$-0	\$-0	\$-0	\$-0	

2009-2024 Actual NSS above Required NSS Growth Rates by Cohort

These box plots reveal whether districts of similar size increased local effort at similar rates (narrow boxes) or diverged over time (wide boxes). Comparing medians shows which cohorts increased spending above requirements faster or slower.

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-200 FTE)	15	\$-9	\$4	\$5	\$7	\$18	
Small (201-800 FTE)	10	\$-10	\$2	\$5	\$9	\$19	
Medium (801-1600 FTE)	8	\$-1	\$3	\$4	\$6	\$16	
Large (1601-10K FTE)	11	\$-1	\$4	\$7	\$10	\$14	
Outliers (Springfield >10K FTE)	1	\$60	\$60	\$60	\$60	\$60	

Section 1 — Western MA traditional public school district trends (continued)

Statistical Test Results for Actual NSS above Required NSS (per foundation pupil)

Statistical Test Results for Actual NSS above Required NSS

What Associates with Current Local Funding Effort (2024 Actual NSS above Required NSS)?

District size (cohort) — MODERATE ASSOCIATION

- ANOVA: $F(3,36) = 4.73$, $p = 0.007$, $\eta^2 = 0.283$
- Enrollment cohort explains ~28% of local funding effort variation
- Effect size: medium (wealth and tax base often correlate with district size)

Enrollment growth rate — NO ASSOCIATION

- Linear regression: $r = 0.087$, $R^2 = 0.008$, $p = 0.607$
- Explains less than 1% of funding effort variation
- Not statistically significant at $\alpha = 0.05$

Combined model — MODEST PREDICTIVE POWER

- ANCOVA: $R^2 = 0.291$, Adjusted $R^2 = 0.210$, $F(4,35) = 4.12$, $p = 0.008$
- Size + growth together explain 29% of variation
- Enrollment growth adds no predictive value once cohort is accounted for

What Associates with Local Funding Effort Growth (2009-2024 CAGR)?

District size (cohort) — NO ASSOCIATION

- ANOVA: $F(3,36) = 0.52$, $p = 0.669$, $\eta^2 = 0.042$
- Effect size: negligible
- All cohorts show similar growth rate distributions in local funding effort

Enrollment growth rate — NO ASSOCIATION

- Linear regression: $r = -0.068$, $R^2 = 0.005$, $p = 0.689$
- Weak negative correlation (not significant)
- Declining enrollment doesn't predict changes in funding effort

Combined model — NO PREDICTIVE POWER

- ANCOVA: $R^2 = 0.047$, Adjusted $R^2 = -0.062$, $F(4,35) = 0.51$, $p = 0.730$
- Negative adjusted R^2 indicates model performs worse than sample mean
- Enrollment characteristics don't predict local funding trajectories

Interpretation

These analyses examine statistical associations, not causal relationships. The weak associations demonstrate that:

- Local funding effort varies widely among districts of similar size
- Enrollment change doesn't systematically drive changes in spending above requirements
- Community wealth, property values, governance decisions, and historical commitments matter more than enrollment characteristics

Section 2 — Western MA cohort details

Section 2 provides detailed per-pupil expenditure (PPE) and Net School Spending (NSS) analysis for Western Massachusetts enrollment cohorts: Tiny, Small, Medium, Large, and Springfield. Each cohort section includes:

- Aggregate **expenditure trends by category** (instruction, administration, operations, etc.)
- Aggregate **Chapter 70 aid** and **NSS** patterns over time
- **Enrollment-weighted averages** that represent typical spending across all districts in each cohort

These cohort pages serve as regional baselines for understanding cost patterns by district size.

To skip to **Section 3 — Selected districts**, turn to page 36.

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Section 3: Selected Districts	36

Section 2 — Western MA cohort details

Tiny (0-200 FTE) — PPE and Enrollment — Weighted average per district.

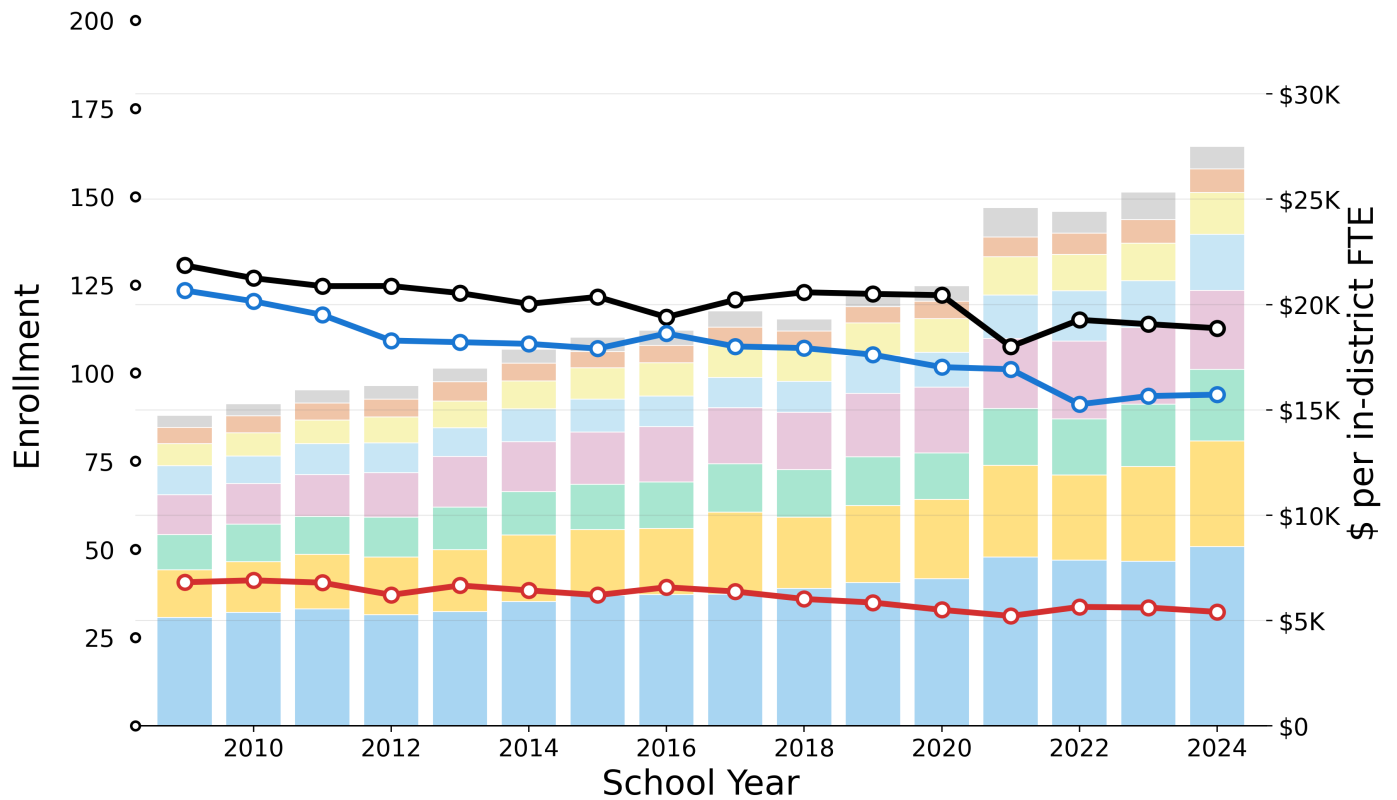


Figure 11

Table 11

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$577	+4.1%	+4.7%	+9.4%	\$1,057
Administration	\$770	+2.6%	+2.8%	+7.9%	\$1,126
Instructional Leadership	\$1,042	+4.4%	+4.2%	+5.2%	\$1,991
Operations and Maintenance	\$1,374	+4.5%	+5.5%	+7.9%	\$2,652
Other Teaching Services	\$1,899	+4.7%	+4.7%	+4.6%	\$3,757
Pupil Services	\$1,669	+4.9%	+5.2%	+8.0%	\$3,406
Insurance, Retirement and Other	\$2,258	+5.5%	+4.8%	+6.5%	\$5,009
Teachers	\$5,148	+3.4%	+3.7%	+4.6%	\$8,507
Total	\$14,738	+4.2%	+4.4%	+6.0%	\$27,504

Table 12

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	131	-1.0%	-0.6%	-1.6%	113
Foundation Enrollment	123	-1.8%	-1.4%	-2.3%	94
Out-of-District FTE Pupils	41	-1.5%	-1.7%	-1.6%	32

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta\text{Enrollment}| \geq 5.0\%$, $|\Delta\text{CAGR}| \geq 0.5\text{pp}$

$\text{CAGR} = (\text{End}/\text{Start})^{(1/\text{years})} - 1$

Above Western MA baseline

Below Western MA baseline

Section 2 — Western MA cohort details

Tiny (0-200 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

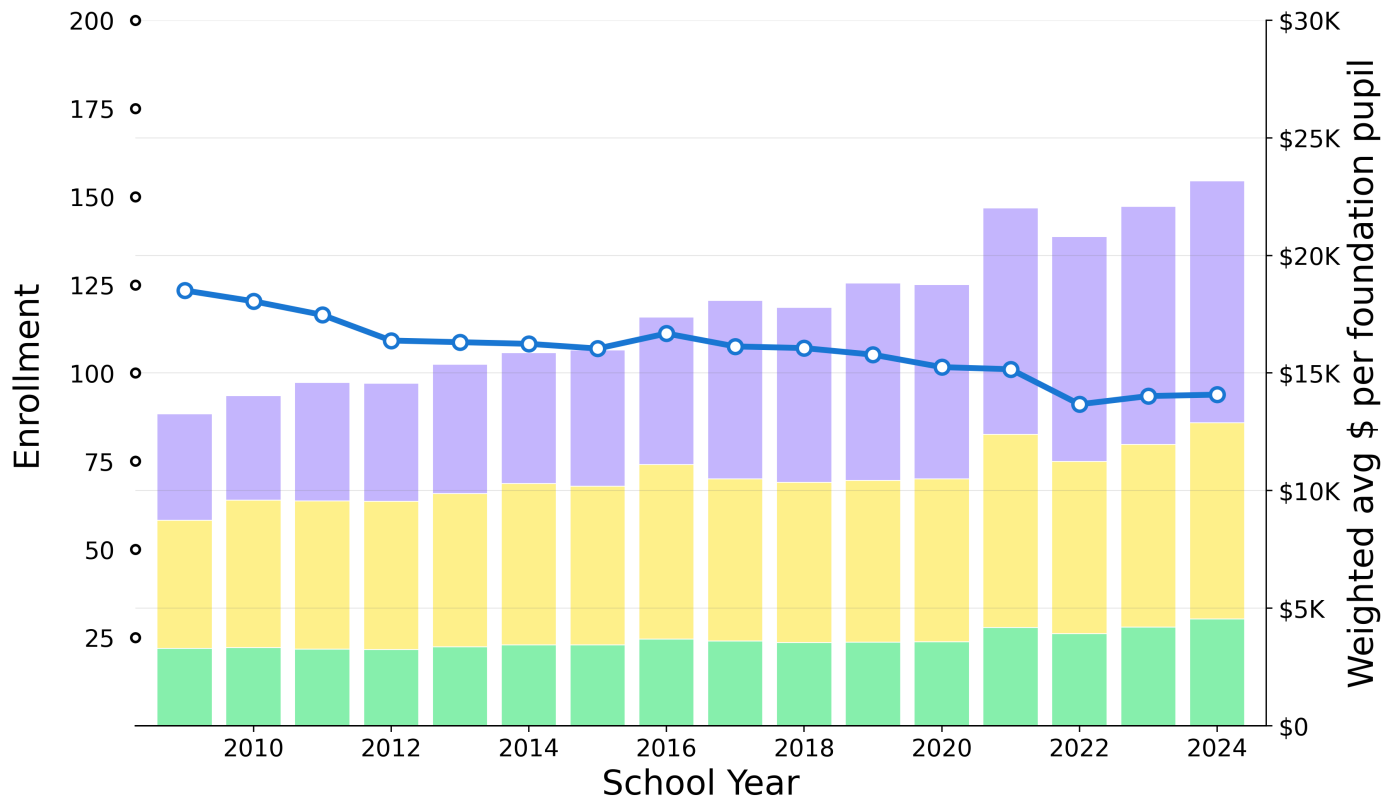


Figure 12

Table 13

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$4,523	5.63%	6.32%	4.13%	\$10,285
Req NSS (minus Ch70)	\$5,438	2.89%	1.98%	3.95%	\$8,338
Ch70 Aid	\$3,297	2.17%	2.82%	5.02%	\$4,547
Total (Actual NSS)	\$13,258	3.79%	3.86%	4.23%	\$23,170
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above Western MA baseline	
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$				Below Western MA baseline	

Tiny (0-200 FTE) includes 17 members:

Districts (15): Conway, Erving, Florida, Hancock, Hawlemont, Leverett, Pelham, Petersham, Richmond, Rowe, Savoy, Shutesbury, Sunderland, Whately, Worthington.

Regions (2): Farmington River Reg, New Salem-Wendell.

Section 2 — Western MA cohort details

Small (201-800 FTE) — PPE and Enrollment — Weighted average per district.

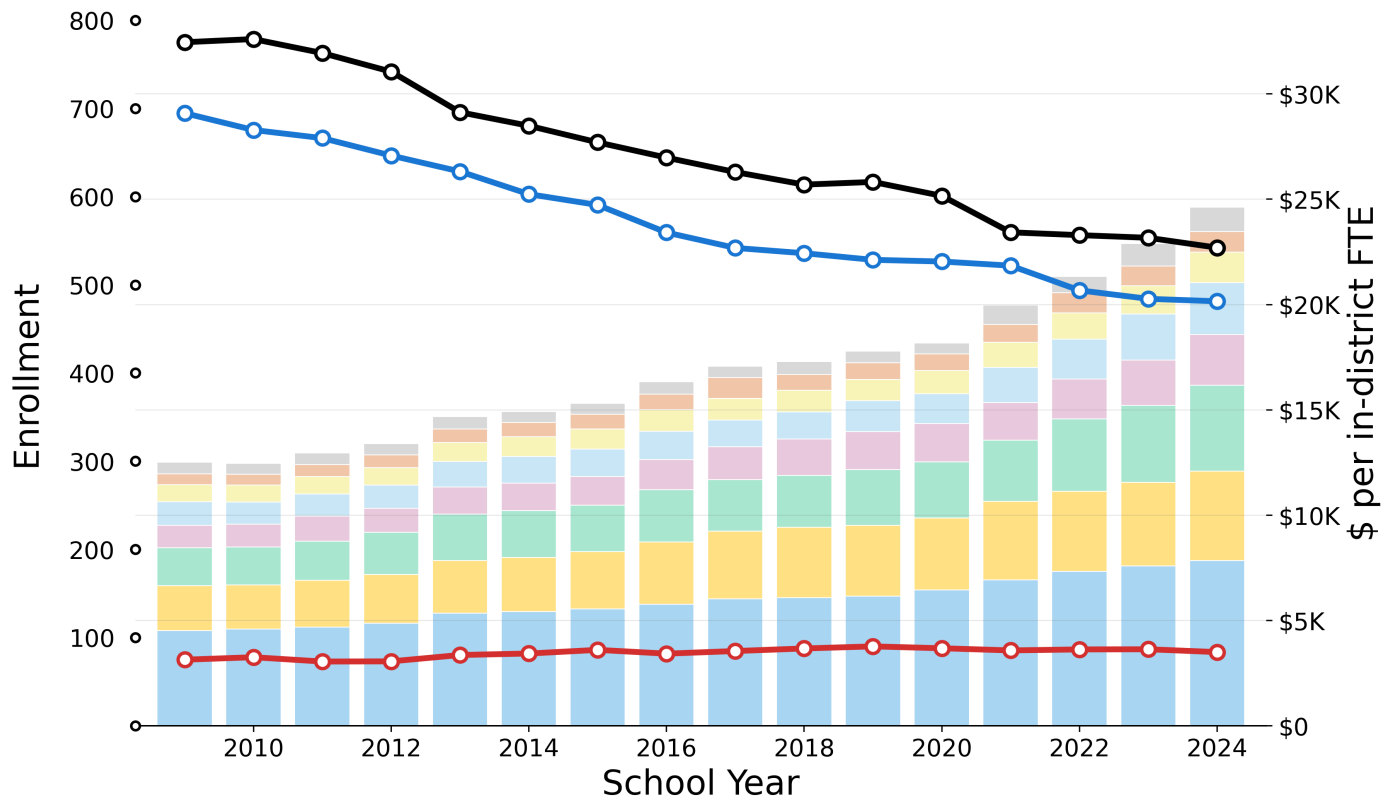


Figure 13

Table 14

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$558	+4.9%	+8.4%	+15.6%	\$1,150
Administration	\$523	+4.3%	+3.9%	+4.4%	\$976
Instructional Leadership	\$811	+4.0%	+4.4%	+7.6%	\$1,463
Operations and Maintenance	\$1,108	+5.4%	+6.9%	+10.8%	\$2,454
Other Teaching Services	\$1,061	+5.6%	+6.2%	+6.0%	\$2,402
Pupil Services	\$1,813	+5.6%	+6.3%	+9.1%	\$4,084
Insurance, Retirement and Other	\$2,128	+4.7%	+5.2%	+4.8%	\$4,254
Teachers	\$4,522	+3.7%	+3.8%	+5.0%	\$7,838
Total	\$12,523	+4.6%	+5.1%	+6.7%	\$24,622

Table 15

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	775	-2.4%	-2.2%	-2.5%	542
Foundation Enrollment	695	-2.4%	-2.2%	-1.8%	481
Out-of-District FTE Pupils	75	+0.7%	+0.2%	-1.5%	84
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta\text{Enrollment} \geq 5.0\%$, $ \Delta\text{CAGR} \geq 0.5\text{pp}$					Above Western MA baseline
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$					Below Western MA baseline

Section 2 — Western MA cohort details

Small (201-800 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

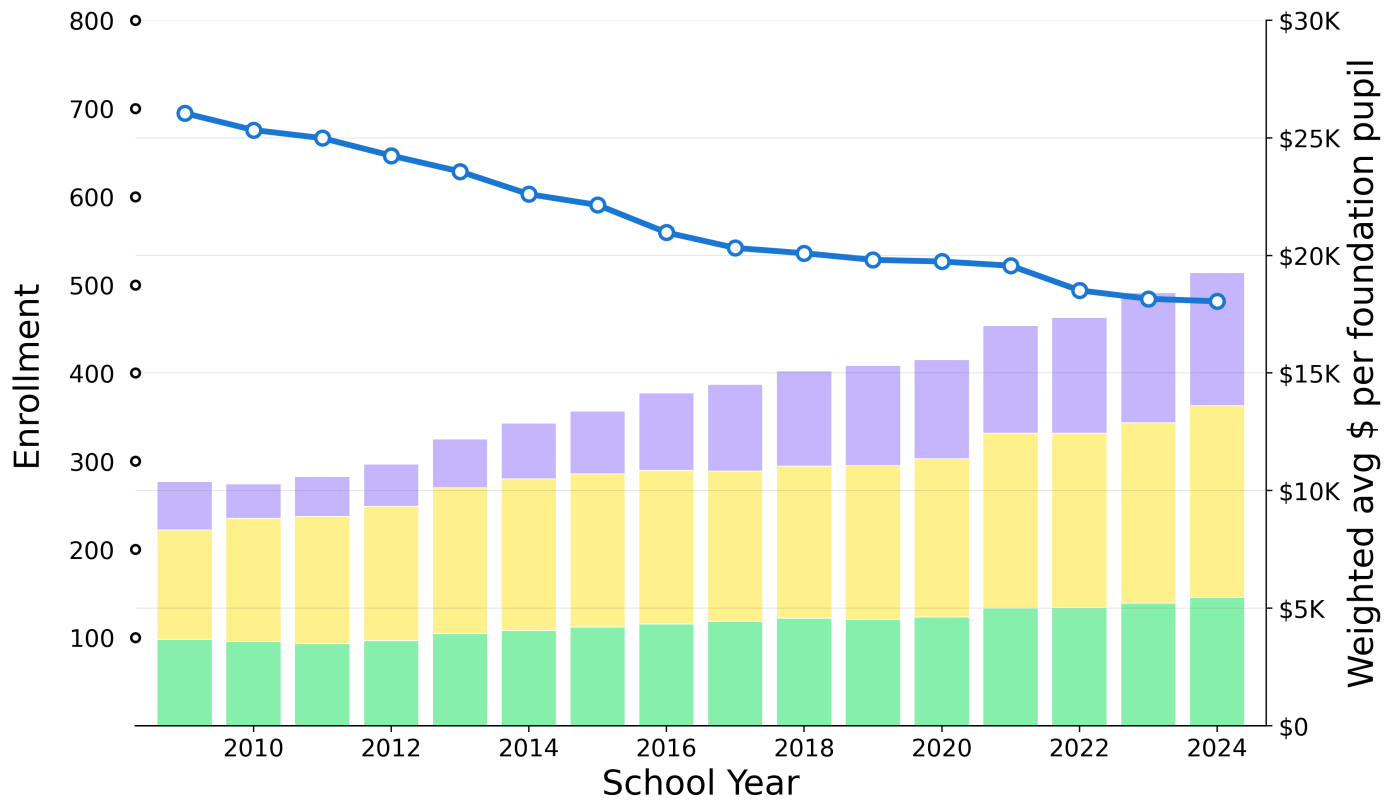


Figure 14

Table 16

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$2,054	6.98%	9.11%	5.86%	\$5,656
Req NSS (minus Ch70)	\$4,657	3.79%	2.35%	4.52%	\$8,139
Ch70 Aid	\$3,672	2.69%	3.05%	3.81%	\$5,465
Total (Actual NSS)	\$10,384	4.20%	4.12%	4.69%	\$19,261
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above Western MA baseline	
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$				Below Western MA baseline	

Small (201-800 FTE) includes 13 members:

Districts (13): Clarksburg, Deerfield, Frontier, Gateway, Granby, Hadley, Hatfield, Lee, Lenox, Orange, Pioneer Valley, Ralph C Mahar, Southern Berkshire.

Section 2 — Western MA cohort details

Medium (801-1600 FTE) — PPE and Enrollment — Weighted average per district.

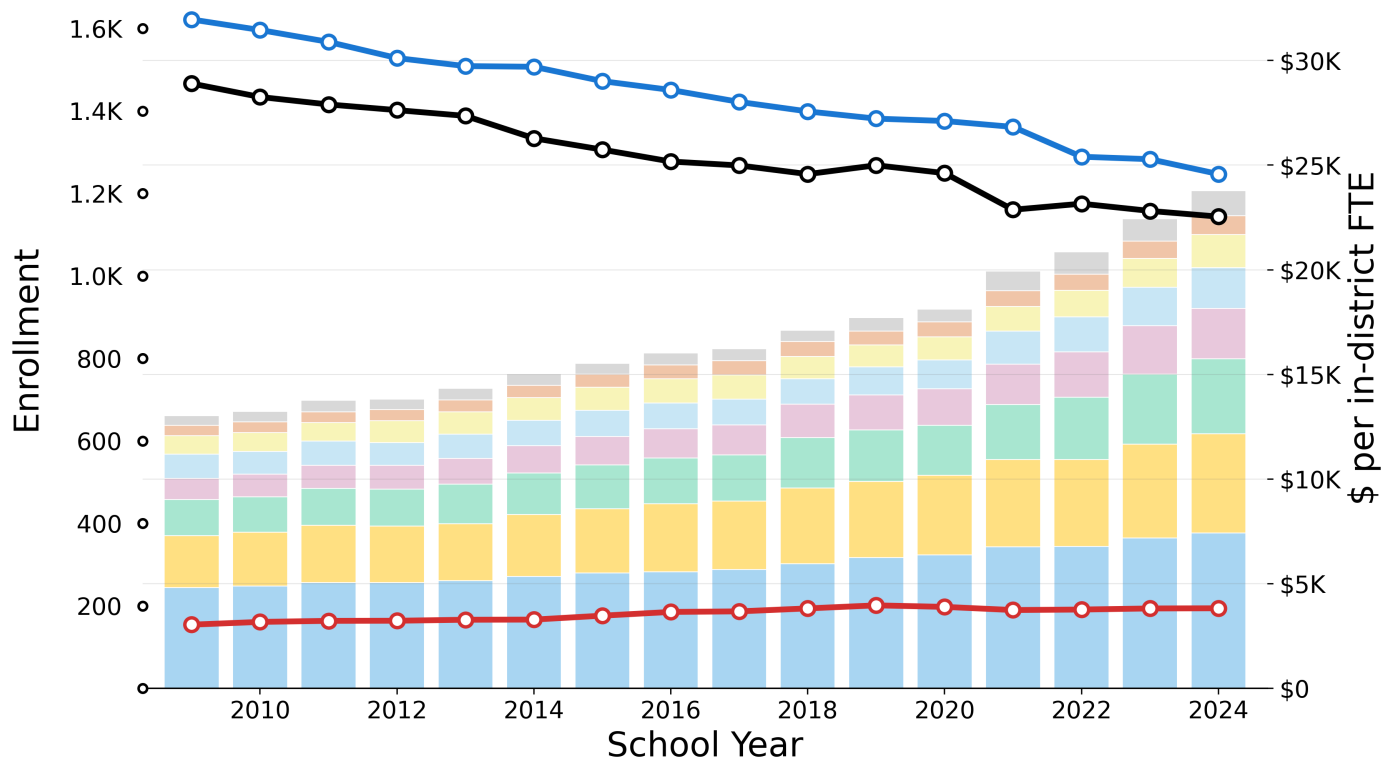


Figure 15

Table 17

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$461	+6.6%	+8.1%	+13.3%	\$1,200
Administration	\$494	+4.2%	+4.7%	+6.9%	\$911
Instructional Leadership	\$893	+3.8%	+3.7%	+8.2%	\$1,573
Operations and Maintenance	\$1,150	+3.6%	+4.9%	+8.0%	\$1,953
Other Teaching Services	\$1,014	+5.9%	+6.2%	+7.3%	\$2,395
Pupil Services	\$1,724	+5.0%	+6.1%	+7.8%	\$3,581
Insurance, Retirement and Other	\$2,483	+4.4%	+4.9%	+5.5%	\$4,740
Teachers	\$4,813	+2.9%	+3.3%	+3.5%	\$7,431
Total	\$13,032	+4.1%	+4.7%	+6.1%	\$23,785

Table 18

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	1,466	-1.6%	-1.5%	-2.0%	1,144
Foundation Enrollment	1,622	-1.7%	-1.9%	-2.0%	1,247
Out-of-District FTE Pupils	154	+1.5%	+1.5%	-0.7%	194
Shading vs baseline: $ \Delta\$/\text{pupil} \geq 5.0\%$, $ \Delta\text{Enrollment} \geq 5.0\%$, $ \Delta\text{CAGR} \geq 0.5\text{pp}$			Above Western MA baseline		
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$			Below Western MA baseline		

Section 2 — Western MA cohort details

Medium (801-1600 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

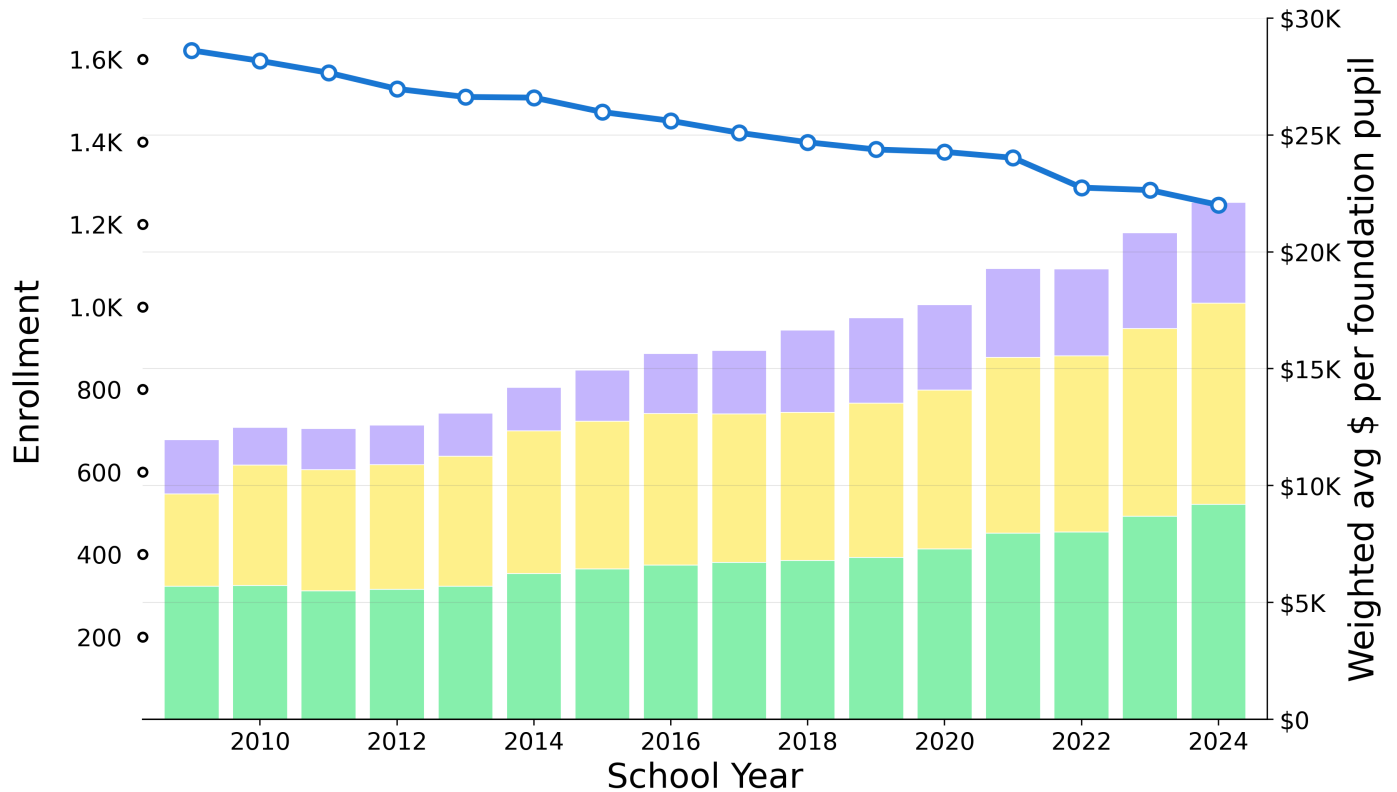


Figure 16

Table 19

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$2,318	4.23%	8.76%	3.35%	\$4,315
Req NSS (minus Ch70)	\$3,945	5.34%	3.49%	5.50%	\$8,605
Ch70 Aid	\$5,698	3.25%	3.97%	5.82%	\$9,202
Total (Actual NSS)	\$11,961	4.18%	4.53%	5.19%	\$22,121
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above Western MA baseline	
CAGR = $(End/Start)^{(1/years)} - 1$				Below Western MA baseline	

Medium (801-1600 FTE) includes 15 members:

Districts (11): Amherst, Berkshire Hills, Central Berkshire, Easthampton, Greenfield, Mohawk Trail, Monson, Mount Greylock, North Adams, Palmer, Ware.

Regions (4): Amherst-Pelham, Gill-Montague, Hoosac Valley Regional, Southwick-Tolland-Granville Regional School District.

Section 2 — Western MA cohort details

Large (1601-10K FTE) — PPE and Enrollment — Weighted average per district.

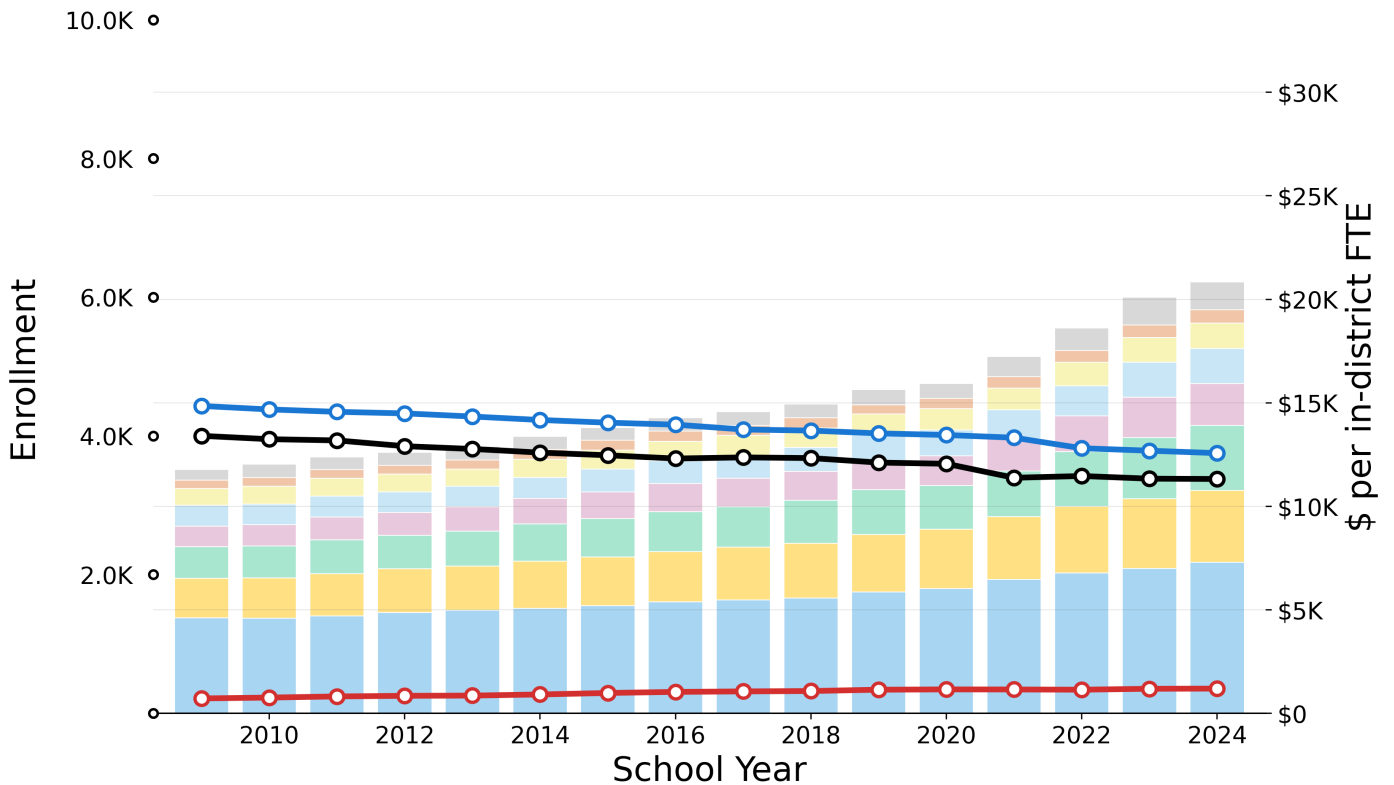


Figure 17

Table 20

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$513	+6.6%	+7.9%	+13.3%	\$1,348
Administration	\$392	+3.4%	+3.5%	+7.5%	\$649
Instructional Leadership	\$811	+2.8%	+3.3%	+4.3%	\$1,231
Operations and Maintenance	\$1,014	+3.4%	+5.1%	+6.2%	\$1,683
Other Teaching Services	\$990	+4.9%	+5.1%	+7.5%	\$2,021
Pupil Services	\$1,531	+4.9%	+5.8%	+7.7%	\$3,148
Insurance, Retirement and Other	\$1,911	+4.1%	+4.3%	+4.6%	\$3,469
Teachers	\$4,613	+3.1%	+3.7%	+4.5%	\$7,292
Total	\$11,775	+3.9%	+4.5%	+5.9%	\$20,841

Table 21

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
— In-District FTE Pupils	4,001	-1.1%	-1.1%	-1.3%	3,379
— Foundation Enrollment	4,432	-1.1%	-1.2%	-1.5%	3,752
— Out-of-District FTE Pupils	215	+3.4%	+2.7%	+0.9%	357

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta\text{Enrollment}| \geq 5.0\%$, $|\Delta\text{CAGR}| \geq 0.5\text{pp}$

$\text{CAGR} = (\text{End}/\text{Start})^{(1/\text{years})} - 1$

Above Western MA baseline

Below Western MA baseline

Section 2 — Western MA cohort details

Large (1601-10K FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

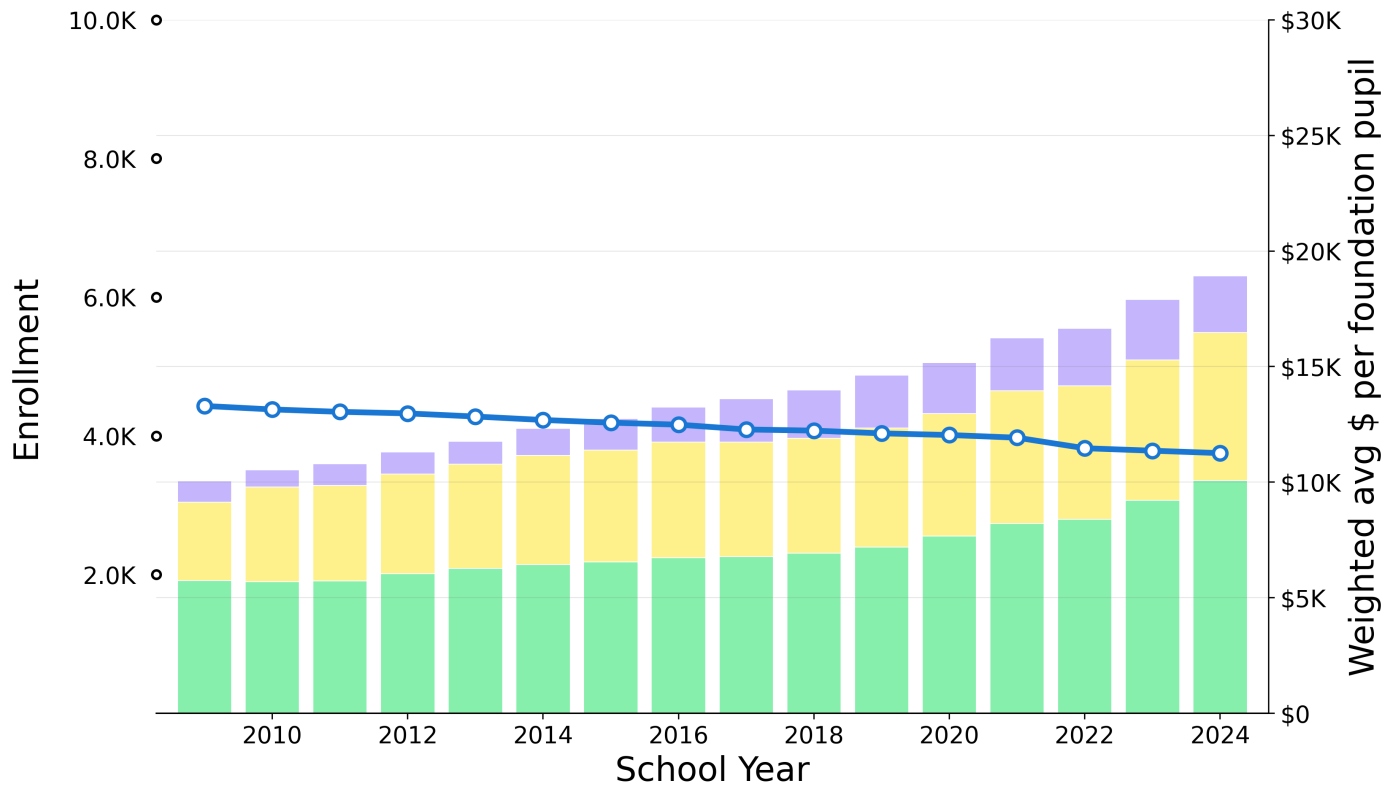


Figure 18

Table 22

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$934	6.66%	7.71%	1.41%	\$2,455
Req NSS (minus Ch70)	\$3,385	4.35%	3.11%	4.46%	\$6,407
Ch70 Aid	\$5,742	3.81%	4.57%	6.96%	\$10,065
Total (Actual NSS)	\$10,060	4.30%	4.39%	5.29%	\$18,927
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above Western MA baseline	
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$				Below Western MA baseline	

Large (1601-10K FTE) includes 14 members:

Districts (12): Agawam, Belchertown, Chicopee, East Longmeadow, Holyoke, Longmeadow, Ludlow, Northampton, Pittsfield, South Hadley, West Springfield, Westfield.

Regions (2): Athol-Royalston, Hampden-Wilbraham.

Section 2 — Western MA cohort details

Outliers (Springfield >10K FTE) — PPE and Enrollment — Weighted average per district.

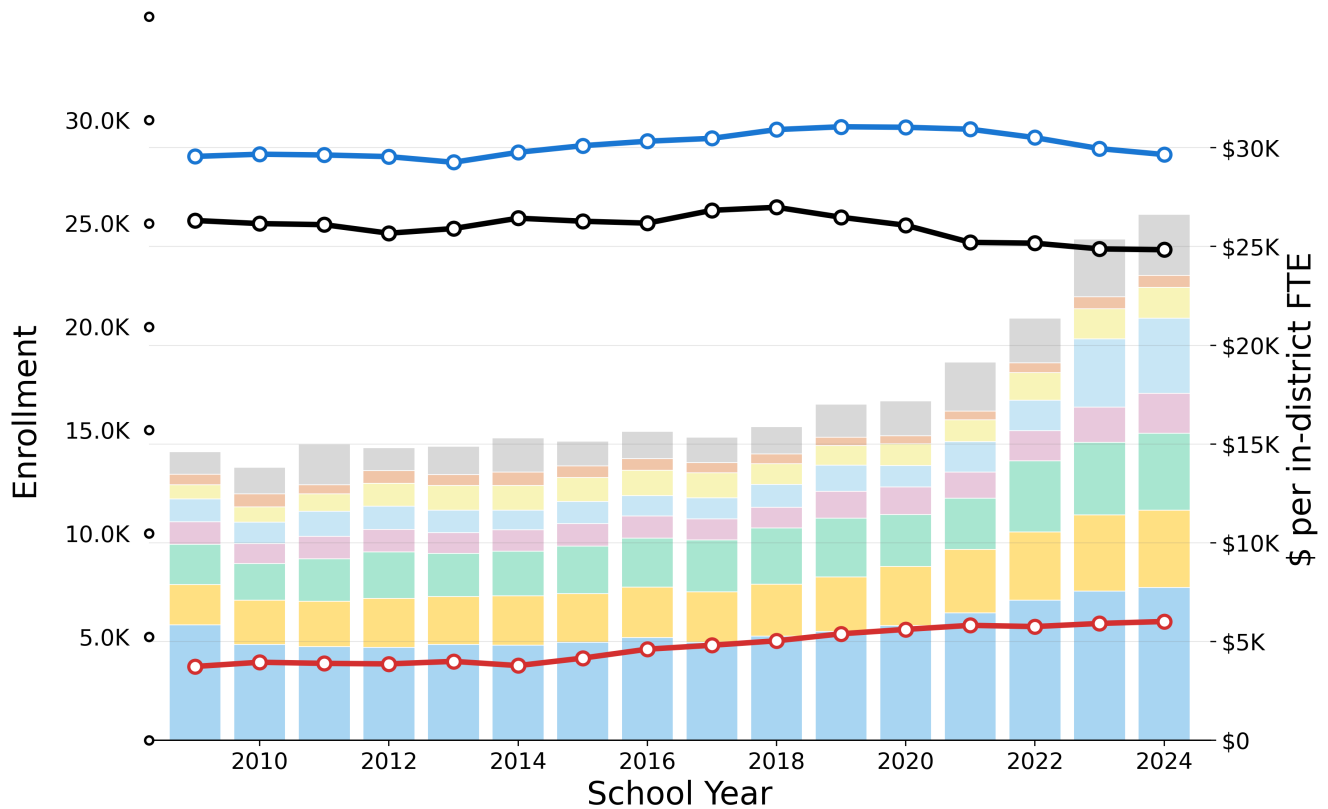


Figure 19

Table 23

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$1,143	+6.8%	+5.9%	+12.9%	\$3,071
Administration	\$533	+0.9%	-1.1%	+7.9%	\$609
Instructional Leadership	\$719	+5.3%	+2.4%	+9.8%	\$1,571
Operations and Maintenance	\$1,150	+8.3%	+14.3%	+23.3%	\$3,806
Other Teaching Services	\$1,143	+3.8%	+6.4%	+8.2%	\$2,014
Pupil Services	\$2,031	+4.4%	+5.6%	+5.6%	\$3,901
Insurance, Retirement and Other	\$2,050	+4.4%	+4.6%	+7.2%	\$3,906
Teachers	\$5,839	+1.9%	+4.8%	+7.0%	\$7,737
Total	\$14,608	+4.1%	+5.7%	+9.4%	\$26,615

Table 24

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	25,137	-0.4%	-0.6%	-1.3%	23,723
Foundation Enrollment	28,235	+0.0%	-0.0%	-0.9%	28,326
Out-of-District FTE Pupils	3,571	+3.2%	+4.7%	+2.2%	5,745
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta\text{Enrollment} \geq 5.0\%$, $ \Delta\text{CAGR} \geq 0.5\text{pp}$					Above Western MA baseline
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$					Below Western MA baseline

Section 2 — Western MA cohort details

Outliers (Springfield >10K FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

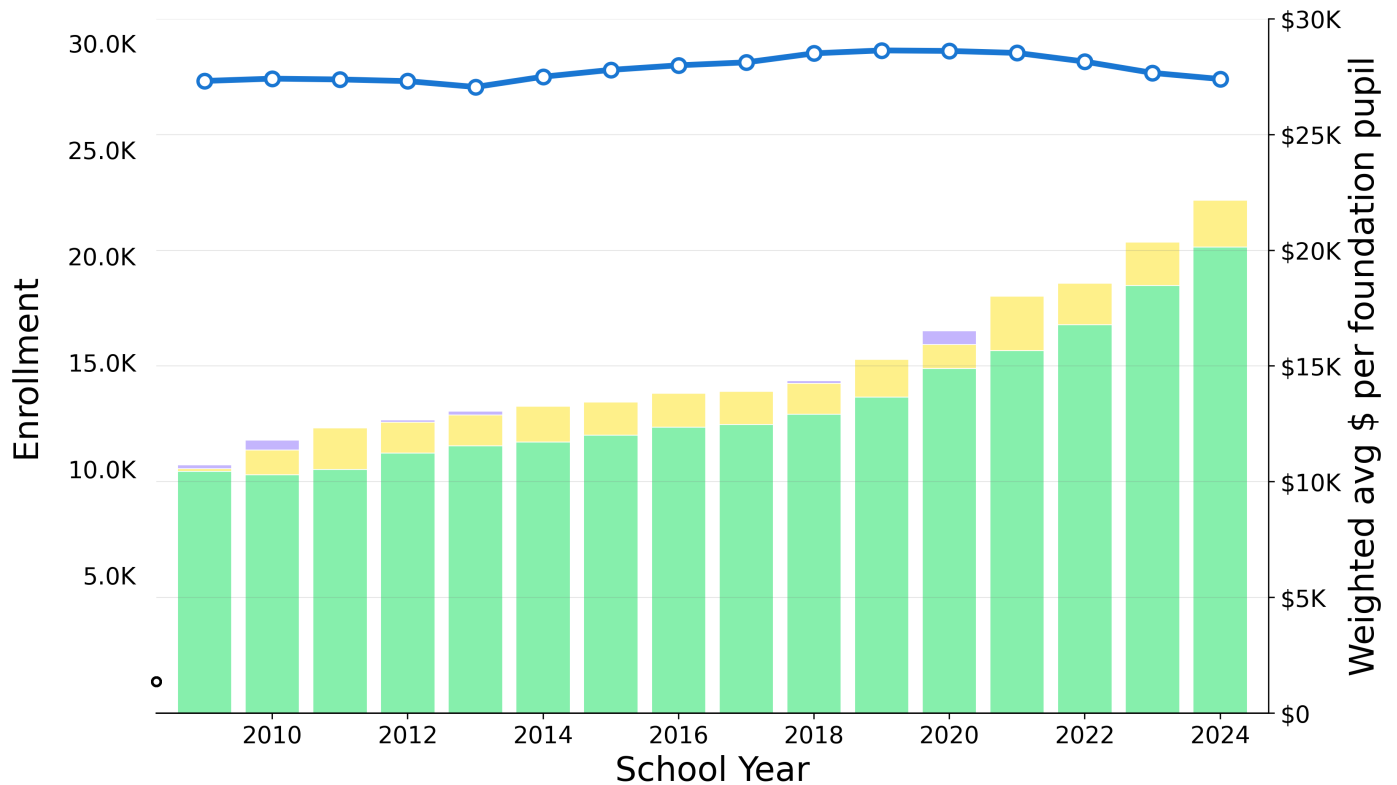


Figure 20

Table 25

Component		2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
	Actual NSS (minus Req NSS)	\$-169	-59.51%	-52.40%	24.01%	\$-0
	Req NSS (minus Ch70)	\$278	14.15%	2.69%	4.35%	\$2,024
	Ch70 Aid	\$10,452	4.47%	5.57%	8.08%	\$20,138
Total (Actual NSS)		\$10,561	5.07%	5.26%	7.70%	\$22,163
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$					Above Western MA baseline	
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$					Below Western MA baseline	

Outliers (Springfield >10K FTE) includes 1 members:

Districts (1): Springfield.

Section 2 — Western MA cohort details

Western MA (all, excl. Springfield) — PPE and Enrollment

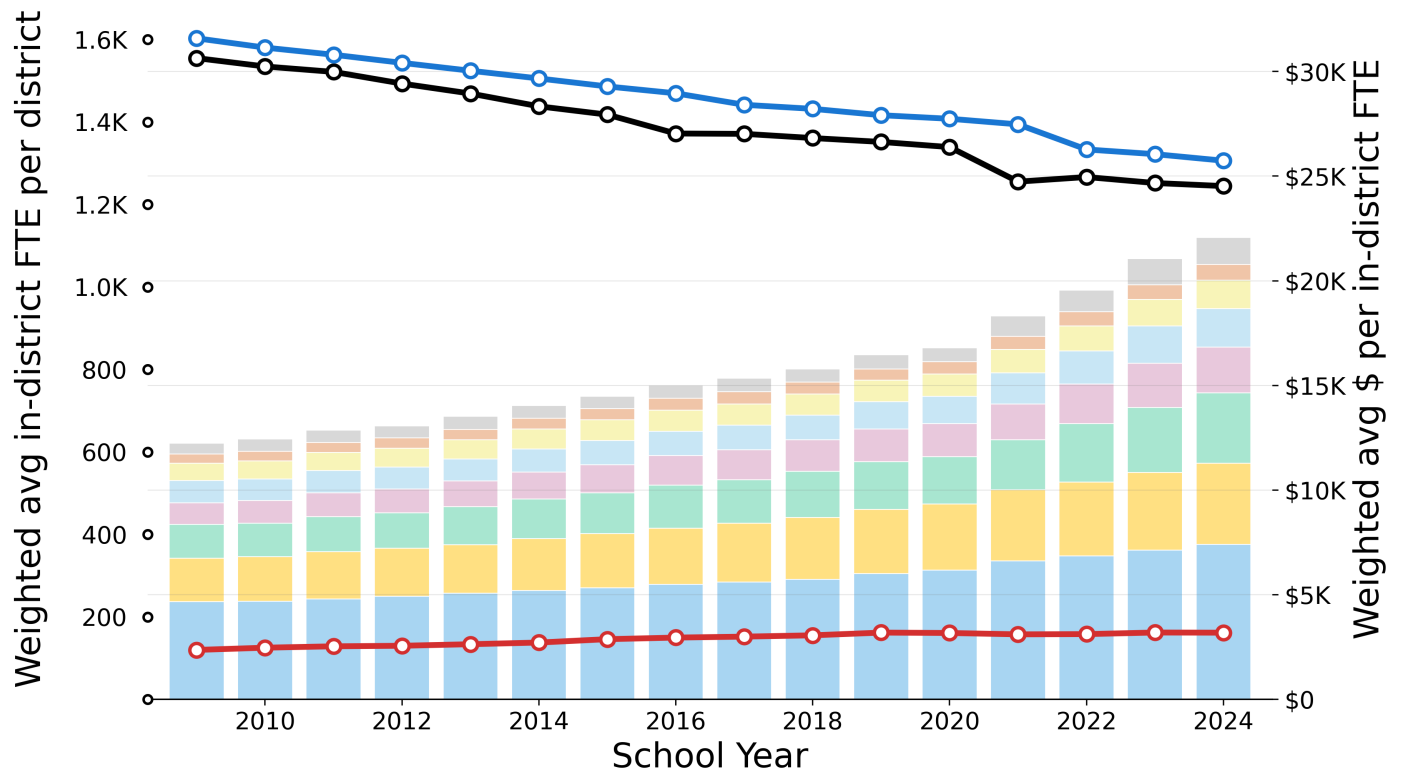


Figure 21

Table 26

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$507	+6.4%	+7.9%	+13.4%	\$1,287
Administration	\$440	+3.6%	+3.8%	+6.8%	\$754
Instructional Leadership	\$836	+3.3%	+3.6%	+5.6%	\$1,353
Operations and Maintenance	\$1,066	+3.7%	+5.3%	+7.2%	\$1,845
Other Teaching Services	\$1,025	+5.2%	+5.5%	+7.1%	\$2,190
Pupil Services	\$1,613	+5.0%	+5.9%	+7.9%	\$3,346
Insurance, Retirement and Other	\$2,083	+4.2%	+4.5%	+4.9%	\$3,881
Teachers	\$4,664	+3.1%	+3.6%	+4.3%	\$7,409
Total	\$12,234	+4.0%	+4.6%	+6.0%	\$22,065

Table 27

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
—●— In-District FTE Pupils	1,555	-1.5%	-1.4%	-1.6%	1,245
—○— Foundation Enrollment	1,603	-1.4%	-1.4%	-1.6%	1,306
—○— Out-of-District FTE Pupils	120	+2.0%	+1.6%	-0.0%	162

Section 2 — Western MA cohort details

Western MA (all, excl. Springfield) — Chapter 70 Aid and Net School Spending (per foundation pupil)

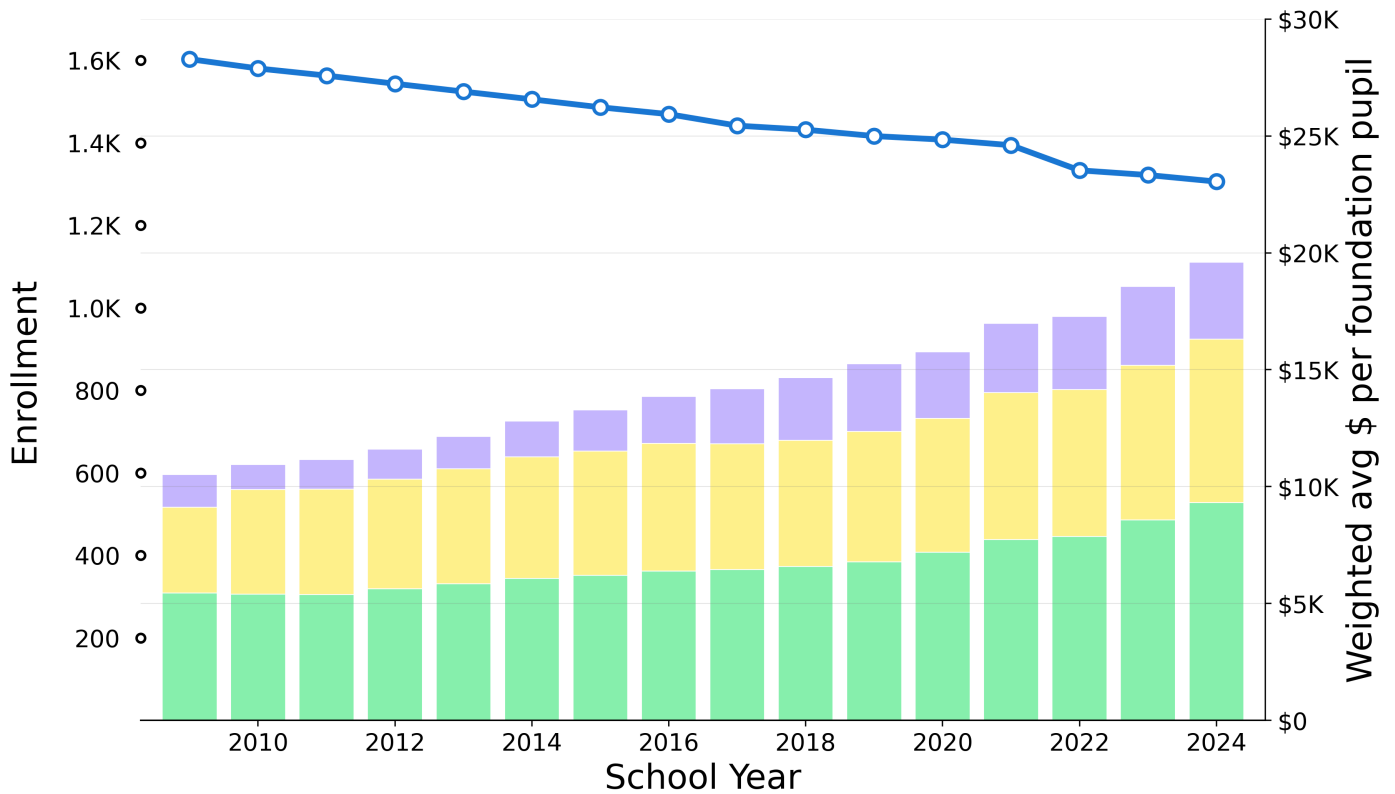


Figure 22

Table 28

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$1,394	5.91%	8.00%	2.64%	\$3,301
Req NSS (minus Ch70)	\$3,676	4.38%	3.02%	4.60%	\$6,988
Ch70 Aid	\$5,444	3.64%	4.36%	6.57%	\$9,313
Total (Actual NSS)	\$10,515	4.24%	4.36%	5.14%	\$19,601

Western MA (all, excl. Springfield) includes 59 members:

Districts (51): Agawam, Amherst, Belchertown, Berkshire Hills, Central Berkshire, Chicopee, Clarksburg, Conway, Deerfield, East Longmeadow, Easthampton, Erving, Florida, Frontier, Gateway, Granby, Greenfield, Hadley, Hancock, Hatfield, Hawlemont, Holyoke, Lee, Lenox, Leverett, Longmeadow, Ludlow, Mohawk Trail, Monson, Mount Greylock, North Adams, Northampton, Orange, Palmer, Pelham, Petersham, Pioneer Valley, Pittsfield, Ralph C Mahar, Richmond, Rowe, Savoy, Shutesbury, South Hadley, Southern Berkshire, Sunderland, Ware, West Springfield, Westfield, Whately, Worthington.

Regions (8): Amherst-Pelham, Athol-Royalston, Farmington River Reg, Gill-Montague, Hampden-Wilbraham, Hoosac Valley Regional, New Salem-Wendell, Southwick-Tolland-Granville Regional School District.

Section 3 — Selected districts

Section 3 examines five selected districts in depth: Amherst-Pelham Regional, Amherst, Leverett, Pelham, and Shutesbury. For each district, three detailed pages show:

- Total PPE breakdown by **expenditure category** with shaded comparison to cohort baseline
- **Chapter 70 aid** and **Net School Spending** breakdown per pupil with shaded comparison to cohort baseline
- **CAGR over 5, 10, and 15-year periods** with shaded comparison to cohort baseline

These district-level pages enable direct comparison between individual district spending patterns and their respective cohorts. Note that although this section focuses on the five selected districts and Western MA, the same visualizations can be produced for other MA districts and regions.

To skip to **Appendix A: Data Sources & Calculation Methodology**, turn to page TBD.

Report Navigation:

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Section 1: Western MA Traditional District Trends	8
Section 2: Western MA Cohort Details	23
Section 3: Selected Districts	36

Section 3 — Specific districts

Amherst-Pelham Regional — PPE and Enrollment

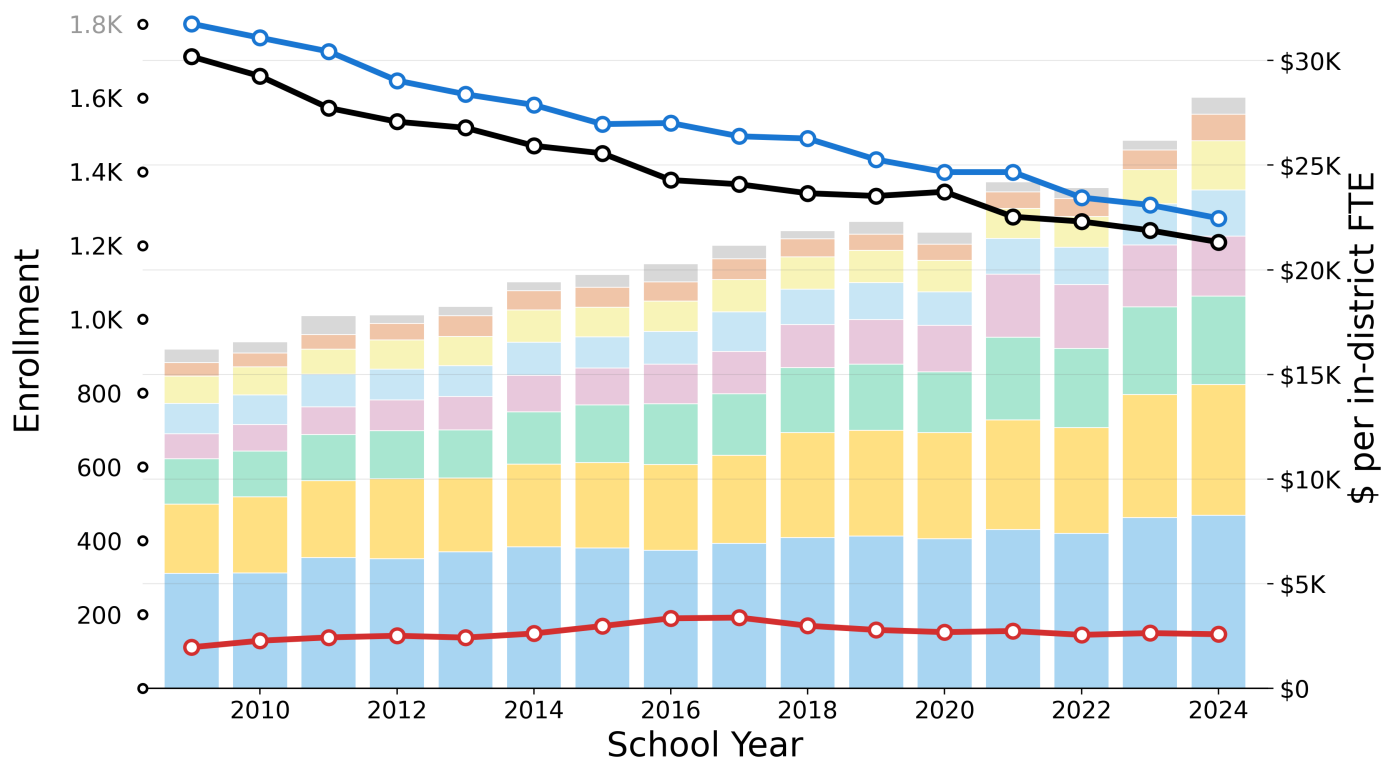


Figure 23

Table 29

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$646	+1.4%	+6.5%	+5.5%	\$799
Administration	\$652	+4.6%	+3.5%	+11.0%	\$1,277
Instructional Leadership	\$1,289	+4.0%	+4.2%	+8.6%	\$2,336
Operations and Maintenance	\$1,459	+2.8%	+3.4%	+4.6%	\$2,221
Other Teaching Services	\$1,199	+5.9%	+5.1%	+6.1%	\$2,851
Pupil Services	\$2,170	+4.6%	+5.4%	+6.0%	\$4,231
Insurance, Retirement and Other	\$3,305	+4.3%	+4.7%	+4.4%	\$6,248
Teachers	\$5,491	+2.8%	+2.0%	+2.6%	\$8,270
Total	\$16,211	+3.8%	+3.8%	+4.8%	\$28,233

Table 30

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	1,712	-2.3%	-1.9%	-1.9%	1,209
Foundation Enrollment	1,801	-2.3%	-2.1%	-2.3%	1,274
Out-of-District FTE Pupils	112	+1.8%	-0.1%	-1.5%	147

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta\text{Enrollment}| \geq 5.0\%$, $|\Delta\text{CAGR}| \geq 0.5\text{pp}$

$\text{CAGR} = (\text{End}/\text{Start})^{(1/\text{years})} - 1$

Above cohort baseline (Medium (801-1600))

Below cohort baseline (Medium (801-1600))

Section 3 — Selected districts

Amherst-Pelham Regional — Chapter 70 Aid and Net School Spending (per foundation pupil)

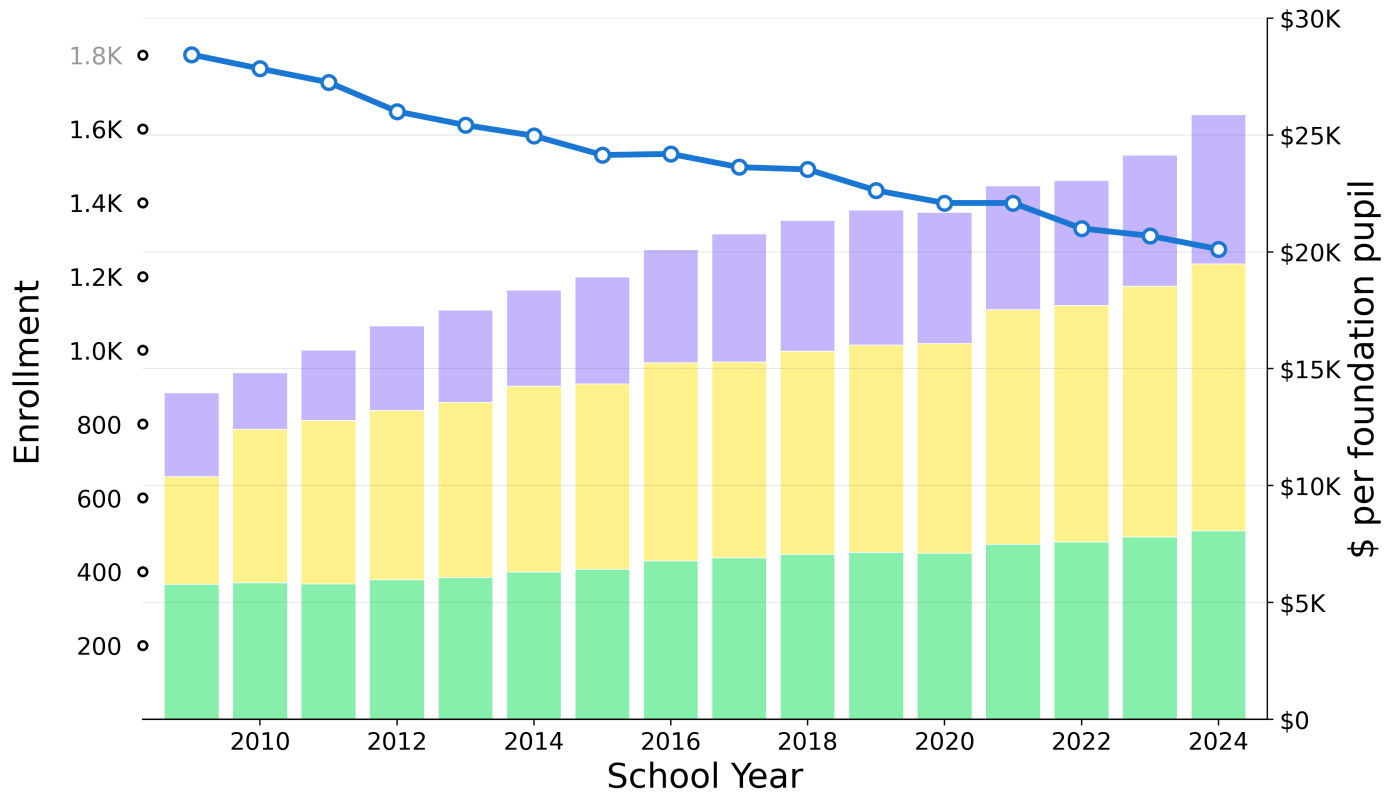


Figure 24

Table 31

Component		2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
	Actual NSS (minus Req NSS)	\$3,584	3.92%	4.46%	2.07%	\$6,383
	Req NSS (minus Ch70)	\$4,611	6.23%	3.70%	5.14%	\$11,414
	Ch70 Aid	\$5,773	2.26%	2.49%	2.49%	\$8,068
Total (Actual NSS)		\$13,968	4.19%	3.48%	3.50%	\$25,865
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$						Above cohort baseline (Medium (801-1600))
CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$						Below cohort baseline (Medium (801-1600))

Section 3 — Specific districts

Amherst — PPE and Enrollment

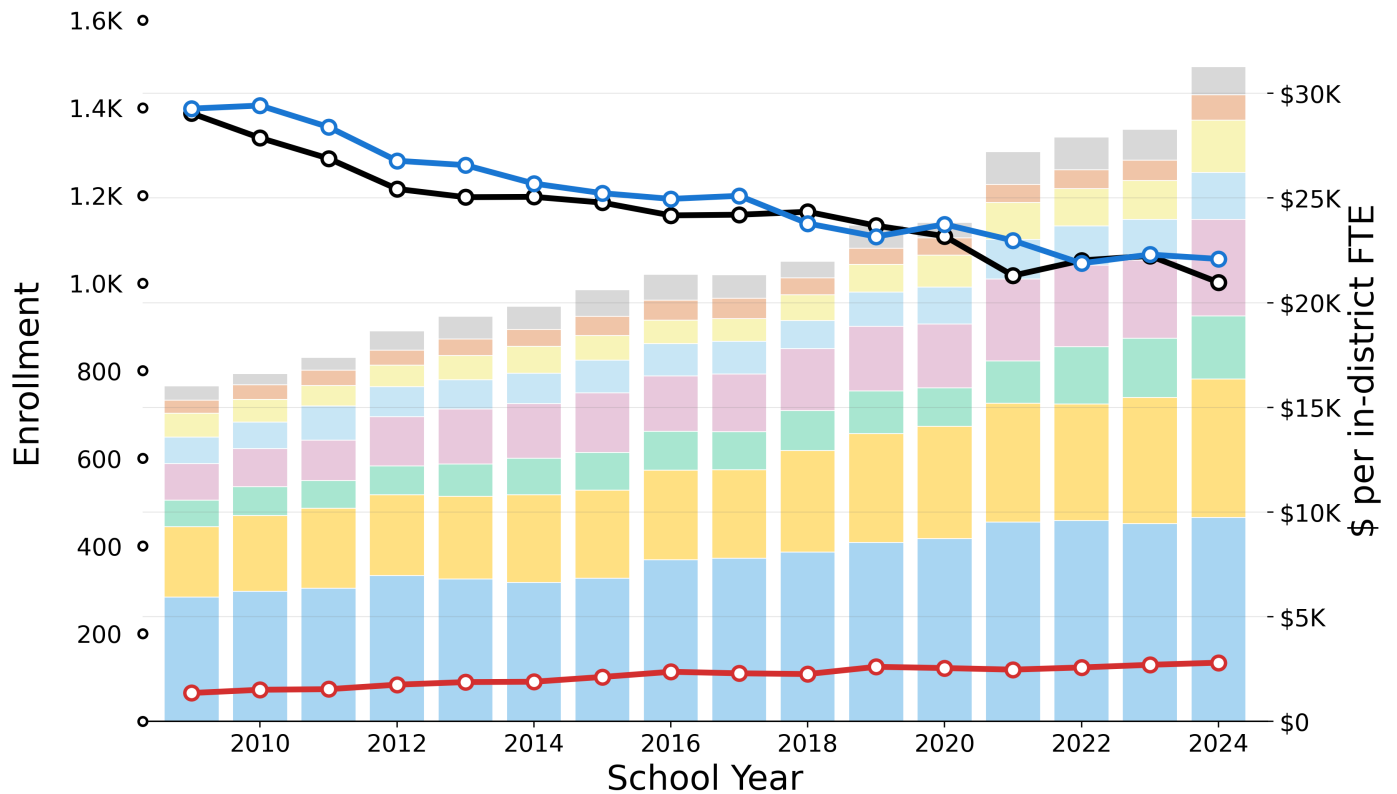


Figure 25

Table 32

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$684	+4.5%	+1.9%	+4.0%	\$1,328
Administration	\$627	+4.5%	+4.1%	+9.6%	\$1,213
Instructional Leadership	\$1,134	+5.4%	+7.0%	+13.5%	\$2,498
Operations and Maintenance	\$1,264	+3.9%	+4.5%	+6.4%	\$2,248
Other Teaching Services	\$1,762	+6.6%	+5.9%	+8.4%	\$4,612
Pupil Services	\$1,261	+6.0%	+5.5%	+8.3%	\$3,018
Insurance, Retirement and Other	\$3,353	+4.6%	+4.7%	+4.9%	\$6,605
Teachers	\$5,944	+3.4%	+3.9%	+2.7%	\$9,745
Total	\$16,029	+4.6%	+4.7%	+5.7%	\$31,267

Table 33

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	1,387	-2.1%	-1.8%	-2.4%	1,002
Foundation Enrollment	1,398	-1.9%	-1.5%	-0.9%	1,055
Out-of-District FTE Pupils	65	+5.0%	+4.0%	+1.5%	134

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta Enrollment| \geq 5.0\%$, $|\Delta CAGR| \geq 0.5pp$

$CAGR = (End/Start)^{(1/years)} - 1$

Above cohort baseline (Medium (801-1600))

Below cohort baseline (Medium (801-1600))

Section 3 — Selected districts

Amherst — Chapter 70 Aid and Net School Spending (per foundation pupil)

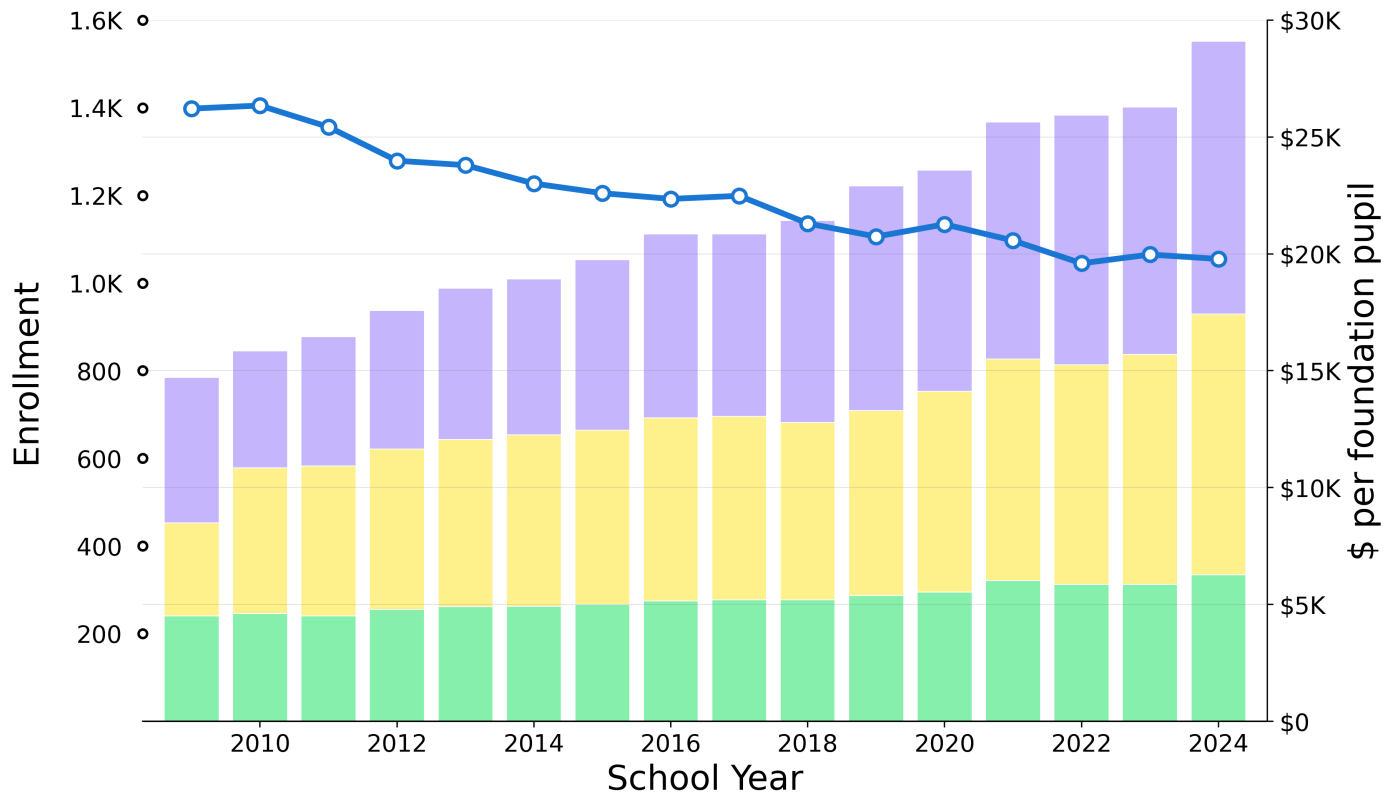


Figure 26

Table 34

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$6,217	4.29%	5.77%	3.94%	\$11,670
Req NSS (minus Ch70)	\$3,977	7.11%	4.27%	7.11%	\$11,151
Ch70 Aid	\$4,518	2.21%	2.44%	3.10%	\$6,271
Total (Actual NSS)	\$14,712	4.65%	4.39%	4.89%	\$29,093
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above cohort baseline (Medium (801-1600))	
CAGR = $(End/Start)^{(1/years)} - 1$				Below cohort baseline (Medium (801-1600))	

Section 3 — Specific districts

Leverett — PPE and Enrollment

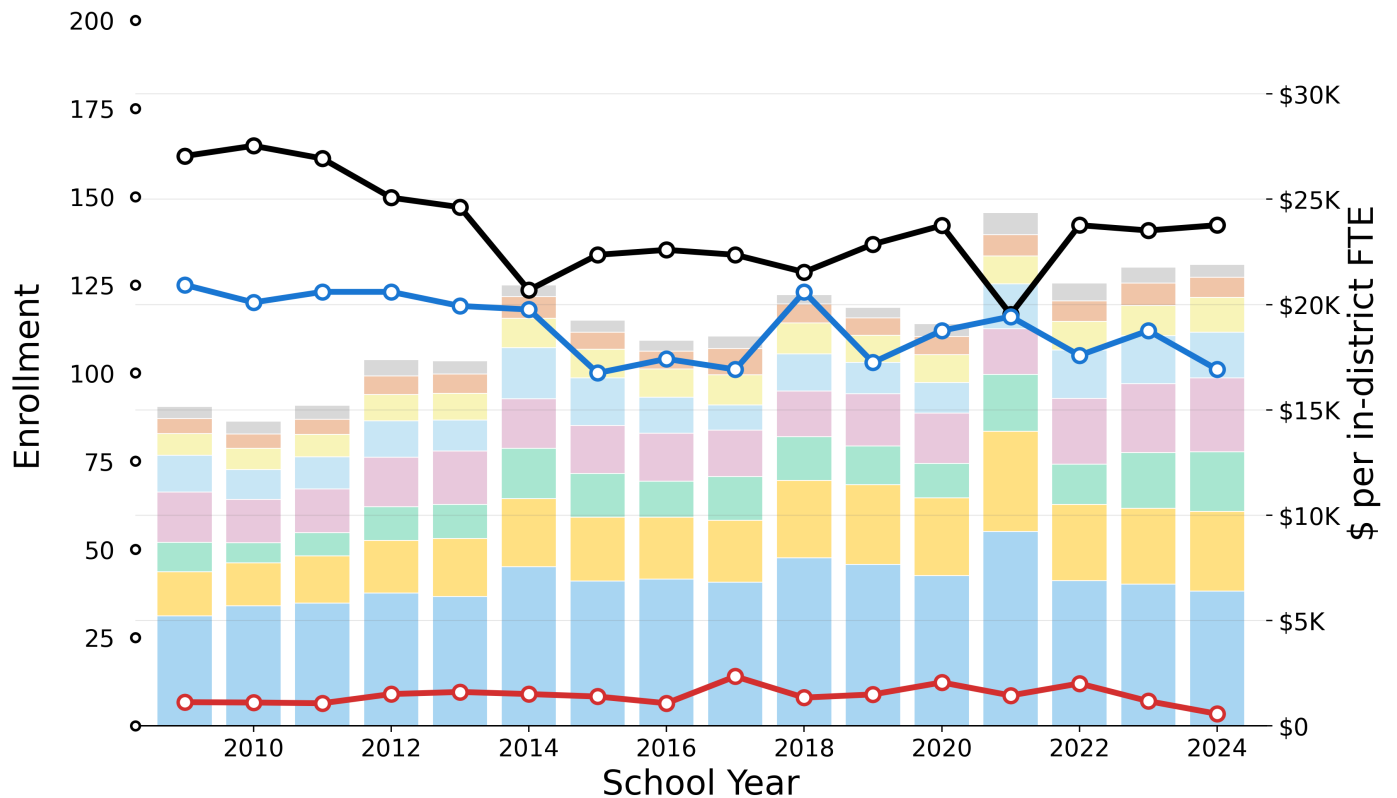


Figure 27

Table 35

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$575	+0.4%	+0.9%	+3.9%	\$608
Administration	\$715	+1.9%	-0.8%	+2.9%	\$953
Instructional Leadership	\$1,015	+3.3%	+1.8%	+5.4%	\$1,663
Operations and Maintenance	\$1,755	+1.4%	-1.1%	+7.7%	\$2,158
Other Teaching Services	\$2,385	+2.6%	+4.1%	+7.3%	\$3,526
Pupil Services	\$1,395	+4.8%	+1.6%	+8.8%	\$2,814
Insurance, Retirement and Other	\$2,098	+4.0%	+1.6%	+0.1%	\$3,793
Teachers	\$5,218	+1.4%	-1.6%	-3.6%	\$6,395
Total	\$15,156	+2.5%	+0.5%	+2.0%	\$21,910

Table 36

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	162	-0.9%	+1.4%	+0.8%	142
Foundation Enrollment	125	-1.4%	-1.5%	-0.4%	101
Out-of-District FTE Pupils	7	-4.4%	-9.3%	-17.5%	3

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta\text{Enrollment}| \geq 5.0\%$, $|\Delta\text{CAGR}| \geq 0.5\text{pp}$

CAGR = $(\text{End}/\text{Start})^{(1/\text{years})} - 1$

Above cohort baseline (Tiny (0-200))

Below cohort baseline (Tiny (0-200))

Section 3 — Selected districts

Leverett — Chapter 70 Aid and Net School Spending (per foundation pupil)

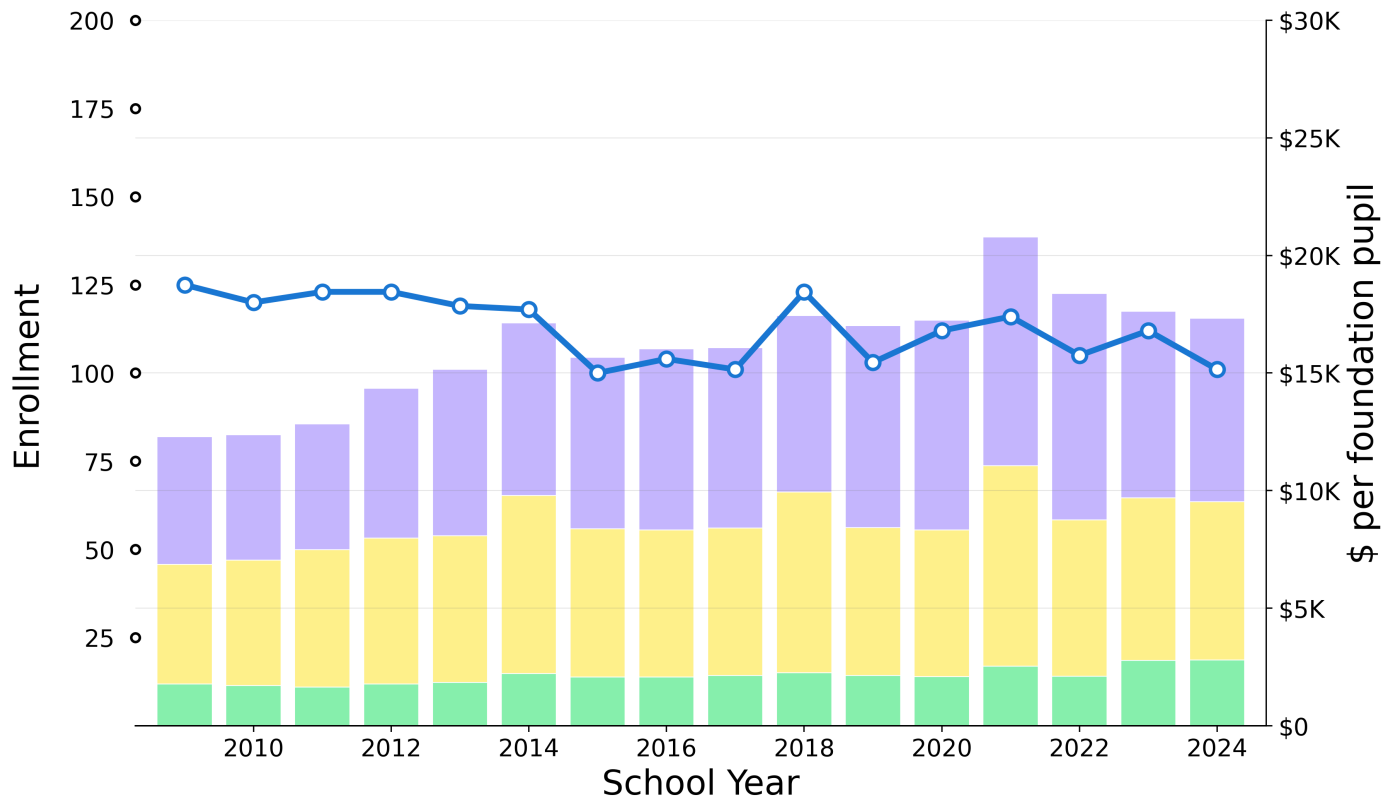


Figure 28

Table 37

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$5,429	2.44%	0.62%	-1.89%	\$7,799
Req NSS (minus Ch70)	\$5,074	1.90%	-1.17%	1.36%	\$6,728
Ch70 Aid	\$1,782	3.05%	2.32%	5.50%	\$2,798
Total (Actual NSS)	\$12,285	2.32%	0.12%	0.37%	\$17,325
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above cohort baseline (Tiny (0-200))	
CAGR = $(End/Start)^{(1/years)} - 1$				Below cohort baseline (Tiny (0-200))	

Section 3 — Specific districts

Pelham — PPE and Enrollment

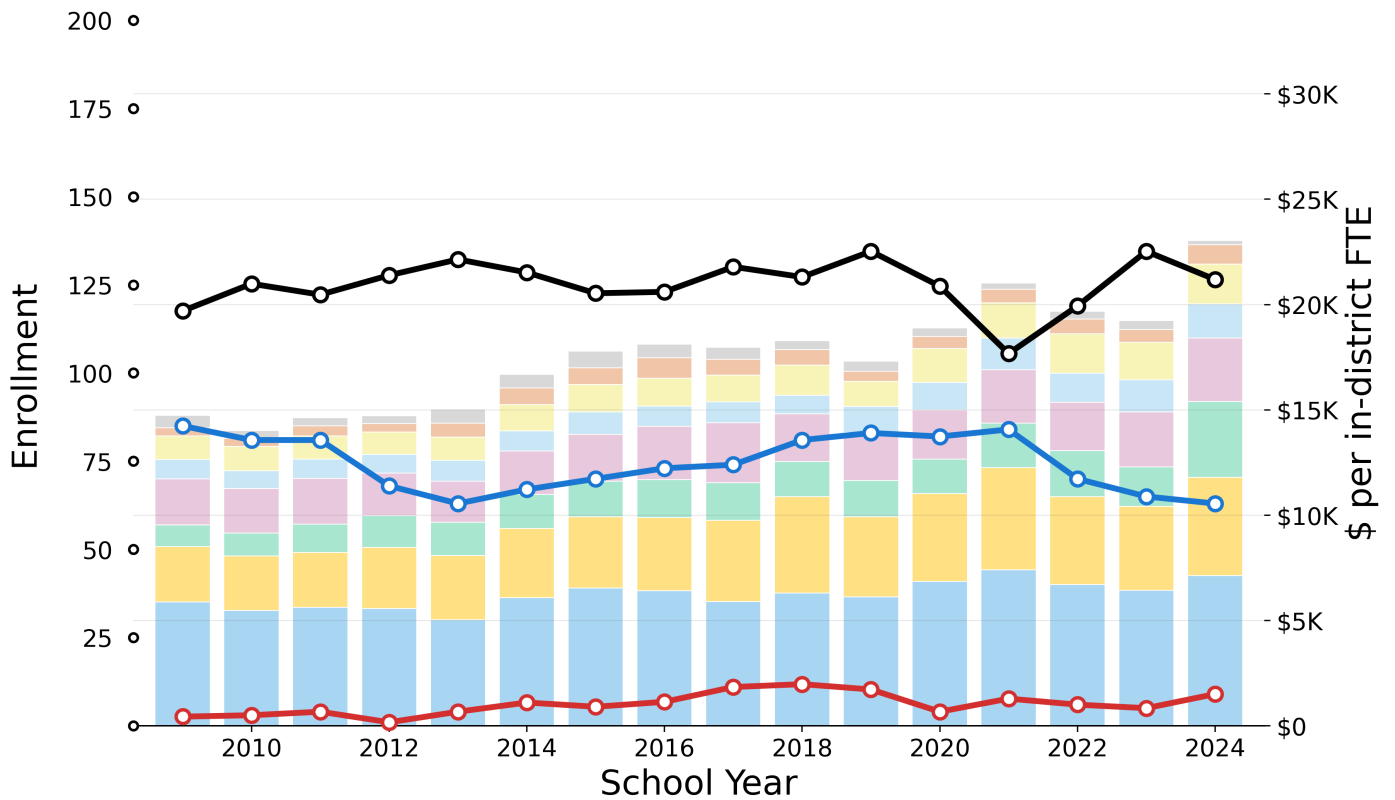


Figure 29

Table 38

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$580	-6.6%	-10.7%	-15.0%	\$207
Administration	\$390	+5.9%	+1.4%	+13.4%	\$917
Instructional Leadership	\$1,123	+3.5%	+4.2%	+9.6%	\$1,885
Operations and Maintenance	\$920	+3.8%	+5.5%	+5.6%	\$1,614
Other Teaching Services	\$2,184	+2.2%	+3.9%	+5.8%	\$3,010
Pupil Services	\$1,032	+8.7%	+8.4%	+15.8%	\$3,610
Insurance, Retirement and Other	\$2,615	+3.9%	+3.6%	+4.3%	\$4,670
Teachers	\$5,889	+1.3%	+1.6%	+3.1%	\$7,127
Total	\$14,733	+3.0%	+3.3%	+5.9%	\$23,040

Table 39

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	118	+0.5%	-0.2%	-1.2%	126
Foundation Enrollment	85	-2.0%	-0.6%	-5.4%	63
Out-of-District FTE Pupils	3	+8.6%	+3.2%	-2.7%	9

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta\text{Enrollment}| \geq 5.0\%$, $|\Delta\text{CAGR}| \geq 0.5\text{pp}$

$\text{CAGR} = (\text{End}/\text{Start})^{(1/\text{years})} - 1$

Above cohort baseline (Tiny (0-200))

Below cohort baseline (Tiny (0-200))

Section 3 — Selected districts

Pelham — Chapter 70 Aid and Net School Spending (per foundation pupil)

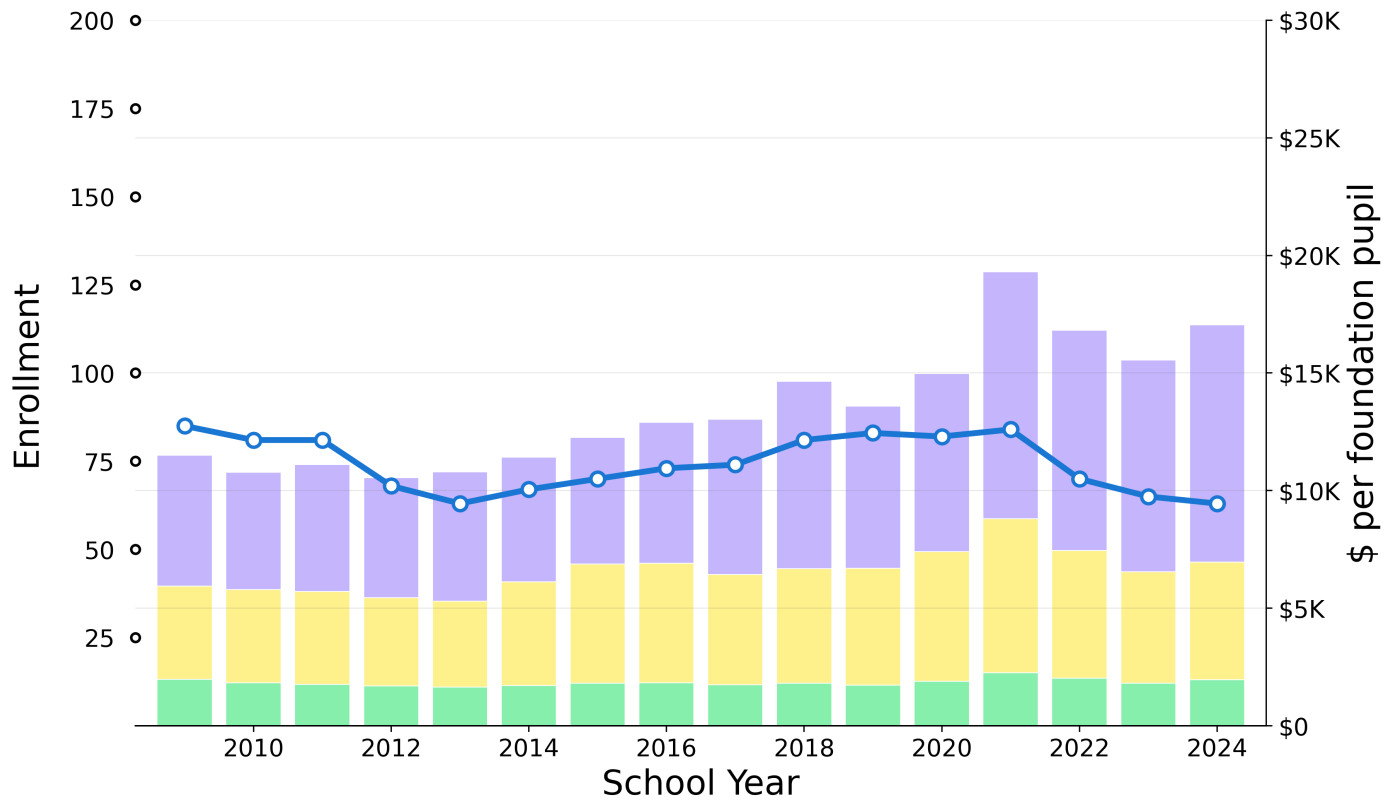


Figure 30

Table 40

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$5,561	4.05%	6.64%	7.91%	\$10,086
Req NSS (minus Ch70)	\$3,963	1.56%	1.26%	0.10%	\$4,996
Ch70 Aid	\$1,983	-0.07%	1.35%	2.52%	\$1,963
Total (Actual NSS)	\$11,507	2.65%	4.08%	4.62%	\$17,044
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above cohort baseline (Tiny (0-200))	
CAGR = $(End/Start)^{(1/years)} - 1$				Below cohort baseline (Tiny (0-200))	

Section 3 — Specific districts

Shutesbury — PPE and Enrollment

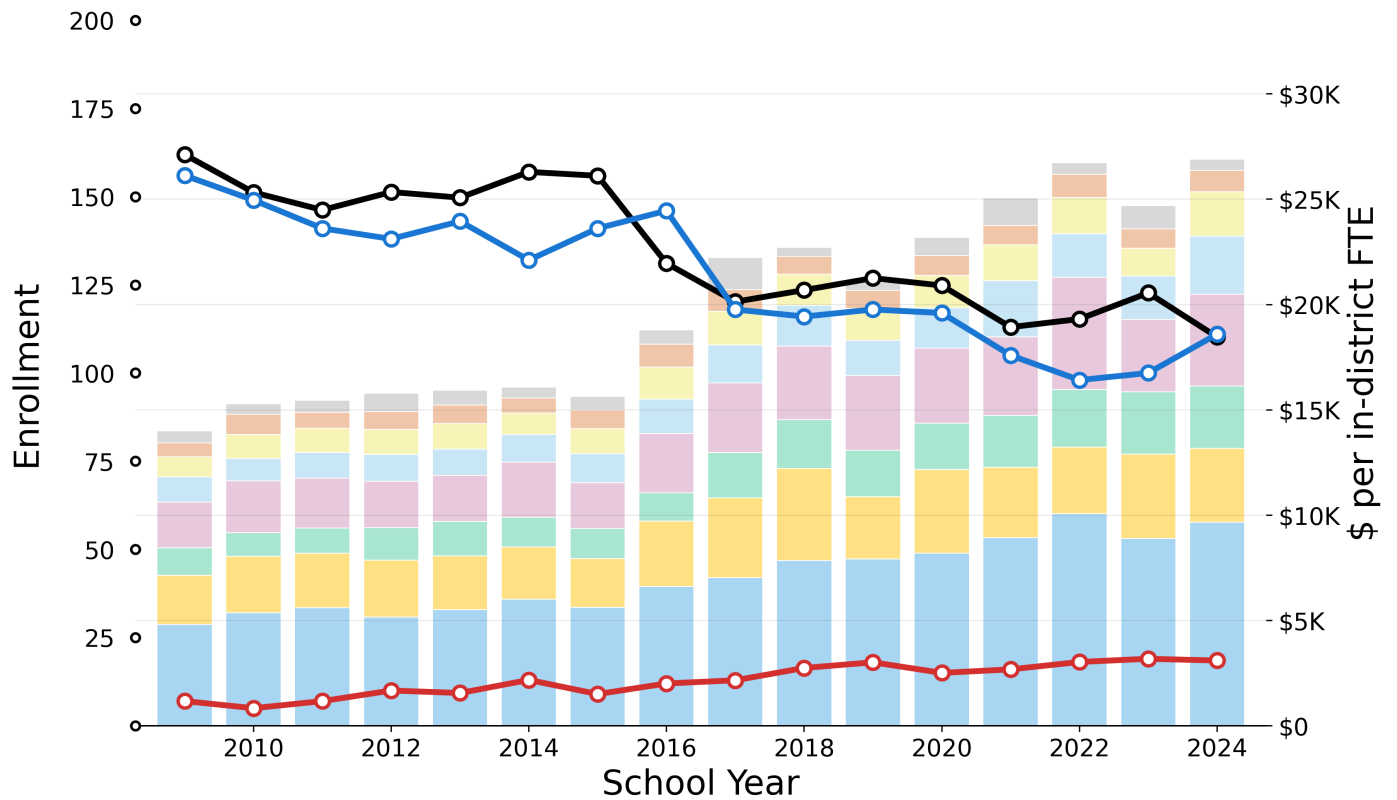


Figure 31

Table 41

Category	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Other	\$580	-0.4%	+0.6%	+7.2%	\$545
Administration	\$652	+3.0%	+3.4%	+3.5%	\$1,015
Instructional Leadership	\$969	+5.3%	+7.7%	+6.8%	\$2,103
Operations and Maintenance	\$1,180	+5.8%	+7.5%	+10.6%	\$2,754
Other Teaching Services	\$2,185	+4.7%	+5.3%	+4.2%	\$4,359
Pupil Services	\$1,292	+5.7%	+7.6%	+6.1%	\$2,962
Insurance, Retirement and Other	\$2,332	+2.7%	+3.5%	+3.3%	\$3,501
Teachers	\$4,821	+4.7%	+4.9%	+4.1%	\$9,669
Total	\$14,011	+4.4%	+5.3%	+5.0%	\$26,908

Table 42

In-district FTE	2009	CAGR 15y	CAGR 10y	CAGR 5y	2024
In-District FTE Pupils	162	-2.5%	-3.5%	-2.8%	110
Foundation Enrollment	156	-2.2%	-1.7%	-1.2%	111
Out-of-District FTE Pupils	7	+6.7%	+3.6%	+0.5%	18

Shading vs baseline: $|\Delta\$/pupil| \geq 5.0\%$, $|\Delta\text{Enrollment}| \geq 5.0\%$, $|\Delta\text{CAGR}| \geq 0.5\text{pp}$

$\text{CAGR} = (\text{End}/\text{Start})^{(1/\text{years})} - 1$

Above cohort baseline (Tiny (0-200))

Below cohort baseline (Tiny (0-200))

Section 3 — Selected districts

Shutesbury — Chapter 70 Aid and Net School Spending (per foundation pupil)

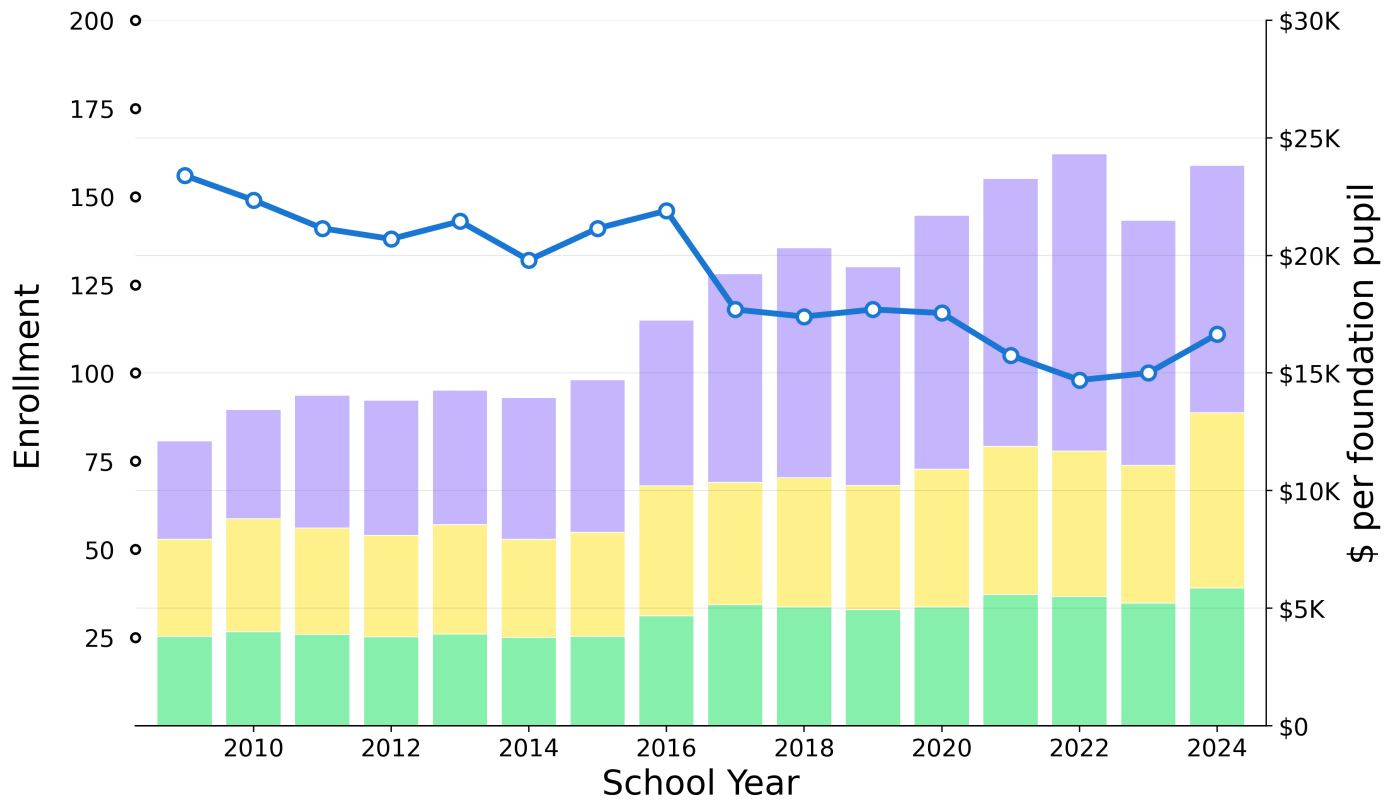


Figure 32

Table 43

Component	2009 \$/pupil	CAGR 15y	CAGR 10y	CAGR 5y	2024 \$/pupil
Actual NSS (minus Req NSS)	\$4,173	6.36%	5.74%	2.51%	\$10,526
Req NSS (minus Ch70)	\$4,124	4.02%	5.92%	7.14%	\$7,445
Ch70 Aid	\$3,808	2.92%	4.58%	3.48%	\$5,862
Total (Actual NSS)	\$12,105	4.62%	5.50%	4.08%	\$23,833
Shading vs baseline: $ \Delta\$/pupil \geq 5.0\%$, $ \Delta CAGR \geq 0.5pp$				Above cohort baseline (Tiny (0-200))	
CAGR = $(End/Start)^{(1/years)} - 1$				Below cohort baseline (Tiny (0-200))	

Appendix A. Data Sources & Calculation Methodology (continued)

5. Orange/Blue Shading Logic (District Comparison Tables)

District pages include tables comparing each district's per-pupil expenditures (PPE) and growth rates (CAGR) to their enrollment-based peer group aggregate (Tiny, Small, Medium, Large, or Springfield).

Two independent tests determine shading:

Test 1 - Dollar Amount (2024 \$/pupil column): Compares the district's 2024 PPE to the baseline's 2024 PPE using relative difference: $(\text{District} - \text{Baseline}) / \text{Baseline}$ • **Orange shading:** District spending is $\geq 5.0\%$ higher than baseline • **Blue shading:** District spending is $\geq 5.0\%$ lower than baseline • **No shading:** Difference is less than 5.0%

Test 2 - CAGR (5y, 10y, 15y columns): Compares the district's CAGR to the baseline's CAGR using absolute percentage point difference: $\text{District_CAGR} - \text{Baseline_CAGR}$ • **Orange shading:** District CAGR is ≥ 0.5 percentage points higher than baseline • **Blue shading:** District CAGR is ≥ 0.5 percentage points lower than baseline • **No shading:** Difference is less than 0.5 percentage points

Shading Intensity: Both metrics use graduated shading where darker colors indicate larger differences: • PPE/Enrollment bins: 5% (lightest), 10% , 15% , $20\%+$ (darkest) • CAGR bins: 0pp (lightest), 1pp , 2pp , $2\text{pp}+$ (darkest)

Key insight: The tests are independent. An orange 2024 \$/pupil with unshaded CAGRs typically means: • The district started at a higher baseline 15 years ago, AND • The district has been growing at roughly the same rate as peers • Therefore it remains higher in absolute dollars but isn't growing faster

Statistical Rationale: The $5\%/0.5\text{pp}$ thresholds were selected through analysis of variation across all Western MA districts. See the Threshold Analysis page for detailed methodology and alternative scenarios considered.

Appendix A. Data Sources & Calculation Methodology (continued)

6. Chapter 70 Aid and Net School Spending (NSS) Calculations

Chapter 70 is Massachusetts' primary state aid program for K-12 education. Net School Spending (NSS) is the total amount a district spends on education from local and state sources.

Data Sources: • **Chapter 70 Aid (c70aid):** State aid received by the district (DESE profile_DataC70 sheet) • **Required NSS (rqdnss2):** Minimum spending required by state law, adjusted (DESE profile_DataC70 sheet) • **Actual NSS (actualNSS):** Total district spending on education (DESE profile_DataC70 sheet)

Important Note: NSS/Ch70 values are reported in **dollars per pupil**, consistent with PPE reporting. All absolute dollar amounts are divided by in-district FTE enrollment to produce per-pupil values.

Stacked Components (bottom to top in plots): 1. **Ch70 Aid (\$/pupil):** $c70aid / enrollment$ • Green bar in plots • State funding received per pupil

2. **Req NSS (adj) (\$/pupil):** $\max(0, rqdnss2 - c70aid) / enrollment$ • Amber bar in plots • Required local contribution per pupil (after subtracting Ch70) • Uses $\max(0, \dots)$ to handle rare cases where Ch70 > Required NSS

3. **Actual NSS (adj) (\$/pupil):** $(actualNSS - rqdnss2) / enrollment$ • Purple bar in plots • Spending above minimum requirement per pupil • Represents discretionary local spending beyond state mandates

Total NSS (\$/pupil): Sum of all three components = $actualNSS / enrollment$

Example Calculation (Amherst, FY2024): Enrollment: 1,721 in-district FTE • Ch70 Aid: $\$6,791,000 / 1,721 = \$3,946/pupil$ • Req NSS (adj): $(\$18,859,000 - \$6,791,000) / 1,721 = \$7,014/pupil$ • Actual NSS (adj): $(\$31,511,000 - \$18,859,000) / 1,721 = \$7,353/pupil$ • Total NSS: $\$3,946 + \$7,014 + \$7,353 = \$18,313/pupil$

Aggregate Calculation (Weighted Per-Pupil Average): For aggregate enrollment groups (Tiny, Small, Medium, Large, Springfield): 1. Convert each district's absolute dollars to per-pupil values 2. Compute enrollment-weighted average per-pupil across all member districts

This methodology produces per-pupil values that are weighted by enrollment size, enabling fair comparison across different-sized district aggregates.

Important: Only districts with valid PPE (per-pupil expenditure) data are included in NSS/Ch70 aggregates to ensure data quality and consistency across all analyses.

Shading: NSS/Ch70 comparison tables use the same orange/blue shading logic as PPE tables (5% dollar threshold, 0.5pp CAGR threshold).

Appendix A. Data Sources & Calculation Methodology (continued)

Rationale for 5% / 0.5pp Shading Thresholds

Table 44

Scenario	PPE/ Enrollment	CAGR	PPE Sensitivity	CAGR Sensitivity	Balance Ratio	Assessment
Previous (2%/2pp)	2.0%	2.00pp	0.09 SD	0.62 SD	6.97x	Unbalanced
Equal SD (5%/0.72pp)	5.0%	0.72pp	0.22 SD	0.22 SD	1.00x	Too tight for CAGR
Selected (5%/0.5pp)	5.0%	0.50pp	0.22 SD	0.15 SD	0.70x	Well-balanced
Proportional (5%/5pp)	5.0%	5.00pp	0.22 SD	1.55 SD	6.97x	CAGR too loose

Rationale for 5% / 0.5pp Shading Thresholds

Problem Statement: Different metrics have different natural variation. Using a uniform threshold (e.g., 2% for everything) creates unbalanced sensitivity where some comparisons over-flag minor differences while others under-flag meaningful ones.

Data Analysis: Examining all 59 Western MA traditional districts:

- **PPE variation:** Mean = \$24,237, SD = \$5,462, CV = 0.225 (22.5% typical variation)
- **CAGR variation:** Mean = 6.00pp, SD = 3.24pp, CV = 0.540 (54% typical variation)
- **Key insight:** CAGR naturally varies 2.4x more than PPE relative to their means

Methodology:

1. Calculated how many standard deviations each threshold represents
2. Compared 'sensitivity' (how strict each threshold is relative to typical variation)
3. Tested scenarios ranging from 2%/2pp (original) to 5%/5pp (proportional loosening)
4. Selected thresholds that balance statistical rigor with practical communication

Scenario Comparison:

- **Previous (2% / 2pp):** PPE=0.09 SD, CAGR=0.62 SD → CAGR 7x more sensitive (unbalanced)
- **Equal SD (5% / 0.72pp):** Both=0.22 SD → Perfect balance but 0.72pp too precise to communicate
- **Selected (5% / 0.5pp):** PPE=0.22 SD, CAGR=0.15 SD → 0.7x ratio (well-balanced, simple numbers)
- **Proportional (5% / 5pp):** PPE=0.22 SD, CAGR=1.55 SD → CAGR 7x looser (under-flags growth differences)

Selected Thresholds (5% / 0.5pp):

- **5% for PPE and enrollment:** Flags differences $\geq$ \$1,212 (at mean), or $\sim 1/5$ of typical variation
- **0.5pp for CAGR:** Flags growth rate differences $\geq$ 0.5pp, or $\sim 1/6$ of typical variation
- **Balance ratio:** 0.7x (CAGR more sensitive, reflecting compound growth impact)
- **Flagging rates:** $\sim 82\%$ for PPE, $\sim 88\%$ for CAGR (higher sensitivity for growth rates)

Design Philosophy: Why $\sim 80\%$ Flagging Rates Work with Gradient Shading

With **gradient shading** (not binary on/off), having $\sim 80\%$ of comparisons show some level of shading is actually ideal. The threshold acts as a **noise floor** that filters trivial differences while the graduated intensity creates three distinct levels of information:

- **No shading (white background):** Districts are statistically similar to their cohort (differences $< 5\%$ / $< 0.5\text{pp}$)
- **Light shading (subtle color):** Notable differences worth attention but not alarming
- **Intense shading (saturated color):** Exceptional outliers that immediately grab the eye

This creates a natural visual hierarchy where readers' eyes are drawn to the most intense colors (true outliers) while still being able to see the full pattern of variation across the lighter-shaded cells. The threshold filters out noise while the gradient intensity tells you *how much* a district differs from its peers.

The Goldilocks Solution (5% / 0.5pp):

This threshold pair hits the sweet spot across multiple dimensions:

Statistical balance:

- High flagging rates (~82% vs ~88%) despite very different natural variation (CV = 22.5% vs 54%)
- CAGR more sensitive to reflect compound growth impact - small differences become large over time

Practical communication:

- Round, memorable numbers (5% and 0.5pp)
- Easy to explain: "We flag 5% differences in dollars/enrollment, half a percentage point differences in growth rates"
- Contrast with alternatives: 0.72pp would be statistically perfect but impractically precise to communicate

Appropriate sensitivity:

- Loosens overly-tight previous thresholds (2%/2pp had 93% PPE flagging)
- Filters trivial noise while capturing meaningful differences
- Reserves intense shading for truly exceptional cases

Shading Intensity: Both metrics use graduated shading with darker colors for larger differences:

- PPE/Enrollment: 5% (lightest), 10%, 15%, 20%+ (darkest)
- CAGR: 0.5pp (lightest), 1pp, 1.5pp, 2pp+ (darkest)

Note: These thresholds apply to all comparison tables throughout this report. They represent a balance between highlighting meaningful differences and avoiding excessive flagging of normal variation.

Data: All Western MA Traditional Districts: Small (201-800 FTE)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 45

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$558	\$505	\$540	\$533	\$585	\$515	\$503	\$583	\$539	\$596	\$558	\$512	\$915	\$777	\$1,064	\$1,150
Administration	\$523	\$515	\$567	\$601	\$648	\$665	\$701	\$739	\$999	\$752	\$788	\$802	\$854	\$942	\$948	\$976
Instructional Leadership	\$811	\$805	\$838	\$824	\$909	\$947	\$967	\$1,024	\$1,028	\$1,042	\$1,016	\$1,097	\$1,188	\$1,260	\$1,336	\$1,463
Operations and Maintenance	\$1,108	\$1,056	\$1,043	\$1,101	\$1,198	\$1,264	\$1,295	\$1,338	\$1,254	\$1,285	\$1,473	\$1,422	\$1,676	\$1,878	\$2,176	\$2,454
Other Teaching Services	\$1,061	\$1,089	\$1,193	\$1,151	\$1,289	\$1,311	\$1,353	\$1,446	\$1,575	\$1,718	\$1,799	\$1,806	\$1,779	\$1,899	\$2,151	\$2,402
Pupil Services	\$1,813	\$1,788	\$1,859	\$1,993	\$2,216	\$2,221	\$2,216	\$2,477	\$2,438	\$2,470	\$2,643	\$2,670	\$2,903	\$3,433	\$3,656	\$4,084
Insurance, Retirement and Other	\$2,128	\$2,107	\$2,230	\$2,317	\$2,486	\$2,568	\$2,711	\$2,955	\$3,219	\$3,345	\$3,362	\$3,420	\$3,734	\$3,801	\$3,969	\$4,254
Teachers	\$4,522	\$4,588	\$4,679	\$4,870	\$5,353	\$5,423	\$5,557	\$5,771	\$6,026	\$6,075	\$6,152	\$6,450	\$6,922	\$7,338	\$7,596	\$7,838
Total	\$12,523	\$12,453	\$12,949	\$13,388	\$14,684	\$14,916	\$15,302	\$16,333	\$17,077	\$17,283	\$17,791	\$18,177	\$19,972	\$21,330	\$22,896	\$24,622

FTE Enrollment

Table 46

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	775	779	763	742	696	680	661	644	628	614	617	601	560	556	553	542
Foundation Enrollment	695	676	666	646	628	603	590	559	542	536	528	526	522	494	484	481
Out-of-District FTE Pupils	75	78	73	73	80	82	86	82	85	88	90	88	85	87	87	84

NSS/Ch70 Funding Components (\$)

Table 47

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$4K	\$4K	\$3K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K
Req NSS (minus Ch70)	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$7K	\$7K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K
Actual NSS (minus Req NSS)	\$2K	\$1K	\$2K	\$2K	\$2K	\$2K	\$3K	\$3K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$6K	\$6K
Total NSS	\$10K	\$10K	\$11K	\$11K	\$12K	\$13K	\$13K	\$14K	\$15K	\$15K	\$15K	\$16K	\$17K	\$17K	\$18K	\$19K

Data: All Western MA Traditional Districts: Medium (801-1600 FTE)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 48

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$461	\$484	\$558	\$494	\$561	\$552	\$524	\$573	\$586	\$534	\$644	\$607	\$930	\$1,064	\$1,055	\$1,200
Administration	\$494	\$521	\$512	\$533	\$575	\$576	\$614	\$653	\$678	\$711	\$653	\$700	\$747	\$785	\$835	\$911
Instructional Leadership	\$893	\$902	\$885	\$1,039	\$1,062	\$1,091	\$1,113	\$1,169	\$1,150	\$1,071	\$1,063	\$1,113	\$1,191	\$1,263	\$1,373	\$1,573
Operations and Maintenance	\$1,150	\$1,085	\$1,155	\$1,095	\$1,146	\$1,208	\$1,233	\$1,218	\$1,228	\$1,203	\$1,326	\$1,383	\$1,564	\$1,663	\$1,836	\$1,953
Other Teaching Services	\$1,014	\$1,078	\$1,100	\$1,135	\$1,235	\$1,317	\$1,362	\$1,400	\$1,444	\$1,596	\$1,681	\$1,729	\$1,936	\$2,180	\$2,311	\$2,395
Pupil Services	\$1,724	\$1,693	\$1,757	\$1,763	\$1,901	\$1,984	\$2,096	\$2,184	\$2,208	\$2,406	\$2,464	\$2,403	\$2,634	\$2,967	\$3,353	\$3,581
Insurance, Retirement and Other	\$2,483	\$2,574	\$2,743	\$2,707	\$2,707	\$2,943	\$3,074	\$3,251	\$3,266	\$3,622	\$3,631	\$3,789	\$4,169	\$4,166	\$4,480	\$4,740
Teachers	\$4,813	\$4,890	\$5,055	\$5,048	\$5,152	\$5,357	\$5,513	\$5,578	\$5,674	\$5,960	\$6,251	\$6,385	\$6,758	\$6,773	\$7,182	\$7,431
Total	\$13,032	\$13,227	\$13,766	\$13,815	\$14,340	\$15,028	\$15,529	\$16,025	\$16,234	\$17,103	\$17,713	\$18,110	\$19,930	\$20,859	\$22,426	\$23,785

FTE Enrollment

Table 49

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,466	1,434	1,416	1,402	1,388	1,334	1,306	1,277	1,268	1,247	1,268	1,249	1,160	1,175	1,158	1,144
Foundation Enrollment	1,622	1,596	1,567	1,528	1,509	1,507	1,472	1,451	1,422	1,399	1,381	1,376	1,361	1,289	1,283	1,247
Out-of-District FTE Pupils	154	161	164	164	166	167	176	185	187	194	201	198	190	191	194	194

NSS/Ch70 Funding Components (\$)

Table 50

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$6K	\$6K	\$5K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$9K	\$9K
Req NSS (minus Ch70)	\$4K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$8K	\$8K	\$8K	\$9K
Actual NSS (minus Req NSS)	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$3K	\$3K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K
Total NSS	\$12K	\$12K	\$12K	\$13K	\$13K	\$14K	\$15K	\$16K	\$16K	\$17K	\$17K	\$18K	\$19K	\$19K	\$21K	\$22K

Data: All Western MA Traditional Districts: Large (1601-10K FTE)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 51																
Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$513	\$648	\$614	\$611	\$654	\$631	\$610	\$644	\$664	\$667	\$722	\$716	\$972	\$1,075	\$1,331	\$1,348
Administration	\$392	\$415	\$426	\$437	\$441	\$462	\$484	\$496	\$480	\$494	\$453	\$502	\$537	\$571	\$616	\$649
Instructional Leadership	\$811	\$859	\$865	\$856	\$837	\$886	\$915	\$932	\$956	\$940	\$998	\$1,028	\$1,060	\$1,153	\$1,195	\$1,231
Operations and Maintenance	\$1,014	\$993	\$1,009	\$986	\$984	\$1,023	\$1,093	\$1,098	\$1,111	\$1,152	\$1,244	\$1,238	\$1,401	\$1,450	\$1,685	\$1,683
Other Teaching Services	\$990	\$1,037	\$1,103	\$1,105	\$1,177	\$1,228	\$1,290	\$1,365	\$1,383	\$1,395	\$1,407	\$1,429	\$1,561	\$1,717	\$1,951	\$2,021
Pupil Services	\$1,531	\$1,536	\$1,622	\$1,611	\$1,697	\$1,797	\$1,849	\$1,926	\$1,948	\$2,069	\$2,171	\$2,128	\$2,200	\$2,654	\$2,955	\$3,148
Insurance, Retirement and Other	\$1,911	\$1,943	\$2,045	\$2,123	\$2,115	\$2,282	\$2,366	\$2,419	\$2,543	\$2,652	\$2,773	\$2,855	\$3,037	\$3,212	\$3,364	\$3,469
Teachers	\$4,613	\$4,604	\$4,707	\$4,868	\$4,990	\$5,067	\$5,194	\$5,393	\$5,482	\$5,569	\$5,863	\$6,031	\$6,461	\$6,780	\$7,001	\$7,292
Total	\$11,775	\$12,034	\$12,391	\$12,597	\$12,895	\$13,376	\$13,801	\$14,273	\$14,568	\$14,937	\$15,629	\$15,927	\$17,229	\$18,612	\$20,097	\$20,841

FTE Enrollment

Table 52																
FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	4,001	3,953	3,935	3,852	3,812	3,760	3,721	3,674	3,690	3,680	3,616	3,598	3,397	3,422	3,384	3,379
Foundation Enrollment	4,432	4,382	4,347	4,325	4,280	4,230	4,191	4,163	4,093	4,077	4,037	4,013	3,975	3,823	3,787	3,752
Out-of-District FTE Pupils	215	227	244	253	256	272	293	309	316	322	341	345	344	341	354	357

NSS/Ch70 Funding Components (\$)

Table 53																
Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$8K	\$9K	\$10K
Req NSS (minus Ch70)	\$3K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K
Actual NSS (minus Req NSS)	\$934	\$735	\$932	\$951	\$991	\$1K	\$1K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$3K	\$2K
Total NSS	\$10K	\$11K	\$11K	\$11K	\$12K	\$12K	\$13K	\$13K	\$14K	\$14K	\$15K	\$15K	\$16K	\$17K	\$18K	\$19K

Data: All Western MA Traditional Districts: Outliers (Springfield at 23,723 FTE in 2024)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 54																
Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$1,143	\$1,337	\$2,068	\$1,143	\$1,424	\$1,734	\$1,241	\$1,376	\$1,286	\$1,371	\$1,673	\$1,765	\$2,487	\$2,260	\$2,923	\$3,071
Administration	\$533	\$654	\$446	\$645	\$550	\$677	\$596	\$580	\$507	\$490	\$416	\$407	\$441	\$501	\$599	\$609
Instructional Leadership	\$719	\$778	\$892	\$1,173	\$1,257	\$1,245	\$1,208	\$1,282	\$1,280	\$1,060	\$983	\$1,090	\$1,103	\$1,402	\$1,534	\$1,571
Operations and Maintenance	\$1,150	\$1,081	\$1,262	\$1,176	\$1,132	\$997	\$1,121	\$1,027	\$1,063	\$1,156	\$1,337	\$1,093	\$1,534	\$1,536	\$3,453	\$3,806
Other Teaching Services	\$1,143	\$1,002	\$1,146	\$1,130	\$1,051	\$1,086	\$1,150	\$1,135	\$1,070	\$1,036	\$1,357	\$1,390	\$1,337	\$1,518	\$1,776	\$2,014
Pupil Services	\$2,031	\$1,873	\$2,156	\$2,353	\$2,183	\$2,257	\$2,393	\$2,467	\$2,613	\$2,845	\$2,968	\$2,631	\$2,590	\$3,602	\$3,675	\$3,901
Insurance, Retirement and Other	\$2,050	\$2,226	\$2,280	\$2,474	\$2,437	\$2,496	\$2,467	\$2,566	\$2,562	\$2,635	\$2,765	\$3,001	\$3,217	\$3,461	\$3,852	\$3,906
Teachers	\$5,839	\$4,859	\$4,750	\$4,711	\$4,848	\$4,823	\$4,963	\$5,196	\$4,962	\$5,274	\$5,510	\$5,800	\$6,442	\$7,095	\$7,561	\$7,737
Total	\$14,608	\$13,810	\$15,000	\$14,805	\$14,882	\$15,315	\$15,139	\$15,629	\$15,343	\$15,867	\$17,009	\$17,177	\$19,151	\$21,375	\$25,373	\$26,615

FTE Enrollment

Table 55																
FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	25,137	24,987	24,935	24,521	24,747	25,249	25,101	25,009	25,629	25,776	25,294	24,906	24,079	24,040	23,763	23,723
Foundation Enrollment	28,235	28,343	28,305	28,226	27,951	28,434	28,755	28,970	29,109	29,533	29,670	29,645	29,551	29,144	28,613	28,326
Out-of-District FTE Pupils	3,571	3,775	3,719	3,695	3,817	3,613	3,974	4,406	4,596	4,812	5,150	5,363	5,559	5,500	5,652	5,745

NSS/Ch70 Funding Components (\$)

Table 56																
Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$10K	\$10K	\$11K	\$11K	\$12K	\$12K	\$12K	\$12K	\$12K	\$13K	\$14K	\$15K	\$16K	\$17K	\$18K	\$20K
Req NSS (minus Ch70)	\$278	\$1K	\$2K	\$1K	\$2K	\$2K	\$1K	\$1K	\$1K	\$1K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K
Actual NSS (minus Req NSS)	\$-169	\$-435	\$-40	\$-97	\$-163	\$0	\$2	\$2	\$-0	\$-122	\$-0	\$-594	\$0	\$0	\$-0	\$-0
Total NSS	\$11K	\$11K	\$12K	\$13K	\$13K	\$13K	\$13K	\$14K	\$14K	\$14K	\$15K	\$16K	\$18K	\$19K	\$20K	\$22K

Data: All Western MA Traditional Districts: All Western MA (excl. Springfield)

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 57																
Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$507	\$590	\$593	\$575	\$624	\$600	\$580	\$622	\$636	\$627	\$686	\$671	\$967	\$1,043	\$1,240	\$1,287
Administration	\$440	\$461	\$471	\$488	\$507	\$520	\$544	\$566	\$589	\$580	\$542	\$587	\$628	\$669	\$713	\$754
Instructional Leadership	\$836	\$869	\$873	\$906	\$910	\$951	\$981	\$1,013	\$1,023	\$996	\$1,029	\$1,070	\$1,122	\$1,205	\$1,266	\$1,353
Operations and Maintenance	\$1,066	\$1,030	\$1,058	\$1,035	\$1,055	\$1,105	\$1,158	\$1,160	\$1,161	\$1,185	\$1,301	\$1,302	\$1,482	\$1,566	\$1,783	\$1,845
Other Teaching Services	\$1,025	\$1,073	\$1,133	\$1,141	\$1,232	\$1,284	\$1,342	\$1,413	\$1,449	\$1,508	\$1,553	\$1,582	\$1,714	\$1,895	\$2,100	\$2,190
Pupil Services	\$1,613	\$1,608	\$1,685	\$1,697	\$1,810	\$1,893	\$1,953	\$2,051	\$2,068	\$2,192	\$2,292	\$2,249	\$2,383	\$2,803	\$3,117	\$3,346
Insurance, Retirement and Other	\$2,083	\$2,124	\$2,247	\$2,301	\$2,318	\$2,491	\$2,589	\$2,689	\$2,816	\$2,966	\$3,060	\$3,157	\$3,404	\$3,516	\$3,715	\$3,881
Teachers	\$4,664	\$4,689	\$4,806	\$4,921	\$5,078	\$5,194	\$5,331	\$5,496	\$5,601	\$5,735	\$6,009	\$6,181	\$6,614	\$6,861	\$7,123	\$7,409
Total	\$12,234	\$12,444	\$12,865	\$13,063	\$13,532	\$14,039	\$14,479	\$15,010	\$15,343	\$15,789	\$16,472	\$16,798	\$18,314	\$19,558	\$21,055	\$22,065

FTE Enrollment

Table 58																
FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,555	1,534	1,521	1,493	1,469	1,438	1,418	1,372	1,371	1,361	1,352	1,339	1,255	1,266	1,252	1,245
Foundation Enrollment	1,603	1,580	1,563	1,543	1,524	1,505	1,486	1,469	1,442	1,432	1,416	1,408	1,394	1,333	1,322	1,306
Out-of-District FTE Pupils	120	125	129	130	134	138	146	150	152	155	162	161	158	158	162	162

NSS/Ch70 Funding Components (\$)

Table 59																
Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$8K	\$8K	\$9K	\$9K
Req NSS (minus Ch70)	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K
Actual NSS (minus Req NSS)	\$1K	\$1K	\$1K	\$1K	\$1K	\$2K	\$2K	\$2K	\$2K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K	\$3K
Total NSS	\$11K	\$11K	\$11K	\$12K	\$12K	\$13K	\$13K	\$14K	\$14K	\$15K	\$15K	\$16K	\$17K	\$17K	\$19K	\$20K

Data: Amherst-Pelham

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 60

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$646	\$534	\$900	\$417	\$459	\$424	\$599	\$860	\$635	\$384	\$610	\$559	\$477	\$527	\$457	\$799
Administration	\$652	\$664	\$687	\$789	\$987	\$907	\$953	\$926	\$996	\$857	\$759	\$780	\$788	\$859	\$928	\$1,277
Instructional Leadership	\$1,289	\$1,337	\$1,209	\$1,397	\$1,390	\$1,545	\$1,429	\$1,437	\$1,541	\$1,555	\$1,544	\$1,501	\$1,436	\$1,472	\$1,644	\$2,336
Operations and Maintenance	\$1,459	\$1,416	\$1,559	\$1,468	\$1,481	\$1,585	\$1,473	\$1,577	\$1,904	\$1,678	\$1,776	\$1,601	\$1,705	\$1,778	\$1,957	\$2,221
Other Teaching Services	\$1,199	\$1,264	\$1,313	\$1,468	\$1,588	\$1,739	\$1,770	\$1,883	\$2,014	\$2,066	\$2,122	\$2,214	\$3,010	\$3,050	\$2,968	\$2,851
Pupil Services	\$2,170	\$2,196	\$2,205	\$2,302	\$2,291	\$2,497	\$2,753	\$2,908	\$2,931	\$3,109	\$3,161	\$2,902	\$3,951	\$3,780	\$4,177	\$4,231
Insurance, Retirement and Other	\$3,305	\$3,636	\$3,673	\$3,811	\$3,526	\$3,954	\$4,086	\$4,103	\$4,218	\$5,006	\$5,047	\$5,074	\$5,237	\$5,062	\$5,887	\$6,248
Teachers	\$5,491	\$5,511	\$6,254	\$6,199	\$6,533	\$6,765	\$6,704	\$6,590	\$6,927	\$7,208	\$7,285	\$7,155	\$7,592	\$7,404	\$8,158	\$8,270
Total	\$16,211	\$16,558	\$17,800	\$17,851	\$18,255	\$19,416	\$19,767	\$20,284	\$21,166	\$21,863	\$22,304	\$21,786	\$24,196	\$23,932	\$26,176	\$28,233

FTE Enrollment

Table 61

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,712	1,659	1,573	1,536	1,519	1,470	1,450	1,378	1,366	1,342	1,334	1,346	1,278	1,265	1,241	1,209
Foundation Enrollment	1,801	1,763	1,726	1,647	1,610	1,581	1,529	1,532	1,496	1,490	1,433	1,399	1,399	1,330	1,310	1,274
Out-of-District FTE Pupils	112	129	138	143	138	149	169	190	192	170	158	152	155	145	150	147

NSS/Ch70 Funding Components (\$)

Table 62

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$8K
Req NSS (minus Ch70)	\$5K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$8K	\$8K	\$9K	\$9K	\$9K	\$10K	\$10K	\$11K	\$11K
Actual NSS (minus Req NSS)	\$4K	\$2K	\$3K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$5K	\$5K	\$6K	\$6K
Total NSS	\$14K	\$15K	\$16K	\$17K	\$18K	\$18K	\$19K	\$20K	\$21K	\$21K	\$22K	\$22K	\$23K	\$23K	\$24K	\$26K

Data: Amherst

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 63

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$684	\$547	\$606	\$931	\$1,098	\$1,098	\$1,265	\$1,242	\$1,119	\$798	\$1,092	\$733	\$1,563	\$1,561	\$1,475	\$1,328
Administration	\$627	\$698	\$735	\$709	\$787	\$809	\$919	\$944	\$981	\$799	\$767	\$827	\$874	\$896	\$981	\$1,213
Instructional Leadership	\$1,134	\$1,079	\$965	\$1,032	\$1,153	\$1,275	\$1,175	\$1,131	\$1,087	\$1,231	\$1,325	\$1,527	\$1,750	\$1,783	\$1,847	\$2,498
Operations and Maintenance	\$1,264	\$1,271	\$1,636	\$1,436	\$1,399	\$1,451	\$1,577	\$1,530	\$1,555	\$1,340	\$1,646	\$1,764	\$1,897	\$1,874	\$1,941	\$2,248
Other Teaching Services	\$1,762	\$1,813	\$1,923	\$2,345	\$2,621	\$2,610	\$2,840	\$2,652	\$2,751	\$2,973	\$3,084	\$3,053	\$3,908	\$3,898	\$3,746	\$4,612
Pupil Services	\$1,261	\$1,386	\$1,335	\$1,386	\$1,553	\$1,761	\$1,802	\$1,850	\$1,818	\$1,909	\$2,030	\$1,844	\$2,035	\$2,740	\$2,821	\$3,018
Insurance, Retirement and Other	\$3,353	\$3,610	\$3,821	\$3,852	\$3,941	\$4,189	\$4,201	\$4,295	\$4,231	\$4,840	\$5,212	\$5,363	\$5,661	\$5,561	\$6,022	\$6,605
Teachers	\$5,944	\$6,213	\$6,355	\$6,963	\$6,804	\$6,627	\$6,841	\$7,712	\$7,792	\$8,091	\$8,538	\$8,725	\$9,528	\$9,598	\$9,451	\$9,745
Total	\$16,029	\$16,617	\$17,376	\$18,654	\$19,356	\$19,820	\$20,620	\$21,356	\$21,334	\$21,981	\$23,694	\$23,836	\$27,216	\$27,911	\$28,284	\$31,267

FTE Enrollment

Table 64

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	1,387	1,331	1,284	1,215	1,196	1,197	1,184	1,154	1,156	1,163	1,131	1,107	1,017	1,052	1,062	1,002
Foundation Enrollment	1,398	1,405	1,356	1,279	1,269	1,227	1,205	1,192	1,199	1,136	1,106	1,134	1,097	1,045	1,065	1,055
Out-of-District FTE Pupils	65	72	73	84	90	90	101	113	110	108	124	122	118	123	129	134

NSS/Ch70 Funding Components (\$)

Table 65

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K
Req NSS (minus Ch70)	\$4K	\$6K	\$6K	\$7K	\$7K	\$7K	\$7K	\$8K	\$8K	\$8K	\$8K	\$9K	\$9K	\$9K	\$10K	\$11K
Actual NSS (minus Req NSS)	\$6K	\$5K	\$6K	\$6K	\$6K	\$7K	\$7K	\$8K	\$8K	\$9K	\$10K	\$9K	\$10K	\$11K	\$11K	\$12K
Total NSS	\$15K	\$16K	\$16K	\$18K	\$19K	\$19K	\$20K	\$21K	\$21K	\$21K	\$23K	\$24K	\$26K	\$26K	\$26K	\$29K

Data: Leverett

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 66																
Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$575	\$602	\$651	\$766	\$616	\$556	\$563	\$529	\$596	\$453	\$503	\$606	\$1,041	\$840	\$756	\$608
Administration	\$715	\$676	\$710	\$888	\$925	\$1,029	\$809	\$842	\$1,244	\$899	\$826	\$857	\$1,010	\$987	\$1,061	\$953
Instructional Leadership	\$1,015	\$1,008	\$1,072	\$1,244	\$1,257	\$1,393	\$1,357	\$1,344	\$1,431	\$1,461	\$1,278	\$1,317	\$1,318	\$1,354	\$1,430	\$1,663
Operations and Maintenance	\$1,755	\$1,427	\$1,532	\$1,741	\$1,479	\$2,421	\$2,272	\$1,707	\$1,209	\$1,765	\$1,491	\$1,458	\$2,131	\$2,296	\$2,292	\$2,158
Other Teaching Services	\$2,385	\$2,040	\$2,074	\$2,346	\$2,540	\$2,352	\$2,274	\$2,281	\$2,193	\$2,175	\$2,477	\$2,386	\$2,178	\$3,119	\$3,266	\$3,526
Pupil Services	\$1,395	\$970	\$1,096	\$1,599	\$1,613	\$2,392	\$2,073	\$1,701	\$2,082	\$2,081	\$1,845	\$1,641	\$2,690	\$1,910	\$2,635	\$2,814
Insurance, Retirement and Other	\$2,098	\$2,041	\$2,253	\$2,499	\$2,753	\$3,245	\$3,041	\$2,937	\$2,943	\$3,677	\$3,781	\$3,686	\$4,766	\$3,630	\$3,611	\$3,793
Teachers	\$5,218	\$5,690	\$5,818	\$6,301	\$6,138	\$7,547	\$6,868	\$6,967	\$6,814	\$7,967	\$7,665	\$7,134	\$9,224	\$6,888	\$6,724	\$6,395
Total	\$15,156	\$14,454	\$15,206	\$17,384	\$17,321	\$20,935	\$19,257	\$18,308	\$18,512	\$20,478	\$19,866	\$19,085	\$24,358	\$21,024	\$21,775	\$21,910

FTE Enrollment

Table 67																
FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	162	164	161	150	147	124	134	135	134	129	136	142	117	142	140	142
Foundation Enrollment	125	120	123	123	119	118	100	104	101	123	103	112	116	105	112	101
Out-of-District FTE Pupils	7	7	6	9	10	9	8	6	14	8	9	12	9	12	7	3

NSS/Ch70 Funding Components (\$)

Table 68																
Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$3K	\$2K	\$3K	\$3K
Req NSS (minus Ch70)	\$5K	\$5K	\$6K	\$6K	\$6K	\$8K	\$6K	\$6K	\$6K	\$8K	\$6K	\$6K	\$9K	\$7K	\$7K	\$7K
Actual NSS (minus Req NSS)	\$5K	\$5K	\$5K	\$6K	\$7K	\$7K	\$7K	\$8K	\$8K	\$7K	\$9K	\$9K	\$10K	\$10K	\$8K	\$8K
Total NSS	\$12K	\$12K	\$13K	\$14K	\$15K	\$17K	\$16K	\$16K	\$16K	\$17K	\$17K	\$17K	\$21K	\$18K	\$18K	\$17K

Data: Pelham

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 69

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$580	\$361	\$384	\$366	\$658	\$639	\$780	\$652	\$569	\$414	\$466	\$407	\$292	\$365	\$422	\$207
Administration	\$390	\$398	\$478	\$391	\$664	\$796	\$806	\$980	\$739	\$731	\$489	\$574	\$656	\$688	\$614	\$917
Instructional Leadership	\$1,123	\$1,159	\$1,089	\$1,064	\$1,113	\$1,252	\$1,288	\$1,322	\$1,278	\$1,431	\$1,194	\$1,609	\$1,670	\$1,873	\$1,782	\$1,885
Operations and Maintenance	\$920	\$847	\$925	\$884	\$992	\$947	\$1,080	\$942	\$987	\$889	\$1,229	\$1,304	\$1,498	\$1,380	\$1,524	\$1,614
Other Teaching Services	\$2,184	\$2,109	\$2,166	\$2,026	\$1,937	\$2,063	\$2,218	\$2,541	\$2,850	\$2,269	\$2,273	\$2,324	\$2,540	\$2,292	\$2,594	\$3,010
Pupil Services	\$1,032	\$1,090	\$1,347	\$1,507	\$1,584	\$1,615	\$1,696	\$1,797	\$1,774	\$1,672	\$1,732	\$1,642	\$2,112	\$2,203	\$1,888	\$3,610
Insurance, Retirement and Other	\$2,615	\$2,585	\$2,602	\$2,897	\$3,052	\$3,283	\$3,376	\$3,473	\$3,863	\$4,561	\$3,791	\$4,168	\$4,850	\$4,157	\$3,983	\$4,670
Teachers	\$5,889	\$5,474	\$5,627	\$5,576	\$5,035	\$6,090	\$6,540	\$6,418	\$5,904	\$6,310	\$6,129	\$6,860	\$7,406	\$6,714	\$6,430	\$7,127
Total	\$14,733	\$14,023	\$14,618	\$14,711	\$15,035	\$16,685	\$17,784	\$18,125	\$17,964	\$18,277	\$17,303	\$18,888	\$21,024	\$19,672	\$19,237	\$23,040

FTE Enrollment

Table 70

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	118	125	122	128	132	128	123	123	130	127	134	125	106	119	134	126
Foundation Enrollment	85	81	81	68	63	67	70	73	74	81	83	82	84	70	65	63
Out-of-District FTE Pupils	3	3	4	1	4	7	5	7	11	12	10	4	8	6	5	9

NSS/Ch70 Funding Components (\$)

Table 71

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K	\$2K
Req NSS (minus Ch70)	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$7K	\$5K	\$5K	\$5K
Actual NSS (minus Req NSS)	\$6K	\$5K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$7K	\$8K	\$7K	\$8K	\$10K	\$9K	\$9K	\$10K
Total NSS	\$12K	\$11K	\$11K	\$11K	\$11K	\$11K	\$12K	\$13K	\$13K	\$15K	\$14K	\$15K	\$19K	\$17K	\$16K	\$17K

Data: Shutesbury

PPE (\$/pupil), FTE Enrollment, and NSS/Ch70 Funding (\$)

PPE by Category (\$/pupil)

Table 72																
Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	\$580	\$498	\$564	\$865	\$704	\$515	\$636	\$684	\$1,510	\$420	\$385	\$873	\$1,296	\$551	\$1,102	\$545
Administration	\$652	\$943	\$767	\$836	\$877	\$726	\$884	\$1,084	\$1,040	\$846	\$856	\$938	\$925	\$1,100	\$934	\$1,015
Instructional Leadership	\$969	\$1,138	\$1,149	\$1,198	\$1,227	\$1,003	\$1,192	\$1,517	\$1,604	\$1,481	\$1,517	\$1,541	\$1,707	\$1,736	\$1,323	\$2,103
Operations and Maintenance	\$1,180	\$1,079	\$1,221	\$1,276	\$1,251	\$1,338	\$1,384	\$1,637	\$1,791	\$1,941	\$1,663	\$1,916	\$2,665	\$2,048	\$2,051	\$2,754
Other Teaching Services	\$2,185	\$2,447	\$2,361	\$2,193	\$2,179	\$2,591	\$2,164	\$2,817	\$3,301	\$3,491	\$3,545	\$3,560	\$3,722	\$5,320	\$3,441	\$4,359
Pupil Services	\$1,292	\$1,128	\$1,195	\$1,556	\$1,628	\$1,418	\$1,427	\$1,340	\$2,165	\$2,301	\$2,206	\$2,198	\$2,456	\$2,748	\$2,946	\$2,962
Insurance, Retirement and Other	\$2,332	\$2,687	\$2,596	\$2,690	\$2,564	\$2,484	\$2,328	\$3,107	\$3,773	\$4,388	\$2,970	\$3,963	\$3,350	\$3,154	\$4,007	\$3,501
Teachers	\$4,821	\$5,363	\$5,603	\$5,169	\$5,509	\$6,012	\$5,620	\$6,612	\$7,043	\$7,838	\$7,914	\$8,204	\$8,933	\$10,075	\$8,898	\$9,669
Total	\$14,011	\$15,283	\$15,456	\$15,783	\$15,939	\$16,087	\$15,635	\$18,798	\$22,227	\$22,706	\$21,056	\$23,193	\$25,054	\$26,732	\$24,702	\$26,908

FTE Enrollment

Table 73																
FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	162	151	146	151	150	157	156	131	120	124	127	125	113	115	123	110
Foundation Enrollment	156	149	141	138	143	132	141	146	118	116	118	117	105	98	100	111
Out-of-District FTE Pupils	7	5	7	10	9	13	9	12	13	16	18	15	16	18	19	18

NSS/Ch70 Funding Components (\$)

Table 74																
Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$4K	\$5K	\$5K	\$5K	\$5K	\$5K	\$6K	\$5K	\$5K	\$6K
Req NSS (minus Ch70)	\$4K	\$5K	\$5K	\$4K	\$5K	\$4K	\$4K	\$6K	\$5K	\$6K	\$5K	\$6K	\$6K	\$6K	\$6K	\$7K
Actual NSS (minus Req NSS)	\$4K	\$5K	\$6K	\$6K	\$6K	\$6K	\$6K	\$7K	\$9K	\$10K	\$9K	\$11K	\$11K	\$13K	\$10K	\$11K
Total NSS	\$12K	\$13K	\$14K	\$14K	\$14K	\$14K	\$15K	\$17K	\$19K	\$20K	\$20K	\$22K	\$23K	\$24K	\$21K	\$24K

Appendix D. Additional Visualizations (continued)

Geographic map showing district locations and enrollment cohorts (2019)

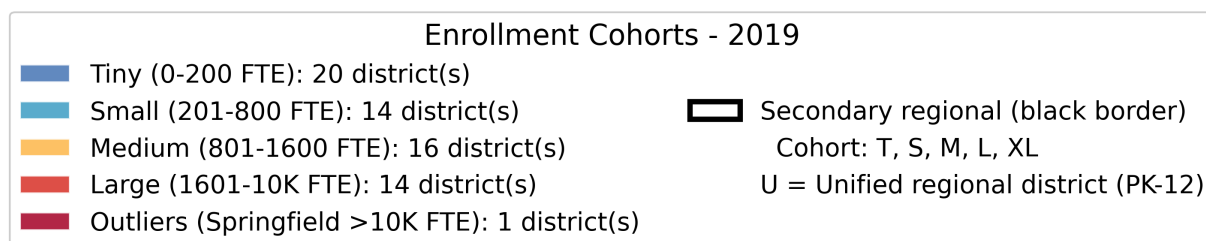
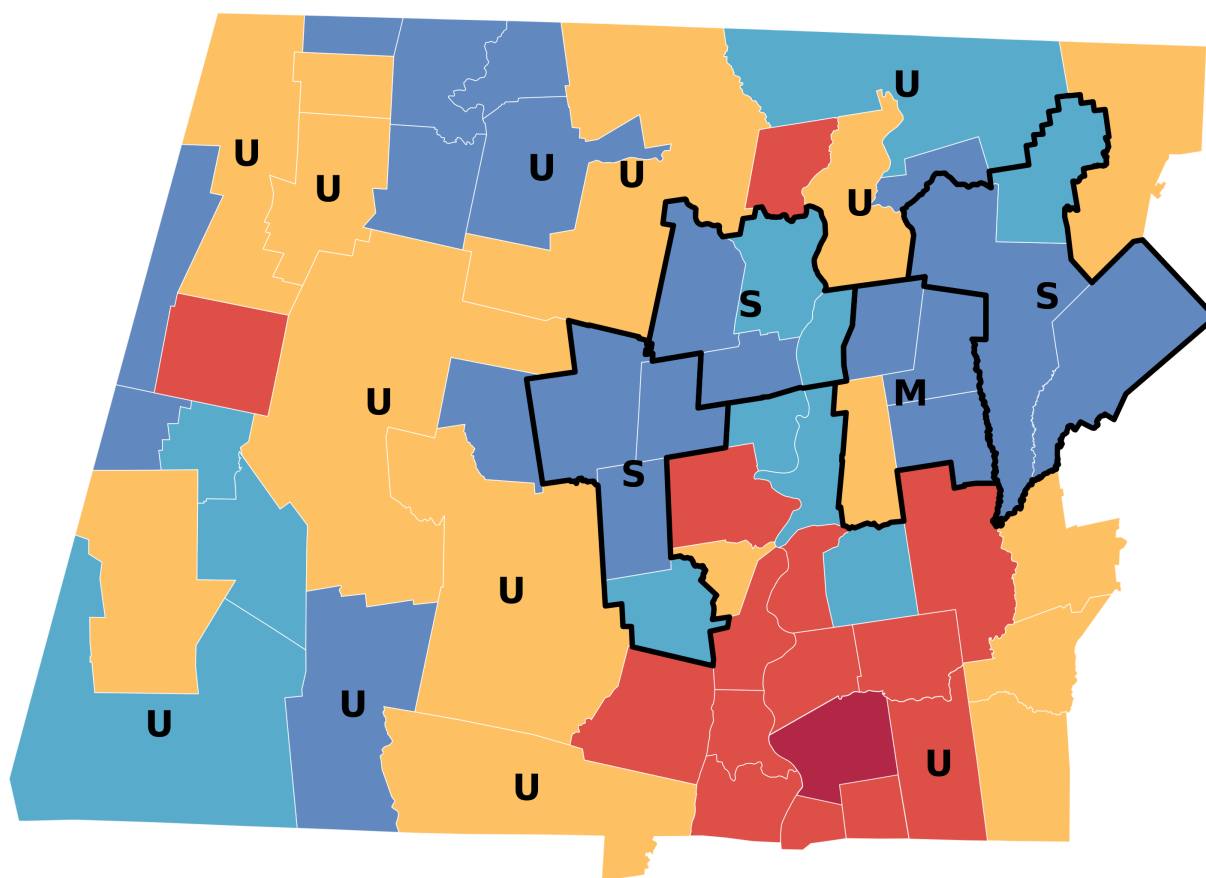


Figure 33

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Appendix D. Additional Visualizations (continued)

Geographic map showing 2019 PPE vs enrollment cohort baseline

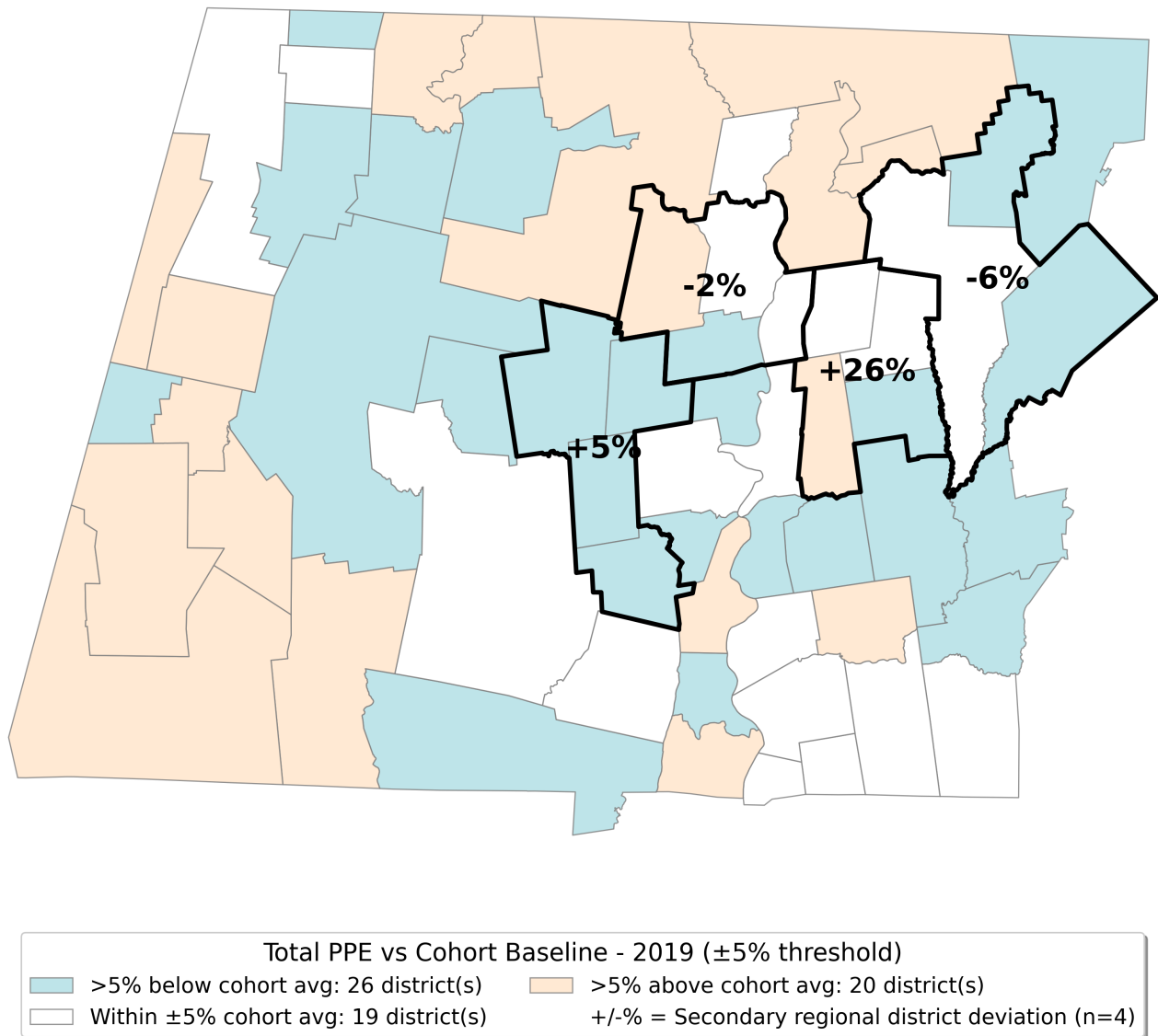


Figure 34

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort).

Appendix D. Additional Visualizations (continued)

Geographic map showing 10-year PPE growth (2009-2019) vs enrollment cohort baseline

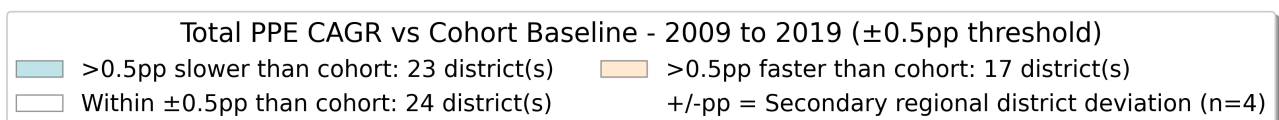
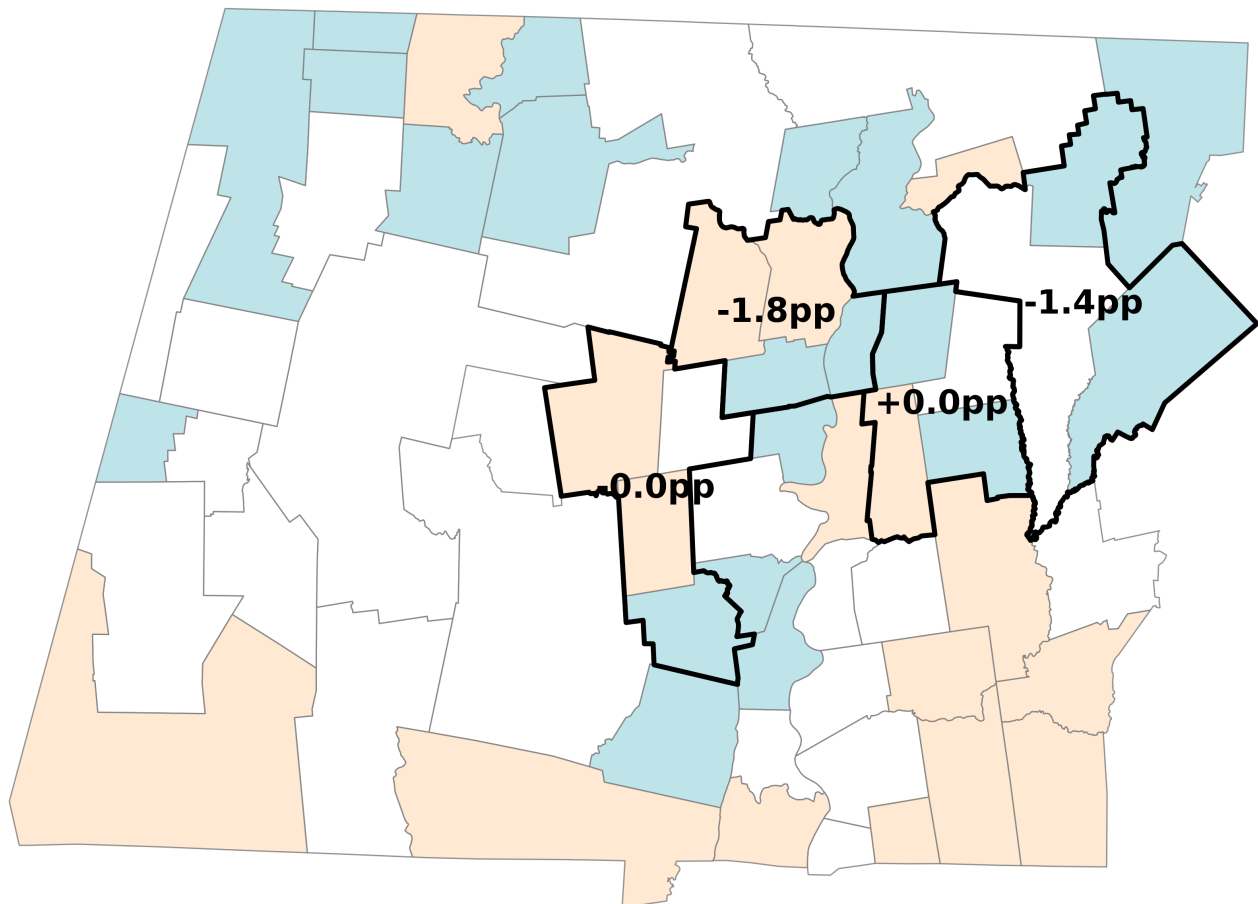


Figure 35

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate.

Appendix D. Additional Visualizations (continued)

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2014)

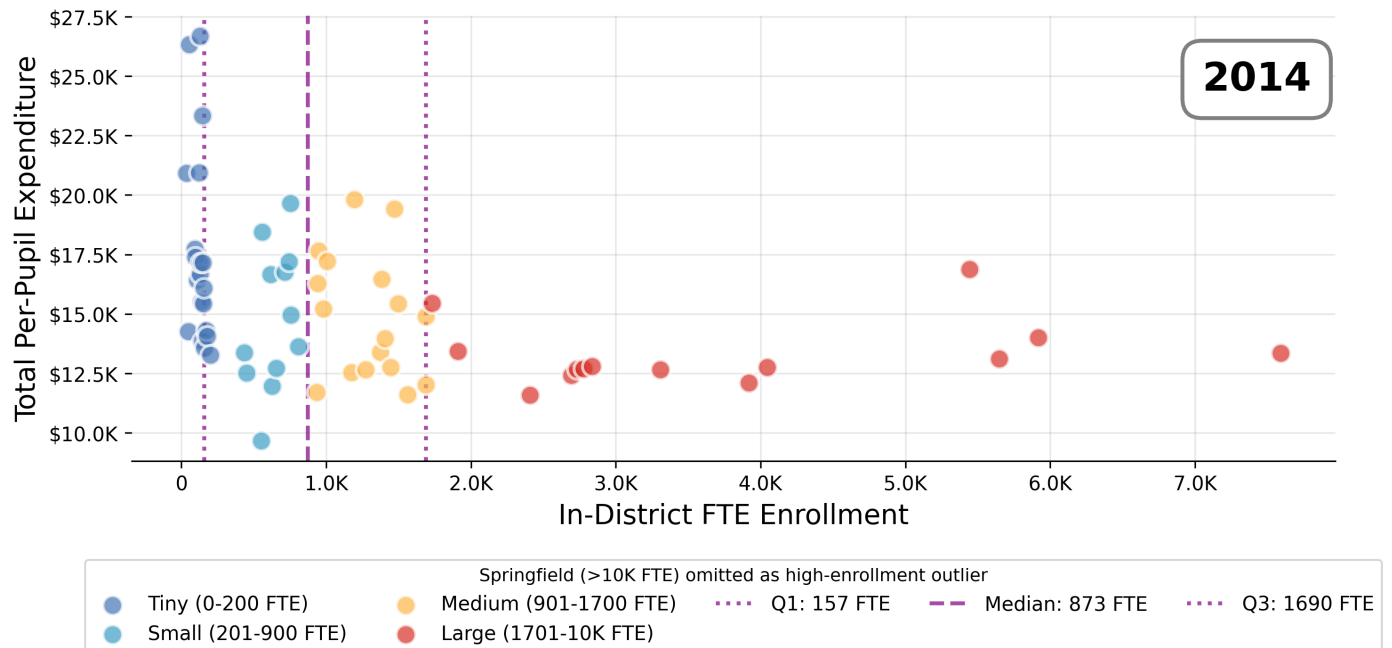


Figure 36

This scatterplot shows the relationship between district enrollment (x-axis) and per-pupil expenditure (y-axis) for Western Massachusetts traditional districts in 2014. Each point represents one district, colored by enrollment cohort. Horizontal lines show quartile boundaries used to define cohorts.

Appendix D. Additional Visualizations (continued)

Geographic map showing district locations and enrollment cohorts (2014)

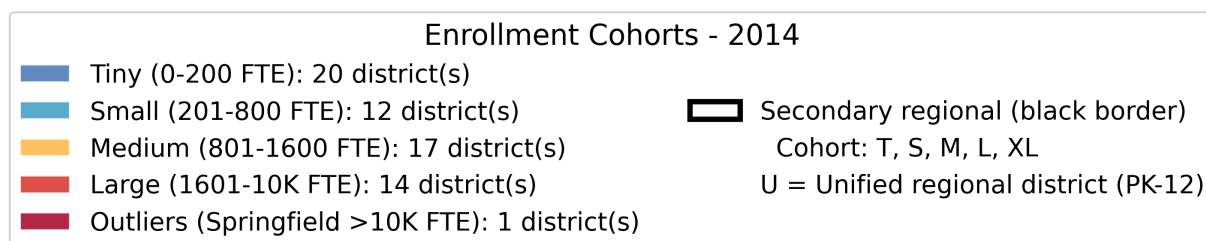
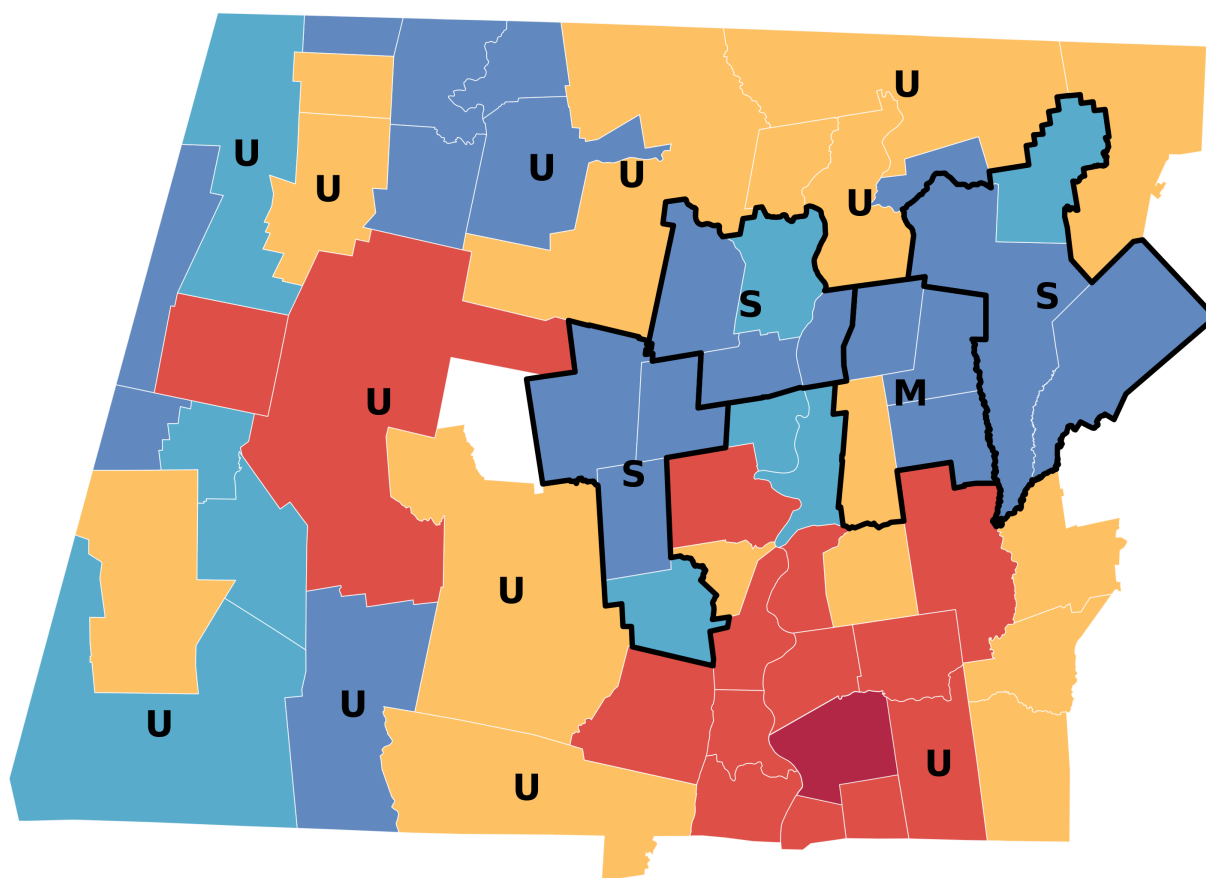


Figure 37

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Appendix D. Additional Visualizations (continued)

Geographic map showing 2014 PPE vs enrollment cohort baseline

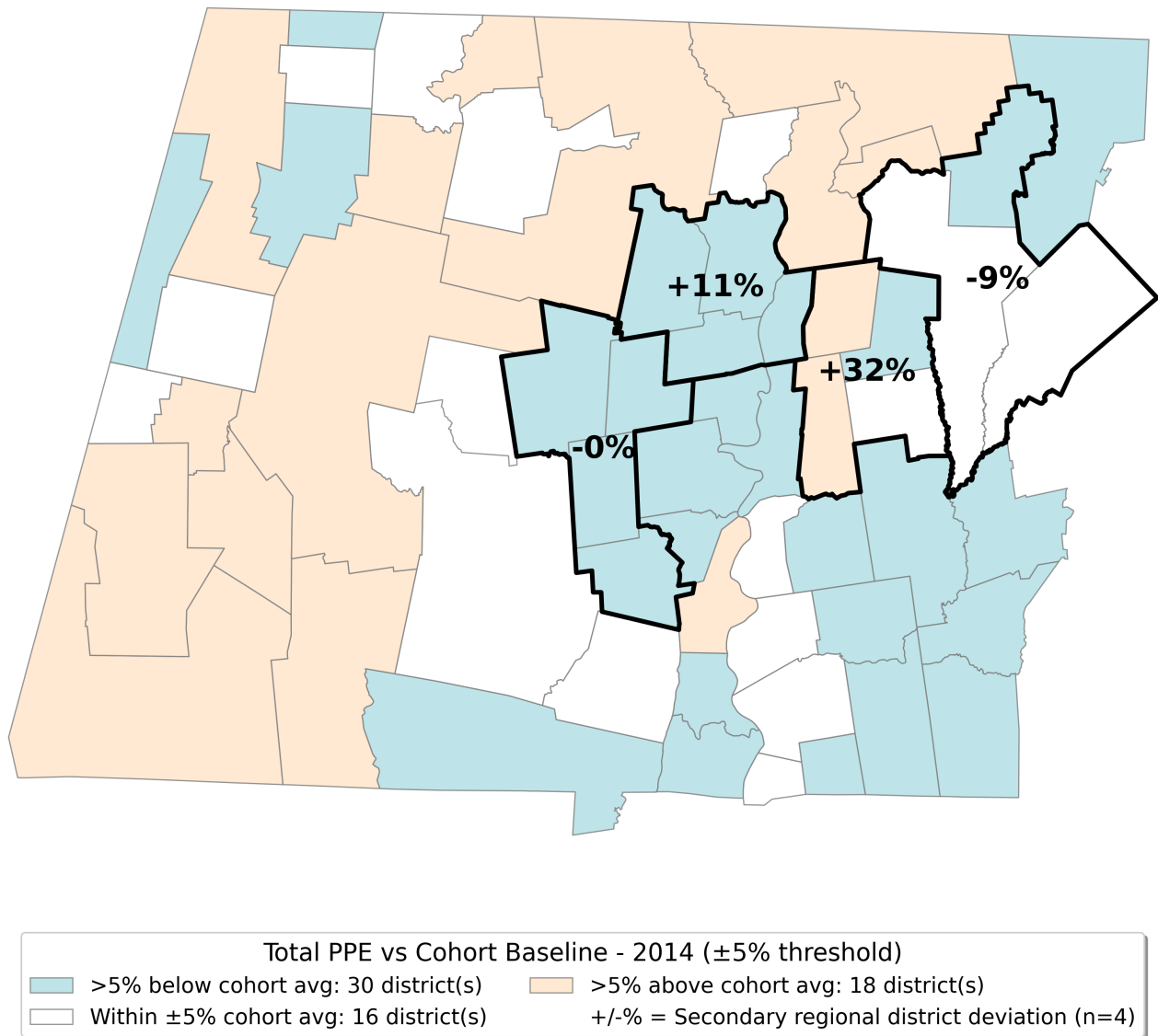


Figure 38

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort).

Appendix D. Additional Visualizations (continued)

Geographic map showing 5-year PPE growth (2009-2014) vs enrollment cohort baseline

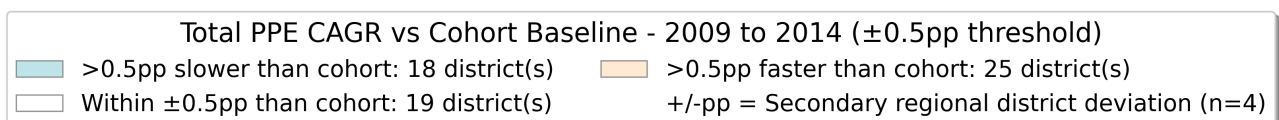
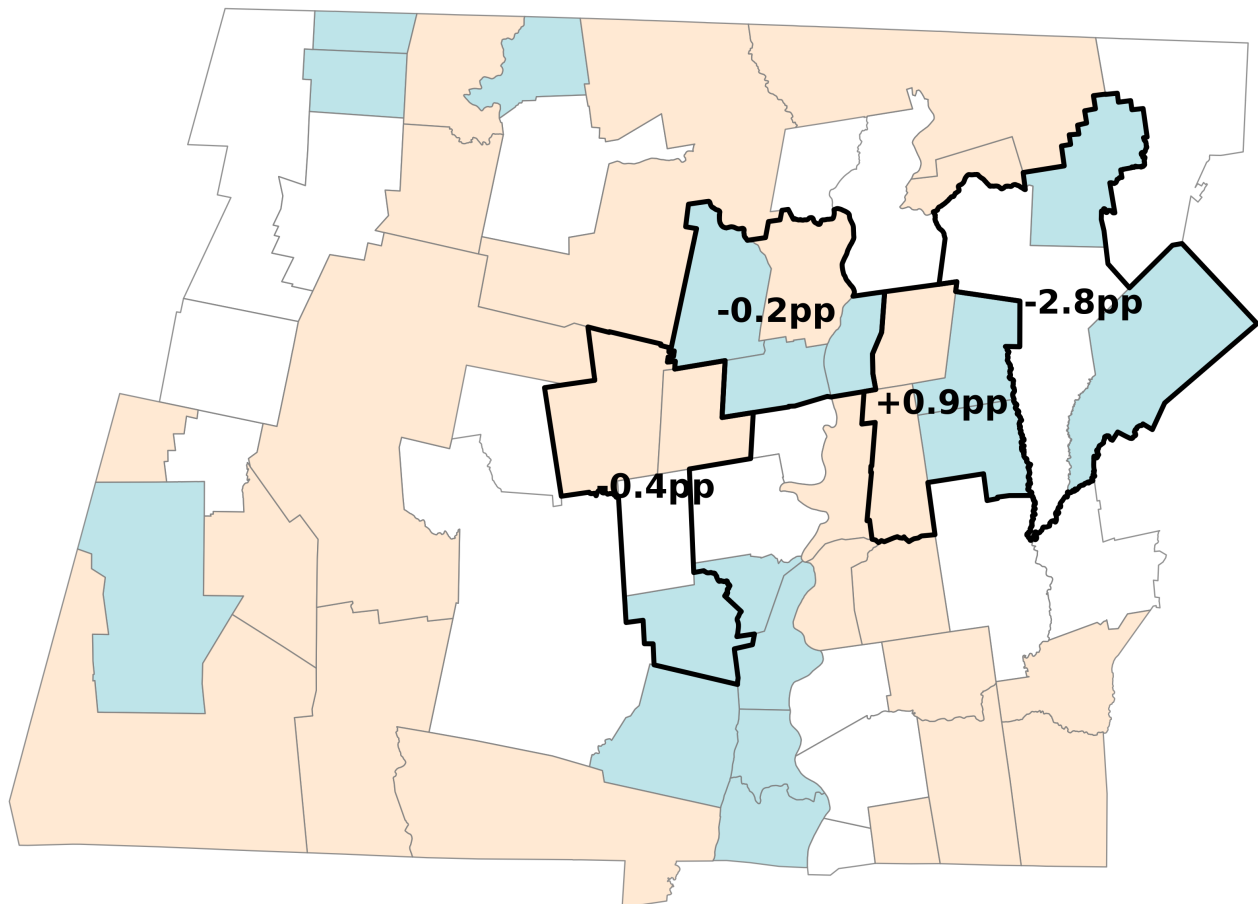


Figure 39

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate.

Appendix D. Additional Visualizations (continued)

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2009)

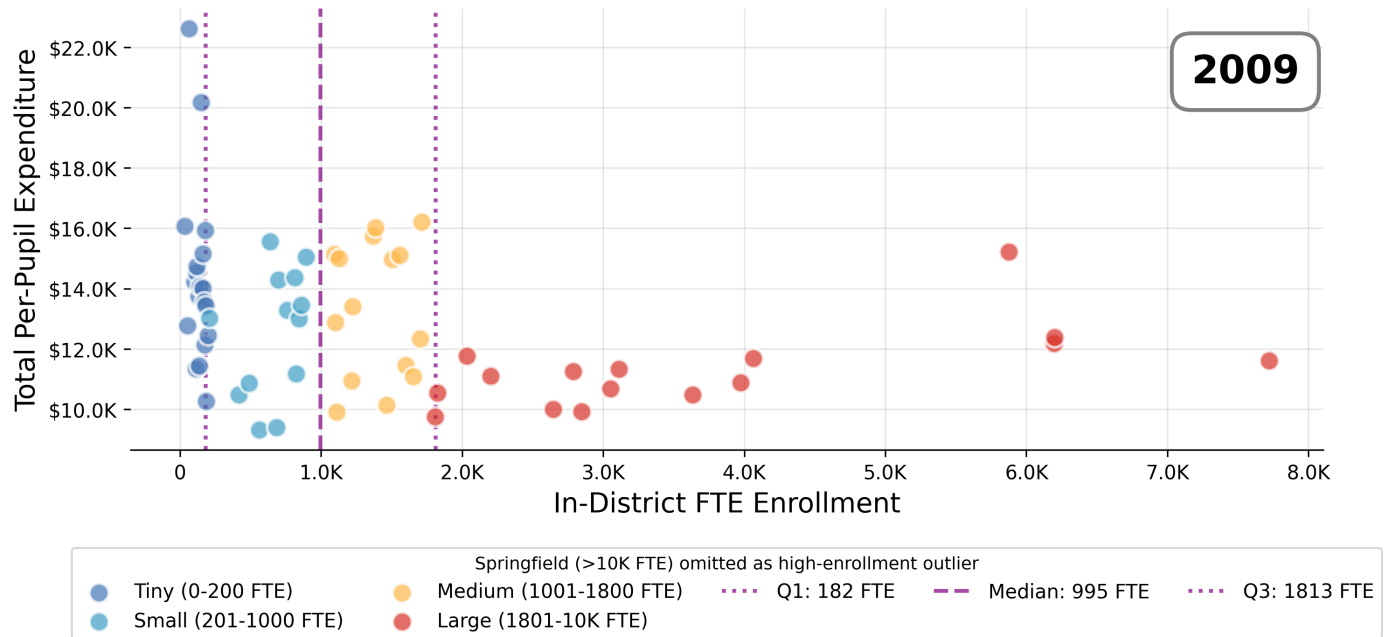


Figure 40

This scatterplot shows the relationship between district enrollment (x-axis) and per-pupil expenditure (y-axis) for Western Massachusetts traditional districts in 2009. Each point represents one district, colored by enrollment cohort. Horizontal lines show quartile boundaries used to define cohorts.

Appendix D. Additional Visualizations (continued)

Geographic map showing district locations and enrollment cohorts (2009)

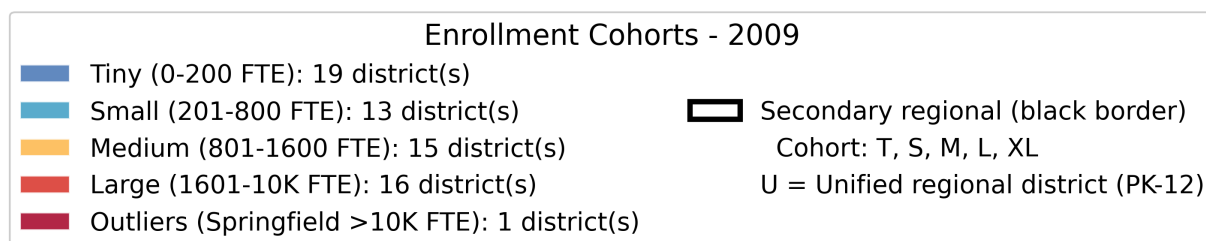
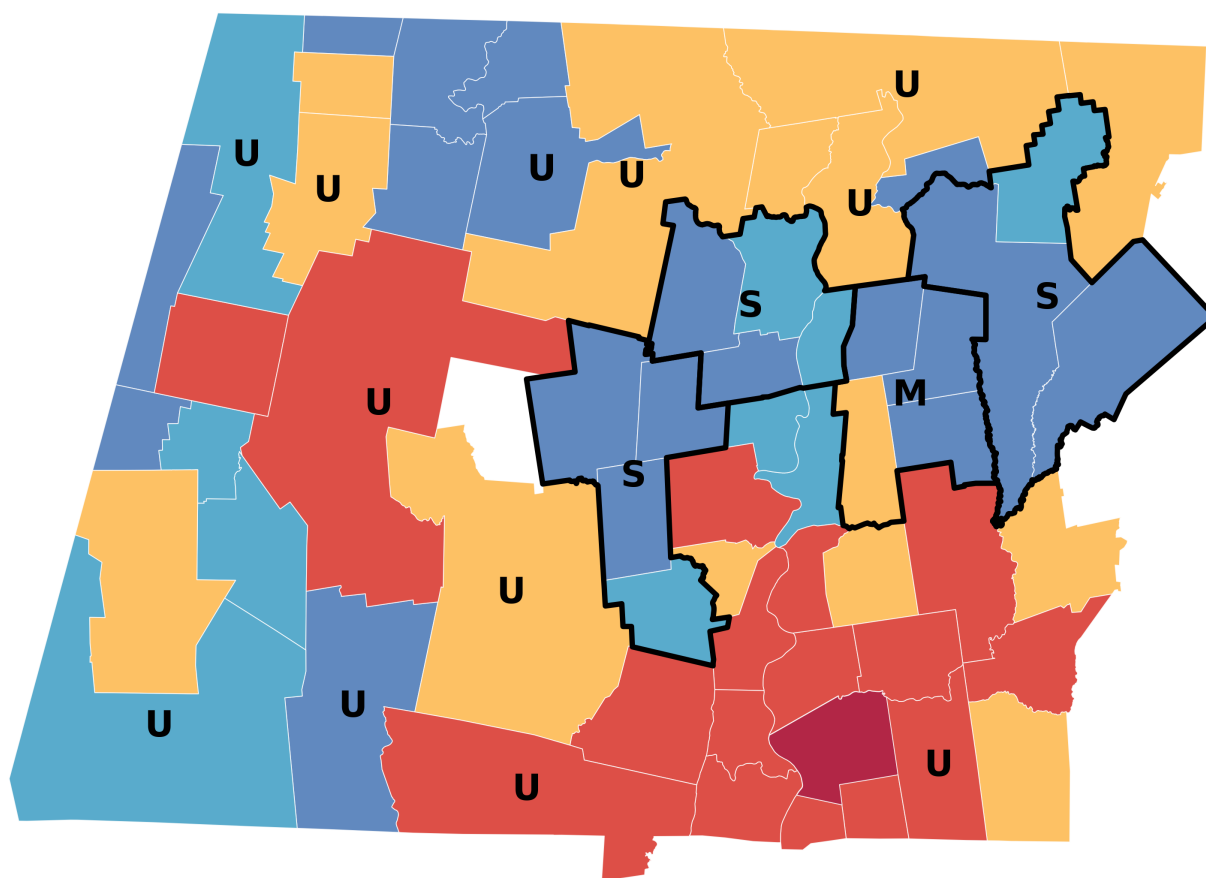


Figure 41

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Appendix D. Additional Visualizations (continued)

Geographic map showing 2009 PPE vs enrollment cohort baseline

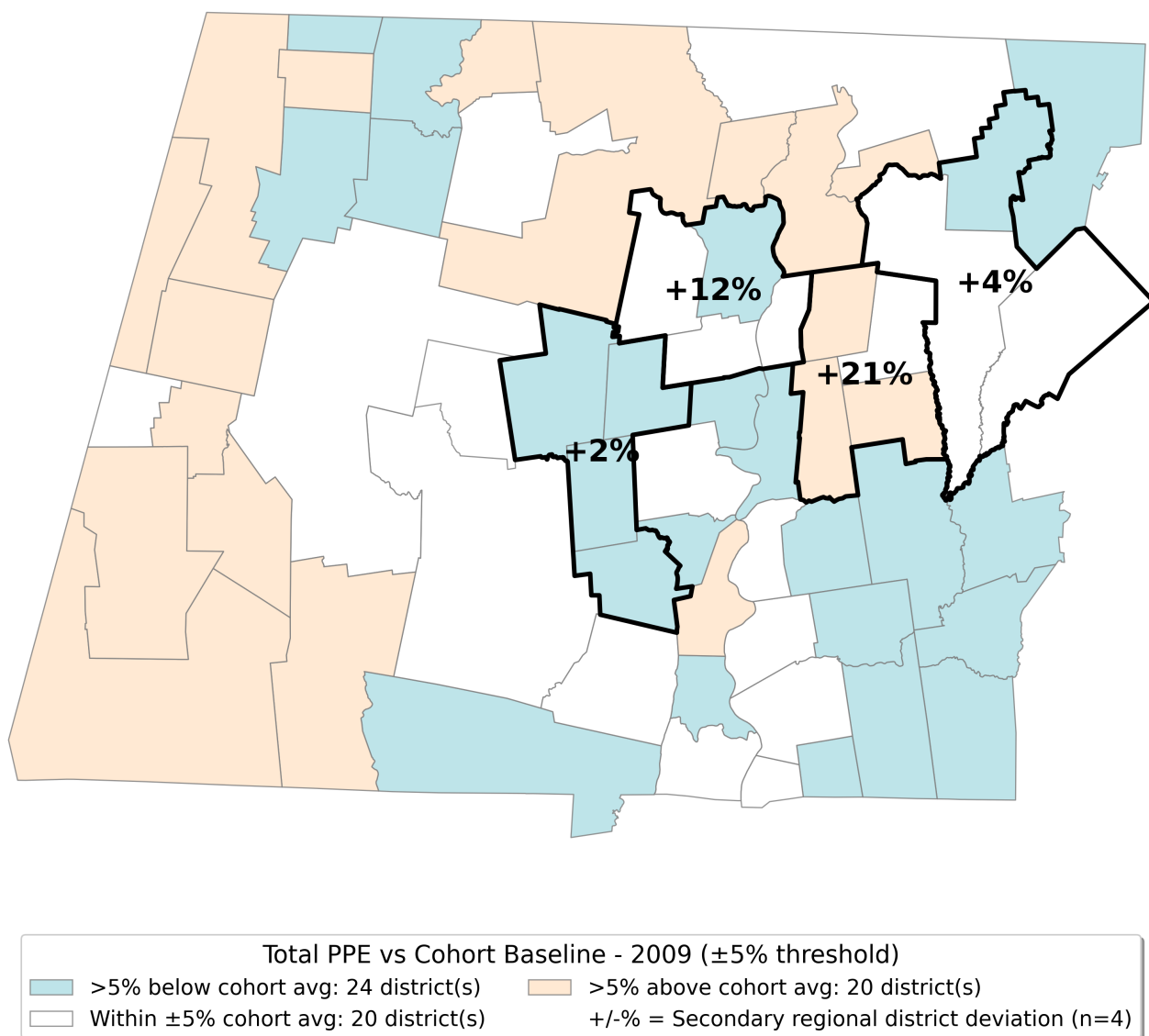


Figure 42

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort).